Flat Galactic Rotation Curves from the Dimensional Reduction of Curvature Transport

Daniel Banasik

April 5, 2026

Abstract

Flat galactic rotation curves are one of the most precisely measured departures from Newtonian gravity. I show they arise as a geometric consequence of curvature transport, with baryonic matter and standard weak-field gravity as the only inputs.

The argument begins with a single observation: in a dynamically evolving spacetime, curvature cannot remain purely local. Under finite-speed causal propagation and differential galactic rotation, orbital phase mixing reorganizes curvature transport from the three-dimensional bulk onto constant-radius two-dimensional shells. Transport on a two-dimensional manifold is governed by a Laplacian whose Green function is logarithmic; the resulting acceleration scales as $1/r$. This dimensional reduction emerges from the spectral structure of the Laplace–Beltrami operator on S^3 and independently from the retarded gravitational field of a differentially rotating source in linearized General Relativity.

The resulting dynamics consists of two coupled channels: a local volumetric $1/r^2$ flux sourced by baryonic matter and a nonlocal $1/r$ shell-transport contribution. Their combination is controlled by a crossfade set entirely by the enclosed baryonic mass. The effective acceleration can be written in compact form as

$$a_{\text{eff}}(r) = a_* \cdot \xi \cdot \frac{\xi^3 + \alpha_0}{1 + \xi^2}, \qquad \xi = \sqrt{\frac{a_b(r)}{a_*}} = \frac{\sqrt{GM_b(r)/a_*}}{r},$$

where $a_* = c^2/(\pi R)$ is a cosmological curvature scale fixed by the Hubble radius, $\xi = \sqrt{a_b/a_*}$ is a dimensionless compactness ratio and α_0 is a morphology-dependent projection efficiency. No fitted length, mass, or acceleration scale is introduced. The formula recovers Newtonian dynamics in the inner regime and reduces to $\alpha_0\sqrt{a_b\, a_*}$ in the outer regime, yielding the baryonic Tully–Fisher relation $V_\infty^4 \propto M_b$.

Applied to the 175-galaxy SPARC archive, the geometry-fixed normalization ($\alpha_0 = \kappa \equiv 1 + \pi^{-2}$, where κ is a geometric constant) achieves median $\chi^2 = 11.5$ and 71 simultaneous wins against both MOND and ΛCDM on the same sample; MOND and ΛCDM each win on the remaining galaxies. Allowing per-galaxy morphology adjustment yields median $\chi^2 = 2.82$ with 151 of 175 wins (86%), with per-galaxy α_0 as the single fitted parameter per galaxy. The framework performs particularly well in diffuse galaxies, where the crossfade responds to local compactness M_b/r^2.

The epistemic structure is made explicit: the $1/r$ scaling and its dimensional origin are derived; the confinement mechanism and crossfade structure are physically argued; the weight function is asserted; and the morphology factor α_0 is empirical. Open problems — including covariant embedding and first-principles prediction of morphology — are identified explicitly.

1 Gravity as Dynamic Curvature Transport

1.1 The problem

Flat galactic rotation curves are among the most precisely measured departures from Newtonian gravity, and their origin remains contested. In the inner regions of galaxies the observed centripetal acceleration follows the inverse-square law

$$a_b(r) = \frac{GM_b(r)}{r^2}, \tag{1}$$

sourced entirely by the observed baryonic mass $M_b(r)$. At larger radii the curves become flat — centripetal acceleration scales as $1/r$ — while the baryonic surface density continues to fall. No additional mass is observed. The discrepancy spans orders of magnitude in galaxy luminosity and surface brightness, and is measured with sufficient precision to distinguish between competing functional forms.

Within a purely local, three-dimensional description of gravity, this transition has no natural origin. The inverse-square law follows directly from flux conservation in three dimensions, and no change in scaling can occur without either modifying the source or introducing additional degrees of freedom.

Two frameworks have addressed this departure most successfully. Modified Newtonian Dynamics (MOND) [2] captures the transition with a single acceleration constant a_0, predicting rotation curves across a wide range of galaxy types with remarkable economy — an empirical achievement that has proved durable for over four decades. The ΛCDM framework accounts for the same observations through a dark matter halo whose density profile adds a velocity contribution on top of the baryonic one; it remains the most successful cosmological framework available, with independent support from the CMB power spectrum, baryon acoustic oscillations, and large-scale structure. Both accomplishments are real, and neither is in dispute here. This work stands on their ground: the SPARC dataset and the MOND benchmarks used throughout are products of that tradition, and the geometric mechanism proposed here is offered as a complementary account — one that, if the derivation holds, would explain *why* the phenomenology takes the form it does, rather than replacing the observation that it does.

This work has a direct intellectual debt to both frameworks. MOND's single parameter a_0 was the first inspiration: the fact that one number, with no adjustment per galaxy, organizes so much observational data demands explanation. This paper is, in one reading, a hunt for the geometric nature of that parameter — and the identification $a_* = c^2/(\pi R)$ is the answer the framework proposes. The second inspiration was the ΛCDM postulate of dark matter itself: the intuition that *something real is out there*, hidden from direct observation yet gravitationally consequential. That intuition is taken seriously here. What is proposed is not that the hidden thing does not exist, but that it may be geometric rather than material — a property of curvature transport rather than an additional substance.

The central question is therefore not how to adjust the law, but *what physical process could reorganize curvature such that its effective propagation changes dimensionality?* The following sections pursue that question geometrically.

1.2 From static fields to dynamic geometry

The analysis in this paper foregrounds the propagation and redistribution of curvature under finite-speed causal constraints — an aspect of gravitational dynamics that is present in General Relativity but not typically the focus of galactic modelling. The emphasis is on three coupled processes:

- curvature generation (local sources),
- causal propagation (finite update speed),
- geometric organization (how transport is structured by orbital dynamics).

In the weak-field, quasi-static limit relevant to galactic dynamics, this perspective recovers standard Newtonian sourcing in the inner regime. The question pursued here is whether the propagation and organizational structure of curvature transport produces observable consequences at large radii — specifically, whether orbital phase mixing is sufficient to reorganize transport onto effectively two-dimensional shells.

1.3 Dimensional consequence of curvature transport

The scaling behavior of a field generated by a localized source is determined by the dimensionality of the space in which it propagates.

For a Laplacian in d spatial dimensions, the Green function satisfies

$$\Delta G_d(r) = -\delta(r), \tag{2}$$

with solutions of the form

$$G_d(r) \sim \begin{cases} r^{2-d}, & d \neq 2, \\ \log r, & d = 2. \end{cases} \tag{3}$$

The corresponding force scales as

$$|\nabla G_d(r)| \sim \begin{cases} r^{1-d}, & d \neq 2, \\ 1/r, & d = 2. \end{cases} \tag{4}$$

Thus:

- three-dimensional propagation produces the inverse-square law $1/r^2$,
- two-dimensional propagation produces an inverse-linear law $1/r$.

This observation has a direct implication:

> *any mechanism that effectively reduces curvature transport from three to two dimensions will necessarily produce a $1/r$ acceleration.*

This statement assumes that the transport on the reduced manifold is governed by a Laplacian-type operator. In the present framework, that this condition is satisfied for shell transport is established explicitly in Section 4, where the bulk dynamics induces a two-dimensional Poisson equation on the shell.

The problem of galactic dynamics is therefore reframed as a geometric and dynamical question:

> *under what conditions does curvature transport become effectively two-dimensional?*

1.4 Two channels of curvature dynamics

I propose that gravitational support in galaxies arises from two complementary curvature channels:

- a local volumetric channel, corresponding to standard three-dimensional flux from baryonic matter,
- a transport channel, in which curvature is redistributed along lower-dimensional manifolds.

The first channel is governed by local field equations and reproduces the inverse-square law. The second does not introduce new sources; instead, it reorganizes existing curvature through transport.

The mechanism described in this work can be formulated entirely within General Relativity in the weak-field limit. The Wave Theory framework provides a geometric language that renders the structure of the transport process explicit.

The key point is that the second channel is not imposed phenomenologically. It is argued to emerge from the dynamics of curvature propagation itself, as the derivations in Sections 3 and 4.5 establish.

1.5 The central claim

The central claim of this work is that:

> *in a dynamically evolving spacetime, curvature cannot remain purely local. Under causal propagation and orbital phase mixing, it is reorganized — by the mechanism established in Sections 3 and 3.4 — into transport along two-dimensional shells.*

Once this occurs, the inverse-linear scaling is no longer an assumption or modification—it is the direct consequence of the Green function on the transport manifold.

The structure of the argument is therefore:

1. curvature is continuously generated in the bulk,
2. causal propagation prevents purely static configurations,
3. orbital dynamics *organizes* transport into shell-like manifolds (established by physical argument in Section 3),
4. transport on these manifolds is intrinsically two-dimensional (derived in Section 4.5 and Appendix A),
5. the resulting Green function produces a logarithmic potential and a $1/r$ acceleration.

1.6 Interpretation

The apparent modification of gravitational scaling at large radii arises within the standard gravitational setting — it reflects the transition from local three-dimensional propagation to nonlocal, effectively two-dimensional transport. The sections that follow establish this transition as a precise geometric and dynamical result, not a qualitative analogy.

1.7 Proof status and epistemic accounting

The framework developed in this paper rests on several distinct types of claim. To make the logical structure transparent, I distinguish four categories throughout:

- **Derived** — follows necessarily from stated assumptions and prior steps, with no additional input.
- **Argued** — established by a physical or mathematical argument that is well-motivated but not fully rigorous; the conclusion is strongly suggested but not compelled.
- **Asserted** — taken as a postulate or definition of the framework; valid as a starting point but not derived within it.
- **Empirical** — requires observational input to fix; the geometric framework constrains the interpretation but not the value.

Table 1 summarises the status of every principal claim in the paper. Readers are encouraged to consult it when evaluating the weight carried by any individual result.

Table 1: Epistemic status of the principal claims in this paper. *Derived*: follows necessarily from stated assumptions. *Argued*: well-motivated but not fully rigorous. *Asserted*: taken as a postulate or framework definition. *Empirical*: value requires observational input.

Claim	Status	Where established
3D flux conservation $\Rightarrow r^{-2}$ local acceleration	Derived	Sec. 4.1, App. A.1
$Q_b^{\mathrm{src}} = 4\pi G M_b$ via Gauss law	Derived	Sec. 4.2
Shell Poisson equation derived from bulk transport	Derived	Sec. 4.5, App. A
Curvature transport is effectively 2D at large r	Derived	Sec. 4.5, App. A
α_0 is the unique undetermined factor in a_{shell}	Derived	Sec. 4.5

Continued on next page

Table 1 continued

Claim	Status	Where established
2D Green function $\Rightarrow$ logarithmic potential	Derived	Sec. 4.5.7, App. B
Logarithmic potential $\Rightarrow$ r^{-1} shell acceleration	Derived	Sec. 4.5.7, App. B
$v_\infty^4 \propto M_b$ (BTFR scaling)	Derived	App. C.1
Quasi-static limit valid ($\varepsilon_1, \varepsilon_2 \ll 1$, screening $\ll 1$; given $\tau_c = T_p$, $\kappa/D \sim 1/R^2$)	Derived	Sec. 4.5.3
Spiral winding confines transport to constant-r shells	Derived	Sec. 3.3, Sec. 3.4, App. F
Effective MOND scale $a_0^{\mathrm{eff}} = \alpha_0^2\, a_*$	Derived	Sec. 5.4
Screening scale $\kappa/D \sim 1/R^2$ (cosmological)	Argued	Sec. 4.5.3
Orbital averaging produces a smooth shell source	Argued	Sec. 3
WTMedium co-moves with baryonic flow (D/Dt substitution)	Argued	Sec. 3.3
Shell occupancy $\eta_s \sim \mathcal{O}(1)$ in transition regime	Argued	App. C.1
Crossfade formula $(1-\sigma)a_b + \sigma\, \alpha_0 \sqrt{a_b a_*}$	Argued	Sec. 5.1
Crossfade weight $\sigma(r) = [1 + (a_b/a_*)]^{-1}$	Argued	Sec. 5.2, 5.3.1
Causal update speed $\nu = \gamma c$ with $\gamma \le 1$	Asserted	Sec. C.1

Continued on next page

Table 1 continued

Claim	**Status**	**Where established**
$G = a_* = c^2/(\pi R)$ (WT definition)	Asserted	App. D
Causal memory time $\tau_c = T_p$ (Planck time)	Asserted	Sec. 4.5.3
Morphology factor α_0; A_{morph} distribution is unimodal, centred near κ	Empirical	Sec. 6.3

The paper proceeds as follows. Section 2 establishes the geometric structure on S^3 and derives the bulk-to-shell projection. Section 4 develops the two-channel curvature model, culminating in the derivation of the shell Poisson equation and the emergence of the r^{-1} scaling in Section 4.5. Subsequent sections determine the geometric coupling, morphology dependence, and observational implications of the model.

1.8 Scope and foundational independence

This paper addresses a single, precisely delimited question: *what geometric mechanism, operating within standard weak-field gravity and baryonic matter alone, produces flat galactic rotation curves?* The derivational chain — causal curvature transport on S^3, orbital phase mixing, shell confinement, two-dimensional Poisson equation, logarithmic Green function, $1/r$ acceleration — is self-contained and does not depend on completing a broader programme. The only inputs required are: (i) finite-speed causal propagation, (ii) differential rotation of the source, and (iii) standard weak-field gravitational sourcing. No additional structure from Wave Theory is required.

Three questions outside this scope are worth naming explicitly, precisely because a reader might otherwise expect them to be answered here.

First, the dynamical equations used in this work — the bulk transport equation, the Gauss projection, the shell Poisson equation, and the mixing function — are assembled from geometric and physical arguments rather than derived from a single covariant action functional. A minimal action formulation exists as a structural proposal, but its completion — including derivation of transport parameters from geometry and variational emergence of shell dynamics — remains open. This does not affect the results derived here: each equation in the chain is independently motivated, and the chain itself is the claim under scrutiny.

Second, the transport equation $\tau_c \partial_t^2 \Phi + \partial_t \Phi = D \Delta_{S^3} \Phi - \kappa \Phi + S_b$ is physically motivated by causality, finite propagation speed, and memory — but its parameters τ_c, D, κ are not derived from a Lagrangian by variation. They are constrained by the quasi-static and large-radius limits in which the shell Poisson equation is recovered, and their geometric interpretation is identified in Section 4.5. A variational derivation would strengthen the framework but is not required for the weak-field predictions presented here.

Third, the projection from bulk to shell dynamics is established through the Gauss flux identity and the spectral structure of the Laplace–Beltrami operator on S^3. The shell is treated as an effective manifold onto which curvature transport is organized by orbital dynamics, not as a variational degree of freedom. Whether a formulation exists in which the shell emerges as a boundary term or brane action is an open problem, identified as such in Section 8.

The appropriate precedent is phenomenological: Milgrom's original MOND formulation [2] identified the correct kinematic relation without a covariant action; Bekenstein–Milgrom AQUAL [3] followed a year later, and the covariant completion TeVeS [4] two decades after. The original result was not weaker for lacking those completions — it was correctly scoped. The present paper makes an analogous claim: that a specific transport mechanism produces a specific observational signature. The completeness of the underlying field theory is a separate question.

A note on the role of Wave Theory in this work. Wave Theory is not presented here as a framework the reader must adopt. It is better understood as a geometric way of thinking about the Universe — one that makes certain structures visible that are harder to see in the standard field-theoretic language. This is not a new ambition: it stands in the tradition of Einstein's original formulation of General Relativity and his later conviction that spacetime geometry, rather than fields propagating through a passive background, is the fundamental ontology. Whether the reader finds that perspective illuminating or unnecessary is an open and unconditional choice. The derivational chain — causal transport, orbital phase mixing, dimensional reduction, $1/r$ acceleration — can be followed and evaluated independently of any commitment to the broader framework.

2 Geometric Framework and Bulk–Shell Interface

2.1 Geometry of the system

The spatial geometry is modelled as a three-dimensional sphere S^3 of radius R. Points on S^3 are parameterized by the dimensionless radial coordinate

$$\chi = \frac{r}{R}, \qquad 0 \le \chi \le \pi, \tag{5}$$

where r denotes the geodesic distance from a chosen origin.

Surfaces of constant χ define two-dimensional spheres S_r embedded in S^3. These surfaces will serve as the manifolds on which curvature transport is organized.

For a given χ, the enclosed region defines a spherical cap $V_{\text{cap}}(\chi)$ with volume

$$V_{\text{cap}}(\chi) = 2\pi R^3 \left(\chi - \frac{1}{2} \sin 2\chi \right), \tag{6}$$

and boundary area

$$A_\partial(\chi) = 4\pi R^2 \sin^2 \chi. \tag{7}$$

In the limit $\chi \ll 1$, these expressions reduce to the familiar Euclidean forms

$$V_{\text{cap}} \approx \frac{4}{3}\pi r^3, \qquad A_\partial \approx 4\pi r^2, \tag{8}$$

ensuring consistency with local geometry.

2.2 Bulk curvature source

The baryonic contribution to curvature is described through a flux field F_b satisfying

$$\nabla \cdot F_b = s_b, \tag{9}$$

where s_b represents the volumetric curvature source associated with the baryonic mass distribution.

The total curvature content enclosed within a region of radius r is then given by

$$Q_b^{\text{src}}(r) = \int_{V_r} s_b \, dV. \tag{10}$$

This quantity represents the conserved curvature budget generated by the baryonic source.

2.3 Bulk–to–shell projection

The transfer of curvature from the bulk to the boundary is naturally described by the flux of F_b through the surface S_r. The surface source density is defined as

$$\sigma_r(\omega) = F_b(r, \omega) \cdot n_r, \tag{11}$$

where n_r is the outward unit normal and ω denotes angular coordinates on S_r.

The total curvature injected into the shell is then

$$Q_b^{\mathrm{src}}(r) = \int_{S_r} \sigma_r(\omega)\, dA. \tag{12}$$

Since $\nabla \cdot F_b = s_b$ holds throughout the bulk, the divergence theorem gives the same result for any surface enclosing the same source region, establishing Q_b^{src} as a conserved quantity independent of the specific projection surface chosen.

This construction shows that the shell is not an additional structure, but the natural boundary on which the conserved curvature flux appears as a source.

2.4 Connection to General Relativity

In the weak-field limit of General Relativity, the gravitational potential Φ satisfies the Poisson equation

$$\nabla^2 \Phi = 4\pi G \rho_b, \tag{13}$$

where ρ_b is the baryonic mass density.

Defining the curvature flux field as

$$F_b \equiv -\nabla \Phi, \tag{14}$$

we obtain

$$\nabla \cdot F_b = -4\pi G \rho_b. \tag{15}$$

The enclosed curvature source becomes

$$Q_b^{\mathrm{src}}(r) = \int_{S_r} F_b \cdot n_r\, dA = 4\pi G M_b(r), \tag{16}$$

where $M_b(r)$ is the enclosed baryonic mass and $Q_b^{\mathrm{src}} \equiv -Q_b = 4\pi G M_b(r)$ denotes the positive source magnitude.

Thus, the bulk-to-shell projection introduced above coincides exactly with the Gauss-law flux of the gravitational field. No modification of the local gravitational law is required. Every subsequent quantity introduced in this paper has an exact GR counterpart; readers who prefer to work entirely within standard GR language are referred to Appendix E, which provides a self-contained restatement of the full framework and an explicit identification of the three elements that are new relative to textbook GR.

2.5 Role of the shell as a transport interface

The geometric construction separates the system into two complementary components:

- the bulk region $V_{\text{cap}}(\chi)$, which contains the curvature source,
- the boundary surface S_r, on which the curvature flux appears as a surface source.

The shell therefore acts as an interface that receives curvature generated in the bulk and provides the manifold on which its subsequent redistribution takes place.

At this stage, no assumption has been made about the dynamics of this redistribution. The shell is defined purely as the geometric locus on which the conserved curvature flux is realized.

The dynamical question of how curvature transported from the bulk becomes organized on the shell, and under what conditions this transport acquires an effectively two-dimensional character, is addressed in the following section.

3 Orbital Organization of Curvature Transport

This section establishes that coherent curvature transport is confined to constant-radius shells S_r as a direct dynamical consequence of causal propagation and differential rotation. The confinement is derived twice, from two independent starting points, and the two derivations are complementary.

Section 3.3 derives the winding mechanism within the Wave Theory framework, using the eikonal limit of the bulk transport equation in the co-moving frame. This derivation uses the geometric language of the WTMedium and provides the foundation for the dimensional reduction developed in Section 4.5.

Section 3.4 and Appendix F establish the same mechanism independently within standard weak-field General Relativity, starting from the retarded gravitational potential of a differentially rotating source. This derivation requires no WT-specific assumptions: the co-moving frame, the telegraph equation, and the constitutive parameters τ_c, κ/D play no role. The Doppler-shifted dispersion relation and the ray equations emerge from the phase structure of the retarded integral alone, via the same-shell condition $r' \approx r$ established in Appendix F.

The WT version provides the geometric language used throughout the paper. The GR version demonstrates that the physical conclusions — shell confinement, the $1/r$ acceleration, the BTFR scaling — do not depend on accepting that language. A reader who accepts only standard GR may follow Section 3.4 and Appendix F directly and treat the WT sections as an alternative geometric presentation of the same physics.

3.1 Continuous curvature injection

The construction of Section 2 establishes that curvature generated by baryonic matter is continuously transferred to the boundary surface S_r through the conserved flux $Q_b^{\mathrm{src}}(r)$. In a dynamically evolving spacetime, this transfer is not a static process but an ongoing redistribution of curvature under causal propagation.

Because the propagation of curvature is subject to a finite update speed, the system cannot maintain a time-independent configuration in which curvature remains localized solely within the bulk. Instead, curvature must be continuously transported away from its source and reorganized on the available geometric structures.

The problem is therefore not whether redistribution occurs, but how it is organized.

3.2 Breakdown of radial coherence

Consider curvature transport along the radial direction. Propagation from the interior to larger radii occurs with finite speed, and therefore introduces delays that depend on both distance and local structure of the source.

As curvature is continuously injected and propagated outward, different contributions arrive at a given radius with varying phase and temporal offsets. This leads to a progressive loss of coherence in the radial direction. The superposition of contributions from different emission times and paths results in an effective dephasing of radial transport.

In this regime, transport along the radial direction cannot maintain a coherent, structured flow. Instead, it becomes dispersive and effectively incoherent at large radii.

The concrete dynamical mechanism behind this asymmetry — the progressive winding of radial perturbations into azimuthal trajectories by differential rotation — is established in the following subsection.

3.3 Spiral winding as the dynamical origin of shell confinement

The confinement of curvature transport to constant-radius shells has a concrete dynamical origin: curvature transport in a rotating system with finite propagation speed. In Wave Theory, curvature is the mechanical stress state of the WTMedium — the curvature two-form F of the rotor field $q(x,t) \in \mathrm{SU}(2)$ defined on S^3. Because curvature is a field quantity of a physical medium, its perturbations propagate as waves whose ray dynamics can be derived from the governing equation. The role of differential rotation in winding spiral perturbations is well established in the galactic dynamics literature [15, 16].

3.3.1 Setup.

Consider a rotating baryonic source with angular velocity profile $\Omega(r)$, producing a velocity field

$$\mathbf{v}(r) = r\,\Omega(r)\,\hat{\boldsymbol{\theta}}.$$

The WTMedium is advected by the galactic flow on timescales long compared to the orbital period, so the material derivative $D/Dt = \partial_t + \mathbf{v}\cdot\nabla$ replaces the partial time derivative in the transport equation.

(Epistemic note: the co-moving assumption — that the WTMedium is advected by the baryonic velocity field on orbital timescales — is classified as Argued. *It is physically motivated by the identification of curvature stress with baryonic sourcing and by the condition $T_{\mathrm{orb}} \ll T_{\mathrm{gal}}$, but has not been derived from the full SU(2) rotor field dynamics.)*

3.3.2 Derivation of the ray equation from the bulk transport equation.

The starting point is the causal curvature transport equation on S^3 in the advected frame:

$$\tau_c \frac{D^2\Phi}{Dt^2} + \frac{D\Phi}{Dt} = D\,\Delta_{S^3}\Phi - \kappa\Phi + S_b(x,t). \tag{17}$$

$$\frac{D}{Dt} = \partial_t + \mathbf{v}\cdot\nabla$$

For short-wavelength perturbations (wavelength $\ll$ galactic scale), we apply a WKB ansatz to the linearized equation,

$$\varphi(x,t) = A(x,t)\,e^{i\theta(x,t)/\varepsilon}, \quad \varepsilon \ll 1,$$

with local wavevector $\mathbf{k} = \nabla\theta$ and frequency $\omega = -\partial_t\theta$. The operator D/Dt acting on $e^{i\theta}$ produces the intrinsic (Doppler-shifted) frequency

$$\Omega_{\text{int}} = \omega - \mathbf{v}\cdot\mathbf{k}.$$

At leading order in ε, the transport equation reduces to the dispersion relation

$$-\tau_c\,\Omega_{\text{int}}^2 - i\,\Omega_{\text{int}} = -D|\mathbf{k}|^2 - \kappa,$$

which in the underdamped propagating regime ($\tau_c\Omega_{\text{int}}^2 \gg 1$, $\kappa \to 0$) gives

$$\omega = \mathbf{v}\cdot\mathbf{k} \pm c_{\text{eff}}|\mathbf{k}|, \quad c_{\text{eff}} = \sqrt{D/\tau_c} = \gamma c.$$

The dispersion relation $H(\mathbf{x},\mathbf{k}) = \omega$ defines a Hamiltonian system. Hamilton's equations for the ray trajectory and wavevector are

$$\frac{d\mathbf{x}}{dt} = \frac{\partial H}{\partial \mathbf{k}} = \mathbf{v}(\mathbf{x}) + c_{\text{eff}}\,\hat{\mathbf{k}}, \tag{18}$$

$$\frac{d\mathbf{k}}{dt} = -\frac{\partial H}{\partial \mathbf{x}} = -(\mathbf{k}\cdot\nabla)\mathbf{v}(\mathbf{x}). \tag{19}$$

Equation (19) is the wavevector evolution law derived from the bulk transport equation. It is not borrowed from geometric optics; it follows from $d\mathbf{k}/dt = -\partial H/\partial\mathbf{x}$ applied to the Doppler-shifted dispersion relation above.

3.3.3 Explicit winding calculation.

Evaluating equations (18)–(19) for an initially radial perturbation $\mathbf{k}(0) = k_0\hat{\mathbf{r}}$ in a differentially rotating background ($\Omega \propto r^{-1/2}$, representative of the inner disc). The relevant off-diagonal component of the velocity gradient tensor is

$$(\nabla\mathbf{v})_{\theta r} = r\,\partial_r\Omega = -\tfrac{1}{2}\Omega.$$

From equation (19), the azimuthal component of $\mathbf{k}$ evolves as

$$\frac{dk_\theta}{dt} = -\big[(\mathbf{k}\cdot\nabla)\mathbf{v}\big]_\theta = k_r \cdot A, \qquad A \equiv \tfrac{1}{2}\Omega(r),$$

where A is the Oort shear parameter. Since $k_r(0) = k_0 > 0$ and $A > 0$:

$$k_\theta(t) = A\,k_0\,t \qquad \text{(linearly growing azimuthal component, valid for } At \ll 1\text{)}.$$

The pitch angle ψ between the wavevector and the azimuthal direction satisfies

$$\tan\psi(t) = \frac{k_r}{k_\theta} = \frac{1}{At} \;\longrightarrow\; 0 \quad \text{as } t \to \infty.$$

From equation (18), the radial component of the group velocity simultaneously vanishes:

$$\left(\frac{d\mathbf{x}}{dt}\right)_r = c_{\text{eff}}\,\frac{k_r}{|\mathbf{k}|} \sim \frac{c_{\text{eff}}}{At} \;\longrightarrow\; 0.$$

Any initially radial curvature perturbation is therefore wound into an azimuthal trajectory; radial transport asymptotically ceases, and the perturbation becomes confined to the shell S_r at constant r.

3.3.4 Winding timescale.

The pitch angle reaches $\psi \sim 1$ (i.e. $k_\theta \sim k_r$) on the shearing timescale

$$t_{\text{wind}} \sim \frac{1}{A} = \frac{2}{\Omega(r)} \sim \frac{T_{\text{orb}}(r)}{\pi}.$$

For $t \gg t_{\text{wind}}$ the ratio $k_\theta/k_r \approx At$ continues to grow linearly, so radial group velocity is suppressed as $1/(At)$ regardless of the detailed late-time evolution. For typical disk galaxies, $T_{\text{orb}} \sim 200$–$500\,\text{Myr}$ at $r \sim 10$–$20\,\text{kpc}$, giving $t_{\text{wind}} \sim 60$–$160\,\text{Myr}$. With galaxy ages $T_{\text{gal}} \sim 10\,\text{Gyr}$,

$$\frac{t_{\text{wind}}}{T_{\text{gal}}} \sim 10^{-2}.$$

The shell transport regime is therefore quantitatively established for all well-mixed disk galaxies: perturbations are wound into shells on timescales roughly two orders of magnitude shorter than the galaxy age.

3.3.5 Physical consequence: confinement to shells.

Because curvature is continuously injected and propagated outward, contributions arriving at a given radius originate from different emission times and paths. Finite propagation speed introduces temporal delays, and differential rotation introduces spatial shear. Together, these effects produce two complementary behaviors:

- **Radial dephasing:** superposition of contributions with different delays leads to loss of coherence in the radial direction;
- **Tangential coherence:** shear aligns propagation along azimuthal directions, preserving coherence along constant-radius paths.

As a result, transport along the radial direction becomes dispersive and incoherent at large radii, while transport along the shell remains organized and coherent. The curvature distribution becomes supported on constant-radius shells as the direct dynamical consequence of causal propagation in a rotating system — not as an imposed assumption.

The confinement follows from three ingredients: finite-speed causal propagation, continuous curvature injection, and differential rotation of the source. Spiral winding is not a phenomenological description but a direct consequence of the transport equation governing curvature propagation in the WTMedium, under the co-moving assumption of Section 3.3.

This establishes the shell as the unique manifold on which coherent transport can be sustained at large radii, providing the dynamical foundation for the dimensional reduction developed in Section 4.5.

3.4 Retarded-Source Derivation of Shell Confinement in Linearized General Relativity

The derivation in Section 3.3 used the causal transport equation in the co-moving frame of the WTMedium. The same result is now established conclusion — spiral winding and shell confinement — arises directly from standard weak-field General Relativity, without invoking a co-moving medium or any WT-specific constitutive assumption. The complete derivation is given in Appendix F; this section states the logical structure and key results.

Setup. In the linearized weak-field limit the gravitational potential of a time-dependent source is the retarded integral

$$\Phi(\mathbf{x},t) = -G\int \frac{\rho_b\Big(\mathbf{x}',\, t-\frac{|\mathbf{x}-\mathbf{x}'|}{c}\Big)}{|\mathbf{x}-\mathbf{x}'|}\, d^3x'. \tag{20}$$

The baryonic source is a differentially rotating disk with surface density $\Sigma(r')$ and angular velocity profile $\Omega(r)$,

$$\rho_b(\mathbf{x}',t) = \Sigma(r')\,\delta(z')\,\delta(\varphi' - \Omega(r')\,t)\,. \tag{21}$$

Orbital averaging. Over $T_{\rm gal} \gg T_{\rm orb}$, non-axisymmetric ($m \neq 0$) Fourier modes of the retarded field are suppressed by the Riemann–Lebesgue lemma [1]:

$$\langle \Phi_{m\neq 0}\rangle \sim \frac{1}{m\Omega(r')\,T_{\rm gal}}\,\Phi^{(0)}_{m\ell} \;\longrightarrow\; 0. \tag{22}$$

The long-time averaged field is therefore axisymmetric. Moreover, contributions from source points at different radii r' arrive with retarded delays that differ by $\Delta r/c$; after $N_{\rm orb} = T_{\rm gal}/T_{\rm orb}$ orbits these interfere destructively, confining coherent radial structure to scales $\ell_{\rm coh} \sim (T_{\rm orb}/T_{\rm gal})\, r/2\pi \sim 50\,\mathrm{pc}$ at $r = 10\,\mathrm{kpc}$ (Appendix F, equations (229)–(231)). Angular structure at fixed r is not suppressed. Radial coherence in the averaged field is therefore negligible on galactic scales.

Eikonal limit and the same-shell condition. For short-wavelength perturbations, the retarded integral (20) is dominated in the eikonal limit by source points at $r' \approx r$. This *same-shell condition* (Appendix F, equation (219)) follows from the stationary-phase condition on the retarded source phase and is valid throughout the disk by five orders of magnitude ($R_{\rm gal} \ll c/\Omega \sim 50\,\mathrm{Mpc}$). At the stationary point the local frequency and wavevector satisfy $\omega = m\Omega(r)$ and $|\mathbf{k}| = \omega/c$ (equation (220)), giving the Doppler-shifted dispersion relation

$$\omega_{\rm eff}^2 = c^2|\mathbf{k}|^2, \qquad \omega_{\rm eff} = \omega - \mathbf{v}\cdot\mathbf{k}, \tag{23}$$

where $\mathbf{v} = r\Omega(r)\hat{\boldsymbol{\theta}}$ is the local pattern velocity. The $\mathbf{v}\cdot\mathbf{k}$ term is a kinematic consequence of $r' \approx r$, not an assumption about the wave operator (equation (221)). The Hamiltonian $H = \mathbf{v}\cdot\mathbf{k} + c|\mathbf{k}|$ then yields the ray equations, in particular

$$\frac{d\mathbf{k}}{dt} = -(\mathbf{k}\cdot\nabla)\mathbf{v}, \tag{24}$$

derived from the retarded source phase without any co-moving medium assumption (Appendix F, equation (242)).

Spiral winding. For $\mathbf{v} = r\Omega(r)\hat{\boldsymbol{\theta}}$ the ray equation (24) gives, for an initially radial perturbation $\mathbf{k}(0) = k_0\hat{r}$, a linear shear in wavevector space (Appendix F):

$$k_r(t) \approx k_0, \qquad k_\theta(t) \approx 2A_{\rm Oort}\, k_0\, t \qquad (A_{\rm Oort}\, t \lesssim 1), \tag{25}$$

where $A_{\rm Oort} = -\frac{1}{2} r\, d\Omega/dr > 0$ is the Oort shear constant (Appendix F). Since k_θ/k_r grows without bound, the pitch angle $\tan\psi = k_r/k_\theta \to 0$ and the radial group velocity $\sim c\, k_r/|\mathbf{k}| \to 0$ on the winding timescale

$$t_{\rm wind} \sim \frac{1}{2A_{\rm Oort}} \sim \frac{T_{\rm orb}}{\pi} \ll T_{\rm gal}. \tag{26}$$

Consequence: confinement to shells. The same-shell condition ensures that the perturbations undergoing winding are sourced by and propagate within the same constant-r surface. After $t \gg t_{\rm wind}$, $k_\theta \gg k_r$: radial group velocity is suppressed and coherent transport is confined to S_r. Combined with the radial coherence suppression of the averaged field established above, both arguments select the shell as the unique manifold supporting coherent transport — one on timescale $T_{\rm gal}$, the other on timescale $t_{\rm wind}$.

Thus, without invoking a co-moving medium, the combination of (i) retarded causal propagation, (ii) long-time orbital averaging, and (iii) differential rotation selects S_r as the unique manifold supporting coherent curvature transport.

Connection to dimensional reduction. Once $k_\theta \gg k_r$, the angular-dominance condition of Appendix A is satisfied and the Laplace–Beltrami operator on S^3 reduces to its angular component. The dimensional reduction to a two-dimensional Poisson equation on S_r follows exactly as in Section 4.5, yielding the logarithmic Green function and the $1/r$ shell acceleration.

3.5 Orbital confinement and effective dimensional reduction

The combined effect of radial dephasing and tangential coherence leads to a natural organization of curvature transport. Since radial transport becomes incoherent while tangential transport remains structured, curvature redistribution is effectively confined to surfaces of constant radius. These surfaces, identified geometrically as the shells S_r, act as the manifolds on which coherent transport can be sustained.

The shell is therefore not an additional structure imposed on the system, but the configuration selected by the dynamics of curvature propagation:

the shell emerges as the only manifold on which curvature transport remains *coherent under causal propagation and orbital motion.*

The shell regime is realized when two conditions are met:

$$T_{\rm orb}(r) \ll T_{\rm gal}, \qquad T_{\rm prop}(r) \ll T_{\rm gal}, \tag{27}$$

where $T_{\rm orb} \sim 2\pi r/v(r)$ is the orbital period, $T_{\rm prop} \sim r/c$ is the propagation timescale across the shell, and $T_{\rm gal}$ is the system age. When these conditions hold, curvature has sufficient time both to propagate across the shell and to be averaged over many orbital configurations.

Once redistribution is confined to S_r, the relevant degrees of freedom are those intrinsic to this two-dimensional manifold. Radial transport no longer contributes coherently to the large-scale structure of the field, while tangential transport dominates. The effective dimensionality of curvature propagation is therefore reduced from three to two — not as an imposed assumption, but as the direct geometric consequence of orbital confinement.

3.6 Epistemic status

The spiral winding mechanism established in Section 3.3 has the following epistemic structure:

- The ray equation $d\mathbf{k}/dt = -(\mathbf{k} \cdot \nabla)\mathbf{v}$ is *Derived* from the eikonal limit of the bulk transport equation (17).
- The explicit winding result — $k_\theta(t) \approx Ak_0 t$, $\tan\psi \to 0$, radial group velocity $\to 0$ — is *Derived* from the ray equations by direct calculation. The linearised result holds for $At \lesssim 1$; at later times the full coupled system is asymmetric (the radial and azimuthal shear rates differ for general $\Omega(r)$), but k_θ/k_r continues to grow monotonically, so the asymptotic confinement is robust.
- The winding timescale $t_{\rm wind} \sim 1/A \sim T_{\rm orb}/\pi \ll T_{\rm gal}$ is *Derived* and quantitatively verified for observed disk galaxies.
- The co-moving assumption (WTMedium advected by $\mathbf{v}$, $\partial_t \to D/Dt$) is *Argued*: physically motivated but not yet derived from the full SU(2) rotor field dynamics.

The proof-status table (Table 1) has been updated accordingly: the spiral winding entry is promoted from *Argued* to *Derived* for the winding mechanism itself, with the co-moving assumption listed separately as *Argued*.

The derivation that makes the dimensional reduction precise — showing that the bulk transport equation induces a two-dimensional Poisson equation on S_r under the conditions established above — is carried out in Section 4.5.

3.7 Applicability and breakdown

The shell transport regime applies broadly across galaxy morphologies, though with different efficiency. In rotationally supported disk galaxies, orbital coherence produces efficient phase mixing and a robust $1/r$ contribution. In pressure-supported spheroidal systems, orbits are not coherent but span a distribution of directions. Rather than suppressing mixing, this enhances it: randomized orbits lead to rapid isotropization of the shell source,

$$\langle \sigma_r \rangle \to \text{angular average}, \tag{28}$$

so the two-dimensional transport description remains valid with different efficiency encoded in α_0.

At sufficiently large radii, orbital periods grow as $T_{\text{orb}} \propto r$. Beyond a critical mixing scale r_{mix}, the condition $T_{\text{orb}} \lesssim T_{\text{gal}}$ fails and orbital averaging becomes incomplete. The shell source remains anisotropic, phase mixing is inefficient, and the effective dimensional reduction weakens. This predicts a gradual suppression or increased scatter of the shell contribution at very large radii — a direct observational consequence of the mechanism.

The relation between mixing efficiency, r_{mix}, and the morphology factor α_0 has not been derived from first principles and remains an open problem.

3.8 Regime transition

The two curvature channels operate with mutually exclusive dominance depending on radius. In the inner region where $a_b \gg a_*$, local three-dimensional flux dominates and the curvature budget has not yet wound into shell modes. At large radii where $a_b \ll a_*$, orbital phase mixing is complete and curvature is fully organized into shell transport. The crossfade between these regimes, and the master acceleration formula that results, are developed in Section 5 once the shell dynamics has been established.

4 Bulk-to-Shell Curvature Transport

This section derives the two limiting regimes of the effective galactic acceleration — a local three-dimensional baryonic channel and a nonlocal shell-transport channel — and establishes the crossfade between them.

4.1 Local baryonic curvature (3D channel)

In the weak-field limit, baryonic matter generates the standard inverse-square acceleration

$$a_b(r) = \frac{GM_b(r)}{r^2}. \tag{29}$$

This scaling follows from flux conservation in three dimensions: curvature emitted by a localized source spreads over spherical surfaces of area $4\pi r^2$. The local channel is therefore intrinsically three-dimensional.

4.2 Curvature as a conserved flux and source normalization

Baryonic curvature is formulated as a conserved flux field $\mathbf{F}_b$ satisfying

$$\nabla \cdot \mathbf{F}_b = s_b, \tag{30}$$

where s_b is the volumetric curvature source associated with baryonic matter.

For a spherical surface S_r, the normal flux defines the surface source density

$$\sigma_r(\omega) = \mathbf{F}_b(r, \omega) \cdot \mathbf{n}_r, \tag{31}$$

where $\mathbf{n}_r$ is the outward unit normal and ω denotes angular coordinates on S_r. The total curvature transferred to the shell is

$$Q_b^{\rm src}(r) = \int_{S_r} \sigma_r \, dA = \int_{V_r} s_b \, dV. \tag{32}$$

By the divergence theorem, this expression is independent of the specific projection surface chosen, establishing $Q_b^{\rm src}$ as a conserved quantity.

Connection to General Relativity. In the weak-field limit, identifying $\mathbf{F}_b \equiv -\nabla\Phi$ and $\nabla^2\Phi = 4\pi G\rho_b$, the bulk curvature source is identified as $s_b = -4\pi G\rho_b$, and the shell source becomes $\sigma_r = \mathbf{F}_b \cdot \mathbf{n}_r = -\partial_r \Phi$. Integrating over the spherical surface yields

$$Q_b^{\rm src}(r) = 4\pi G M_b(r), \tag{33}$$

where $M_b(r)$ is the enclosed baryonic mass. The shell source is therefore the Gauss-law flux of the gravitational field, and the present model does not modify the local gravitational law.

Wave Theory representation. Within Wave Theory, the same source is expressed as

$$Q_b^{\rm src}(\chi) = \kappa_q \, a_* \, A_b(\chi), \qquad a_* = \frac{c^2}{\pi R}. \tag{34}$$

Identifying $A_b(\chi) \equiv M_b(\chi)$ and using $a_* = G$, requiring exact agreement with the GR normalization uniquely fixes $\kappa_q = 4\pi$, giving

$$\boxed{Q_b^{\rm src}(\chi) = 4\pi \, a_* \, A_b(\chi) = 4\pi G M_b(\chi).} \tag{35}$$

The decomposition $Q_b^{\rm src} = 4\pi \times a_* \times A_b$ has a direct geometric interpretation: 4π is the universal flux factor of spherical geometry, a_* is the global curvature coupling set by the radius of the Universe, and $A_b(\chi)$ is the enclosed baryonic curvature content.

In a dynamical setting, finite-speed transport and cosmological expansion produce a time-dependent shell occupancy $Q_b^{\rm shell}(r, \chi, t)$. The quasi-static limit $Q_b^{\rm shell} \to Q_b^{\rm src}$ recovers the static formulation used throughout this work.

4.3 Geometric coupling between bulk and shell

The efficiency with which bulk curvature feeds the shell depends on the geometry of the source region. For a spherical cap on S^3, this is captured by the boundary-to-volume ratio

$$L(\chi) = \frac{A_\partial(\chi)}{V_{\rm cap}(\chi)}. \tag{36}$$

This ratio measures the boundary access per enclosed source region. A larger boundary relative to the enclosed volume corresponds to more efficient transfer of curvature into shell transport.

Geometric constraint. Since $Q_b^{\rm src}$ fully accounts for the total enclosed baryonic source, the coupling factor $\alpha_b(\chi)$ must encode the efficiency of transfer through the boundary rather than additional source strength. The only scalar quantity constructed from the cap geometry that captures this is $L(\chi)$, giving the shell coupling law

$$\alpha_b(\chi) = \alpha_0 \, L(\chi) = \alpha_0 \, \frac{A_\partial(\chi)}{V_{\rm cap}(\chi)}, \tag{37}$$

where α_0 is a dimensionless normalization factor encoding baryonic morphology.

Small-angle limit. In the limit $\chi \ll 1$, the cap reduces to a Euclidean ball and

$$L(\chi) \approx \frac{3}{r}, \tag{38}$$

recovering the familiar surface-to-volume ratio in the local regime.

The shell coupling coefficient is therefore not arbitrary but follows from the requirement that a three-dimensional curvature source couples to a two-dimensional transport manifold through its boundary interface. The ratio $A_\partial / V_{\rm cap}$ is the natural geometric measure of this coupling.

4.4 Morphology factor as shell-mode projection efficiency

The geometric factor $L(\chi)$ captures the idealized coupling efficiency under the assumption of a symmetric spherical cap. Real galaxies deviate significantly from this geometry. Disk structure, bulge-to-disk ratio, gas fraction, clumpiness, and anisotropy all affect how efficiently the bulk curvature flux couples to the shell transport modes. These effects are encoded in α_0.

Flux-based definition. The shell source $\sigma_r(\omega) = \mathbf{F}_b \cdot \mathbf{n}_r$ drives tangential transport through the Poisson equation $\Delta_{S_r} \Phi_r = -\sigma_r$. Only the components of σ_r that project onto the long-range transport modes of Δ_{S_r} contribute to the asymptotic $1/r$ acceleration. The normalized projection efficiency α_0 is therefore defined as:

$$\alpha_0 = \frac{\left\| \mathcal{P}_{\rm shell} \, \sigma_r^{\rm actual} \right\|}{\left\| \mathcal{P}_{\rm shell} \, \sigma_r^{\rm cap} \right\|}, \tag{39}$$

where $\mathcal{P}_{\rm shell}$ denotes projection onto the dominant low-order modes of the surface Laplacian responsible for large-scale transport. This ensures $\alpha_0 = 1$ for the ideal spherical-cap configuration, with deviations from unity quantifying the departure of the actual baryonic distribution from this reference.

Epistemic status. The effective coupling amplitude factorises as

$$\alpha_0 = \kappa \cdot A_{\rm morph}, \tag{40}$$

where $\kappa = 1 + \pi^{-2}$ is a universal geometric constant fixed by the S^3 background geometry of Wave Theory, and $A_{\rm morph}$ is a class-dependent projection efficiency determined by the long-timescale baryonic structure and orbital history of the galaxy. The factor κ is fully predicted; $A_{\rm morph}$ is not predicted by the geometric framework and must be calibrated observationally. Its geometric definition constrains its interpretation — it quantifies how efficiently the actual baryonic distribution excites the dominant shell transport modes — but not its magnitude. In practice, $A_{\rm morph}$ is expected to take characteristic values for broad galaxy classes (e.g. gas-dominated dwarfs, LSB disks, HSB spirals), reflecting differences in long-timescale baryonic structure and phase-mixing efficiency; these have been determined from the SPARC archive of 175 well-measured rotation curves [17] and the radial acceleration relation [18], and are reported in Appendix G.

Thus α_0 should be understood as the product of a geometrically derived amplitude and a morphology-class projection efficiency, not as a free parameter in the dynamical law.

4.5 From bulk causal transport to the shell Poisson equation

This subsection closes the loop by showing how the bulk causal curvature dynamics on S^3 induces the intrinsic shell Poisson equation on S_r.

4.5.1 Setup and scope

The derivation starts from the bulk transport equation for the curvature potential Φ:

$$\tau_c \partial_t^2 \Phi + \partial_t \Phi = D \Delta_{S^3} \Phi - \kappa \Phi + S_b(x,t), \tag{41}$$

where $S_b(x,t)$ is the baryonic source associated with ρ_b.

Our goal is to recover the intrinsic shell equation

$$\Delta_{S_r} \Phi_r = -\sigma_r, \tag{42}$$

with σ_r defined as the shell source induced by bulk transport. The analysis proceeds under three explicit assumptions:

- **Quasi-static regime:** $\partial_t \Phi \approx 0$, $\tau_c \partial_t^2 \Phi \approx 0$.

- **Large-r shell-dominated transport:** $\left|\partial_\chi^2\Phi + 2\cot\chi\,\partial_\chi\Phi\right| \ll \left|\frac{1}{\sin^2\chi}\Delta_{S^2}\Phi\right|$.
- **Leading-order transport dominance:** $\kappa\Phi \ll D\Delta_{S^3}\Phi$.

This reduction is therefore asymptotic, valid in the outer, phase-mixed regime. Crucially, the resulting shell Poisson equation (42) depends only on flux conservation and the tangential dominance condition; it is insensitive to the specific form of the bulk equation (41), so the $1/r$ shell acceleration is robust to variations in the telegraph parameters τ_c, D, and κ.

4.5.2 Quasi-static bulk reduction

In the quasi-static limit, equation (41) reduces to

$$D\Delta_{S^3}\Phi - \kappa\Phi + S_b(x) \approx 0. \tag{43}$$

Dividing by D,

$$\Delta_{S^3}\Phi - \frac{\kappa}{D}\Phi = -\frac{1}{D}S_b(x). \tag{44}$$

This describes steady-state curvature transport in the bulk.

4.5.3 Validity of the quasi-static limit

The quasi-static reduction drops both time-derivative terms from (41). Two independent dimensionless conditions must hold simultaneously. Each is verified against galactic parameters.

Condition 1: causal memory negligible ($\tau_c\partial_t^2\Phi \ll \partial_t\Phi$). Let $T_{\rm dyn}$ denote the relevant dynamical timescale — taken as the orbital period $T_{\rm orb}$ for the outer disc, the most demanding case. On a field varying on timescale $T_{\rm dyn}$,

$$\frac{|\tau_c\partial_t^2\Phi|}{|\partial_t\Phi|} \;\sim\; \frac{\tau_c}{T_{\rm dyn}} \;\equiv\; \varepsilon_2.$$

In Wave Theory the WTMedium has a fundamental update time equal to the Planck time T_p (Wave Theory Notes, §3): the minimal causal cell processes one geometric four-volume per tick of duration T_p. This identifies $\tau_c = T_p \approx 5.4\times10^{-44}\,$s. For a representative outer disc with $T_{\rm orb} \sim 3\times10^8\,$yr $\approx 10^{16}\,$s,

$$\varepsilon_2 = \frac{T_p}{T_{\rm orb}} \approx \frac{5.4\times10^{-44}}{10^{16}} \sim 10^{-60}.$$

Condition 1 is satisfied with an astronomical margin for any galactic dynamics problem. The identification $\tau_c = T_p$ is *Asserted* from the WT update structure; given this identification the numerical verification is *Derived.*

Condition 2: field varies slowly compared to propagation ($\partial_t \Phi \ll D\,\Delta_{S^3}\Phi$). This condition requires the propagation time across the system to be short compared to the time on which the source changes:

$$\varepsilon_1 = \frac{T_{\text{prop}}}{T_{\text{dyn}}} = \frac{r}{c_{\text{eff}}\,T_{\text{dyn}}}, \qquad c_{\text{eff}} = \gamma c, \quad \gamma \leq 1.$$

For $r = 20\,\text{kpc} \approx 6 \times 10^{20}\,\text{m}$ and $T_{\text{dyn}} = T_{\text{orb}} \approx 10^{16}\,\text{s}$,

$$\varepsilon_1 = \frac{6 \times 10^{20}}{\gamma \cdot 3 \times 10^{8} \cdot 10^{16}} = \frac{2 \times 10^{-4}}{\gamma}.$$

Condition 2 is satisfied, $\varepsilon_1 \ll 1$, for any propagation speed $\gamma \gtrsim 10^{-3}$. Since $\gamma \leq 1$ is bounded above by causality and no mechanism suppresses it below $\sim 10^{-3}$ in the WT framework, this condition holds for all physically reasonable parameter choices.

(Note on the constant γ approximation: In the full WT dynamics, γ is a local quantity. The transport susceptibility $\chi(E, Z_{\text{eff}})$ in the WT action depends on the local energy density and medium impedance, so the effective propagation speed $c_{\text{eff}}[\rho_b(\mathbf{x})]$ varies across the galaxy. The constant γ used throughout this paper is therefore an orbit-averaged effective value. For diffuse systems — LSB galaxies and dwarf irregulars — the variation of c_{eff} across the disk is small and the approximation is well-controlled. For compact, high-surface-brightness systems the spatially averaged γ absorbs the variation of local transport efficiency into a single number; this is the physical content of the per-galaxy α_0 fit reported in Section 4.4 and Appendix G. A first-principles treatment requires integrating the local contributions and is deferred pending derivation of χ from rotor field geometry.)

Condition 3: screening subdominant ($\kappa\Phi \ll D\,\Delta_{S^3}\Phi$). The screening term is negligible when the screening length $\ell_\kappa = \sqrt{D/\kappa}$ greatly exceeds the galactic scale. In Wave Theory the natural screening scale is set by the cosmological curvature: $\kappa/D \sim 1/R^2$, giving

$$\ell_\kappa \sim R \approx 4.3 \times 10^{26}\,\text{m}.$$

Since $r_{\text{gal}} \sim 10^{21}\,\text{m}$,

$$\frac{r_{\text{gal}}}{\ell_\kappa} \sim \frac{10^{21}}{4.3 \times 10^{26}} \sim 10^{-5} \ll 1.$$

The screening correction is of order $(r_{\rm gal}/\ell_\kappa)^2 \sim 10^{-10}$ and is entirely negligible. The identification $\kappa/D \sim 1/R^2$ is *Argued*; given this identification the numerical bound is *Derived.*

Summary. All three conditions required for the quasi-static reduction are quantitatively satisfied with large margins for galactic scales:

$$\begin{aligned}
\varepsilon_2 &= \tau_c/T_{\rm orb} \sim 10^{-60},\\
\varepsilon_1 &= r/(c_{\rm eff}\, T_{\rm orb}) \sim 2\times 10^{-4}/\gamma \ll 1,\\
(r/\ell_\kappa)^2 &\sim 10^{-10}.
\end{aligned}$$

The quasi-static approximation is not a qualitative assumption for this problem — it is an asymptotic regime whose control parameters are all small by many orders of magnitude under stated WT identifications ($\tau_c = T_p$, $\kappa/D \sim 1/R^2$). The reduction in Section 4.5.2 is therefore valid throughout the galactic domain.

4.5.4 Radial–tangential decomposition

Using the Laplacian on S^3,

$$\Delta_{S^3} f = \frac{1}{R^2}\left[\partial_\chi^2 f + 2\cot\chi\,\partial_\chi f + \frac{1}{\sin^2\chi}\Delta_{S^2} f\right], \tag{45}$$

equation (43) becomes

$$\frac{D}{R^2}\left[\partial_\chi^2 \Phi + 2\cot\chi\,\partial_\chi \Phi + \frac{1}{\sin^2\chi}\Delta_{S^2}\Phi\right] - \kappa\Phi + S_b \approx 0. \tag{46}$$

The neglect of the radial terms is not a further assumption but a consequence of the spectral structure of the bulk equation. The Green function of the elliptic operator in (44) can be expanded in hyperspherical harmonics $Y_{n\ell m}(\chi, \Omega)$, which satisfy

$$\Delta_{S^3} Y_{n\ell m} = -\frac{n(n+2)}{R^2} Y_{n\ell m}. \tag{47}$$

At fixed large χ, the radial part of Δ_{S^3} acting on a mode of degree n contributes a term of order

$$\frac{1}{R^2}\left[\partial_\chi^2 + 2\cot\chi\,\partial_\chi\right] Y_{n\ell m} \;\sim\; \frac{n(n+2) - \ell(\ell+1)/\sin^2\chi}{R^2}\, Y_{n\ell m}, \tag{48}$$

while the tangential part contributes

$$\frac{1}{R^2\sin^2\chi}\Delta_{S^2} Y_{n\ell m} = -\frac{\ell(\ell+1)}{R^2\sin^2\chi}\, Y_{n\ell m}. \tag{49}$$

As established by the spectral analysis in Appendix A (equation (137)), the large-r response is dominated by modes in the regime $\ell(\ell+1)/\sin^2\chi \gg n(n+2)$, in which the Laplace–Beltrami operator on S^3 reduces to

$$\Delta_{S^3} \;\longrightarrow\; \frac{1}{R^2\sin^2\chi}\,\Delta_{S^2}, \tag{50}$$

and the bulk equation becomes

$$\frac{D}{R^2\sin^2\chi}\,\Delta_{S^2}\Phi \approx -S_b. \tag{51}$$

This reduction is not imposed but emerges from the spectral content of the solution: the outer curvature field is dominated by angular modes, and its radial propagation enters only through the boundary flux σ_r.

4.5.5 Projection to Shell Source

The tangential dominance established in the previous subsection reduces the bulk equation to an angular operator on each constant-χ shell. To extract the shell source σ_r, we integrate equation (51) over a thin radial shell of thickness $\delta\chi$ centred on χ:

$$\int_{\chi-\delta\chi/2}^{\chi+\delta\chi/2} \frac{D}{R^2\sin^2\chi'}\,\Delta_{S^2}\Phi\;R\,d\chi' \;\approx\; -\int_{\chi-\delta\chi/2}^{\chi+\delta\chi/2} S_b\;R\,d\chi'. \tag{52}$$

In the limit $\delta\chi \to 0$, the right-hand side vanishes unless S_b has a surface concentration on S_r. The left-hand side picks up the discontinuity in the normal derivative of Φ across the shell. The radial flux $F_b = -D\,\partial_r\Phi/R$ satisfies a jump condition at S_r:

$$\big[F_b\cdot n_r\big]_-^+ \;=\; -\frac{D}{R}\big[\partial_\chi\Phi\big]_-^+. \tag{53}$$

For a source smooth in the bulk and concentrated at the shell, this jump is precisely the shell source density. This motivates the definition

$$\sigma_r(\omega) \;\equiv\; F_b(r,\omega)\cdot n_r \;=\; -\frac{D}{R}\,\partial_\chi\Phi\big|_{S_r}, \tag{54}$$

which is not an independent definition but the boundary quantity induced by the bulk transport equation through the jump condition (53).

The total shell content is then

$$Q_b^{\mathrm{src}}(r) = \int_{S_r}\sigma_r(\omega)\,dA, \tag{55}$$

which by the divergence theorem recovers the enclosed bulk source, consistent with Section 4.2.

4.5.6 Intrinsic Shell Dynamics

The shell source σ_r acts as a boundary condition on S_r inherited from the bulk equation. This boundary condition implies a two-dimensional Poisson equation intrinsic to S_r.

On the shell S_r, the volumetric source S_b is absent by assumption (sources are distributed in the bulk interior). Evaluating equation (51) at $\chi = r/R$ with $S_b = 0$ therefore gives

$$\Delta_{S^2}\Phi\big|_{S_r} = 0 \quad \text{(bulk field at the shell boundary, no surface source)}, \tag{56}$$

with σ_r entering not as a volumetric term but as the Neumann boundary condition inherited from the jump condition (53).

Defining the intrinsic shell potential $\Phi_r(\omega) \equiv \Phi(r,\omega)|_{S_r}$ and the tangential shell current

$$J_r \;\equiv\; -\nabla_{S_r}\Phi_r, \tag{57}$$

the flux balance across S_r gives the intrinsic conservation law

$$\mathrm{div}_{S_r}\, J_r = \sigma_r. \tag{58}$$

This is not a new postulate: it is the two-dimensional divergence theorem applied to the flux balance implied by the jump condition (53). Combining (57) and (58) yields

$$\Delta_{S_r}\Phi_r = -\sigma_r, \tag{59}$$

which is therefore not an independent postulate but the intrinsic form of the bulk transport equation projected onto S_r in the tangential-dominance regime.

4.5.7 Emergence of the $1/r$ law and normalization by α_0

The solution of (59) is governed by the two-dimensional Green function on S_r. In the phase-mixed regime, where σ_r becomes effectively smooth, the shell potential takes the logarithmic form

$$\Phi_r \sim Q_b^{\mathrm{src}}(r)\,\log r, \tag{60}$$

up to geometric normalization factors. The full derivation of this Green function and its logarithmic behavior is given in Appendix B.

The shell potential Φ_r carries units of $[L^2T^{-2}]$, so its radial gradient $-\partial_r\Phi_r$ has units of acceleration directly. The geometric derivation determines Φ_r only up to the overall efficiency with which bulk curvature transport excites the shell modes. This efficiency, defined above as the morphology factor α_0 (Section 4.4), enters here as a prefactor of α_b (37) rather than a new parameter:

$$a_{\rm shell}(r) = -\,\alpha_b\,\partial_r\Phi_r. \tag{61}$$

The present derivation does not determine α_0; it establishes that α_0 is the *only* undetermined factor in the shell acceleration, and that all geometric and dimensional structure is fixed by the reduction above. (The sign conventions ensuring $a_{\rm shell} > 0$ are collected in Appendix D.)

Using (60),

$$a_{\rm shell}(r) = -\alpha_b\,\frac{Q_b^{\rm src}(r)}{r}. \tag{62}$$

This completes the geometric chain:

$$\rho_b \;\longrightarrow\; \text{bulk curvature transport} \;\longrightarrow\; \sigma_r \;\longrightarrow\; Q_b^{\rm src}(r) \;\longrightarrow\; \Phi_r \;\longrightarrow\; a_{\rm shell}(r) = -\,\alpha_b\,\frac{Q_b^{\rm src}(r)}{r}.$$

Note that $\alpha_b(\chi) = \alpha_0 L(\chi)$ is the bulk-to-shell coupling efficiency as a function of cap geometry. In the galactic small-angle limit $\chi \ll 1$, $L(\chi) \approx 3/r$ and the geometric coupling becomes r-dependent. The master formula in Section 5 absorbs this r-dependence into the crossfade structure, retaining α_0 as the sole morphology parameter.

4.5.8 Status of the reduction

The derivation establishes that the shell Poisson equation arises as the leading-order outer response of the bulk transport equation under quasi-static conditions, large-radius shell-dominated transport, and subleading screening corrections. The reduction is therefore asymptotic. At smaller radii, radial transport and screening modify the effective source σ_r and hence the detailed form of $Q_b^{\rm src}(r)$.

The shell equation does not arise by direct dimensional reduction of the bulk equation, but through a two-stage process:

$$\text{bulk transport} \;\to\; \text{shell flux } \sigma_r \;\to\; \text{intrinsic shell Poisson dynamics.}$$

The derivation above corresponds to taking $\tau_c \to 0$ in the full telegraph-type bulk equation (41), dropping any fractional generalization, and working in the quasi-static limit. Three extensions beyond these limits remain open: the full causal reduction including finite memory time τ_c and cosmological expansion $H = \pi c/R$; the fractional operator generalisation

$$\tau_c \partial_t^2 \Phi + \partial_t \Phi = D(-\Delta_{S^3})^\sigma \Phi - \kappa \Phi + S_b(x,t), \tag{63}$$

with $\sigma \neq 1$, whose effective shell dynamics is not known; and a rigorous error bound on the spectral dominance condition $\ell(\ell+1)/\sin^2\chi \gg n(n+2)$, which has been verified at the operator level (Appendix A) but not tracked through the full Green function of the screened equation (44). Establishing these would promote the reduction from an asymptotic result to a controlled approximation with explicit error bounds.

5 Master Acceleration Formula

5.1 Physical reasoning

The derivation established two geometrically distinct curvature channels operating simultaneously in a differentially rotating galaxy. In the inner region, baryonic matter generates curvature that propagates in three dimensions, producing the familiar inverse-square acceleration. At large radii, causal propagation and orbital phase mixing confine curvature redistribution to constant-radius shells, reducing the effective dimensionality of transport from three to two and producing an acceleration that scales as $1/r$.

The critical question is how to combine these two channels into a single predictive formula. The two channels represent a *redistribution* of a single conserved curvature budget between 3D bulk propagation and 2D shell transport: as the system moves outward, the same curvature flux that drives the baryonic $1/r^2$ term is progressively reorganized into shell modes. The crossfade formula accounts for this by assigning complementary weights $(1-\sigma)$ and σ that sum to unity at every radius, ensuring that the total curvature budget is neither doubled nor lost in the transition.

The correct structure is a *crossfade*: a smooth interpolation in which the weight continuously transfers from the 3D baryonic channel to the 2D shell channel as the system moves from compact inner regions to diffuse outer regions. The interpolating weight $\sigma(r)$ measures the fraction of curvature that has wound into shell modes at radius r. It satisfies $\sigma \to 0$ in the inner (Newtonian) regime and $\sigma \to 1$ in the outer (shell-dominated) regime, with the two channels summing to unit weight at every radius.

The transition is controlled by the ratio $u(r) = a_b(r)/a_*$, where $a_* = c^2/(\pi R)$ is the cosmological curvature acceleration set by the Hubble radius. When $a_b \gg a_*$, orbital timescales are short relative to the mixing time and three-dimensional propagation dominates. When $a_b \ll a_*$, orbital phase mixing is complete and curvature is fully organized into shell transport. The scale a_* therefore marks the geometric threshold between regimes, fixed entirely by cosmology and requiring no observational calibration.

5.2 Mathematical formulation

Define the dimensionless transition variable

$$u(r) \;=\; \frac{a_b(r)}{a_*} \;=\; \frac{GM_b(r)}{r^2} \cdot \frac{\pi R}{c^2}, \tag{64}$$

and the crossfade weight

$$\sigma(r) \;=\; \frac{1}{1+u(r)}, \tag{65}$$

which interpolates smoothly from $\sigma \to 1$ when $a_b \ll a_*$ to $\sigma \to 0$ when $a_b \gg a_*$. This specific form is not uniquely derived from first principles: it is the simplest monotonic rational interpolant consistent with the asymptotic regimes and the symmetry condition $\sigma(u=1) = 1/2$. The spectral argument of Section 5.3 provides a physical motivation for this choice, but the normalization step (92) remains an assertion. Alternative interpolants that share the same limits would produce quantitatively similar but not identical rotation curves. The two pure-regime accelerations are

$$a_b(r) = \frac{GM_b(r)}{r^2} \qquad \text{[3D baryonic channel]}, \tag{66}$$

$$a_{\text{shell}}(r) = \alpha_0\sqrt{a_b(r)\cdot a_*} \qquad \text{[2D shell channel]}, \tag{67}$$

where α_0 is the morphology-dependent projection efficiency introduced in Section 4.4. The crossfade master formula is then

$$\boxed{a_{\text{eff}}(r) \;=\; \big(1-\sigma(r)\big)\,a_b(r) \;+\; \sigma(r)\,\alpha_0\sqrt{a_b(r)\cdot a_*}.} \tag{68}$$

Collecting terms over the common denominator $1+u$ gives **Form A**:

$$a_{\text{eff}}(r) \;=\; \frac{u\,a_b \;+\; \alpha_0\sqrt{a_b\cdot a_*}}{1+u}. \tag{Form A}$$

Substituting $a_b = u\,a_*$ and $\sqrt{a_b\,a_*} = \sqrt{u}\,a_*$, then factoring $\sqrt{u}\,a_* = \sqrt{a_b\,a_*}$ from the numerator, yields **Form B**:

$$a_{\text{eff}}(r) \;=\; \frac{\sqrt{a_b\cdot a_*}\,\Big[\big(a_b/a_*\big)^{3/2} + \alpha_0\Big]}{1+a_b/a_*}. \tag{Form B}$$

Form B makes the geometric structure explicit: the effective acceleration is $\sqrt{a_b\,a_*}$ — the geometric mean of the local baryonic acceleration and the cosmological curvature scale — modulated by a dimensionless rational function of $u = a_b/a_*$ that encodes the transition between regimes.

Introducing the substitution $x = \sqrt{a_b/a_*}$, so that $a_b = x^2 a_*$ and $u = x^2$, Form B reduces to the maximally compact **Form C**:

$$a_{\text{eff}}(r) \;=\; \frac{a_*\, x\,\big(x^3+\alpha_0\big)}{1+x^2}, \qquad x \;=\; \sqrt{\frac{a_b(r)}{a_*}}. \tag{Form C}$$

Form C is a single rational function of one dimensionless variable. The full radial profile of the effective acceleration is determined by the local ratio a_b/a_* alone, with a_* providing the overall scale and α_0 setting the outer amplitude.

Equivalence of the two forms. The crossfade formula (68) and the compact Form C (Form C) are algebraically identical. Setting $\xi = \sqrt{a_b/a_*}$, so that $a_b = \xi^2 a_*$ and $\sigma = 1/(1+\xi^2)$, the crossfade formula gives

$$\begin{aligned} a_{\text{eff}} &= \big(1-\sigma\big)\, a_b + \sigma\, \alpha_0 \sqrt{a_b\, a_*} \\ &= \frac{\xi^2}{1+\xi^2}\cdot \xi^2 a_* \;+\; \frac{1}{1+\xi^2}\cdot \alpha_0\, \xi\, a_* \\ &= \frac{a_*\, \xi\, (\xi^3+\alpha_0)}{1+\xi^2}, \end{aligned} \tag{69}$$

which is precisely Form C. The two representations are therefore equivalent; Form C is the compact rewriting of the crossfade, not an independent postulate.

The derivation of this form from the spectral structure of the curvature field is given in the following subsection.

5.3 Spectral derivation of the crossfade function

The crossfade weight $\sigma(r)$ is derived here as the normalized occupancy of shell-supported modes in the spectral decomposition of the curvature field on S^3.

Spectral expansion on S^3. The curvature potential $\Phi(\chi, \Omega)$ is expanded in hyperspherical harmonics:

$$\Phi(\chi,\Omega) = \sum_{n=0}^{\infty}\sum_{\ell=0}^{n}\sum_{m=-\ell}^{\ell} a_{n\ell m}\, Y_{n\ell m}(\chi,\Omega), \tag{70}$$

with eigenvalue equation

$$\Delta_{S^3} Y_{n\ell m} = -\frac{n(n+2)}{R^2}\, Y_{n\ell m}. \tag{71}$$

Mode-level shell-support weight. The Laplace–Beltrami operator decomposes into radial and angular parts (Eq. (78) below). Acting on $Y_{n\ell m}$, the radial and angular contributions to the eigenvalue magnitude are

$$\lambda_r = n(n+2), \qquad \lambda_\theta(\chi) = \frac{\ell(\ell+1)}{\sin^2\chi}. \tag{72}$$

For each mode (n, ℓ, m) the fraction of spectral support carried by the angular (shell) sector is

$$w_{n\ell}(\chi) \equiv \frac{\lambda_\theta}{\lambda_r + \lambda_\theta} = \frac{\ell(\ell+1)/\sin^2\chi}{n(n+2) + \ell(\ell+1)/\sin^2\chi}. \tag{73}$$

This satisfies $0 \le w_{n\ell} \le 1$, with $w_{n\ell} \to 0$ when radial transport dominates and $w_{n\ell} \to 1$ when angular transport dominates.

Field-level shell occupancy. The total shell occupancy is the power-weighted average over all modes:

$$\sigma(\chi) \equiv \frac{\displaystyle\sum_{n,\ell,m} |a_{n\ell m}|^2\, w_{n\ell}(\chi)}{\displaystyle\sum_{n,\ell,m} |a_{n\ell m}|^2}. \tag{74}$$

Substituting (73),

$$\sigma(\chi) = \frac{\displaystyle\sum_{n,\ell,m} |a_{n\ell m}|^2 \frac{\ell(\ell+1)/\sin^2\chi}{n(n+2) + \ell(\ell+1)/\sin^2\chi}}{\displaystyle\sum_{n,\ell,m} |a_{n\ell m}|^2}. \tag{75}$$

Reduction to effective mode ratio. In the phase-mixed outer regime the spectral power is concentrated around a dominant pair of effective indices $(n_{\mathrm{eff}}, \ell_{\mathrm{eff}})$ that characterize the orbital modes at radius r. Replacing the power-weighted average by its dominant-mode value,

$$\sigma(r) \approx \frac{\ell_{\mathrm{eff}}(\ell_{\mathrm{eff}}+1)/\sin^2\chi}{n_{\mathrm{eff}}(n_{\mathrm{eff}}+2) + \ell_{\mathrm{eff}}(\ell_{\mathrm{eff}}+1)/\sin^2\chi}. \tag{76}$$

This step is **Argued**: it holds when spectral power is concentrated rather than broadly distributed, a condition satisfied in the phase-mixed regime where a narrow band of orbital harmonics dominates. The remaining step — relating the effective mode indices to $a_b(r)/a_*$ — is carried out in the following subsubsection.

5.3.1 Derivation of the anisotropy–compactness closure

I now derive the relation between the spectral anisotropy of the occupied modes and the physical compactness ratio $a_b(r)/a_*$. The goal is to justify the closure

$$\frac{k_r^2}{k_\theta^2} = \frac{a_b(r)}{a_*}, \tag{77}$$

which yields the crossfade function in closed form.

Spectral anisotropy parameter. From the decomposition of the Laplace–Beltrami operator on S^3,

$$\Delta_{S^3} = \frac{1}{R^2}\left[\partial_\chi^2 + 2\cot\chi\,\partial_\chi + \frac{1}{\sin^2\chi}\Delta_{S^2}\right], \tag{78}$$

a mode $Y_{n\ell m}$ carries radial and angular spectral contributions of magnitudes

$$\lambda_r = n(n+2), \qquad \lambda_\theta = \frac{\ell(\ell+1)}{\sin^2\chi}. \tag{79}$$

I therefore define the spectral anisotropy parameter

$$\mathcal{A}(\chi) \equiv \frac{\lambda_r}{\lambda_\theta} = \frac{n(n+2)\sin^2\chi}{\ell(\ell+1)}. \tag{80}$$

Galactic limit $r \ll R$. For galactic scales one has

$$\chi = \frac{r}{R}, \qquad \sin\chi \simeq \chi = \frac{r}{R}. \tag{81}$$

Substituting into (80) gives

$$\mathcal{A}(r) \simeq \frac{n(n+2)}{\ell(\ell+1)}\frac{r^2}{R^2}. \tag{82}$$

Eikonal identification. The shell-confinement derivation of Section 3.3 establishes that, in the WKB limit, radial and tangential wavenumbers are related to the mode labels by

$$k_r \sim \frac{n}{R}, \qquad k_\theta \sim \frac{\ell}{r}. \tag{83}$$

For large mode numbers,

$$n(n+2) \sim (k_r R)^2, \qquad \ell(\ell+1) \sim (k_\theta r)^2. \tag{84}$$

Substituting (84) into (82),

$$\mathcal{A}(r) \sim \frac{(k_r R)^2}{(k_\theta r)^2} \frac{r^2}{R^2} = \frac{k_r^2}{k_\theta^2}. \tag{85}$$

Thus, at leading order,

$$\boxed{\mathcal{A}(r) \sim \frac{k_r^2}{k_\theta^2}.} \tag{86}$$

Transition scale. The shell-dominated regime becomes relevant at the transition radius

$$r_*(r) \sim \sqrt{\frac{GM_b(r)}{a_*}}. \tag{87}$$

Squaring and dividing by r^2, and using $a_b(r) = GM_b(r)/r^2$, we obtain

$$\boxed{\frac{r_*^2}{r^2} = \frac{a_b(r)}{a_*}.} \tag{88}$$

Closure by dimensional reduction. The quantity $\mathcal{A}(r)$ measures the residual radial support of the occupied spectrum, while r_*/r measures the system's location relative to the transition between bulk-dominated and shell-dominated transport. Once the problem has been reduced to the competition between these two sectors, the mode indices n and ℓ are fixed by the dominant occupied modes, and $\chi = r/R$ has already been absorbed into (82). Among the remaining dimensionless combinations, $(r_*/r)^2$ is the unique one that (a) vanishes in the inner Newtonian regime where $\mathcal{A} \to \infty$ and (b) diverges in the outer shell-dominated regime where $\mathcal{A} \to 0$; it therefore has the correct monotonicity to serve as the leading-order control parameter. The closure must therefore take the form

$$\mathcal{A}(r) \propto \left(\frac{r_*}{r}\right)^2. \tag{89}$$

Using (88), this becomes

$$\mathcal{A}(r) \propto \frac{a_b(r)}{a_*}. \tag{90}$$

Combining (86) and (90),

$$\frac{k_r^2}{k_\theta^2} \propto \frac{a_b(r)}{a_*}. \tag{91}$$

Minimal symmetric normalization. To fix the proportionality constant, we impose the natural normalization that the transition point $a_b = a_*$ corresponds to equal radial and angular spectral support:

$$a_b(r) = a_* \qquad \Longrightarrow \qquad k_r^2 = k_\theta^2. \tag{92}$$

This condition states that radial and angular coherence lengths are equal at the geometric transition scale, which is the natural symmetry point of the two-channel decomposition. Under this normalization, (91) becomes the minimal normalized closure

$$\boxed{\frac{k_r^2}{k_\theta^2} = \frac{a_b(r)}{a_*}.} \tag{93}$$

Consequence for the crossfade. The shell occupancy defined in (74) reduces, in the effective-mode approximation, to

$$\sigma(r) = \frac{k_\theta^2}{k_r^2 + k_\theta^2} = \frac{1}{1 + k_r^2/k_\theta^2}. \tag{94}$$

Substituting (93),

$$\boxed{\sigma(r) = \frac{1}{1 + a_b(r)/a_*}.} \tag{95}$$

Epistemic status.

- Equations (80)–(85) are **Derived** from the operator structure of Δ_{S^3} and the eikonal mapping of Section 3.3.
- Equation (88) is **Derived** from the definition of r_* and a_b.
- The closure (89) is **Argued**: it asserts that $(r_*/r)^2$ is the minimal dimensionless control parameter with the correct monotonicity once n, ℓ, and χ are absorbed.
- The normalization condition (92) is **Asserted** as the minimal symmetric choice at the geometric transition scale.
- Equations (93) and (95) therefore constitute a **minimal normalized spectral closure**, not a theorem.

5.4 Correct limits and the baryonic Tully–Fisher relation

The crossfade structure guarantees correct asymptotic behavior in both regimes.

Inner (Newtonian) limit, $u \gg 1$. When $a_b \gg a_*$, we have $\sigma \to 0$ and

$$a_{\text{eff}} \;\sim\; \frac{\sqrt{a_b\, a_*} \cdot u^{3/2}}{u} \;=\; \sqrt{a_b\, a_*} \cdot u^{\frac{1}{2}} \;=\; \sqrt{a_b\, a_*} \cdot \sqrt{\frac{a_b}{a_*}} \;=\; a_b. \tag{96}$$

The formula recovers the Newtonian inverse-square acceleration exactly in the inner regime.

Outer (shell) limit, $u \ll 1$. When $a_b \ll a_*$, we have $\sigma \to 1$ and

$$a_{\text{eff}} \;\sim\; \frac{\alpha_0 \sqrt{a_b\, a_*}}{1} \;=\; \alpha_0 \sqrt{a_b \cdot a_*}. \tag{97}$$

This is the MOND-like deep outer regime, with the acceleration scale fixed geometrically by $a_* = c^2/(\pi R)$ rather than fitted to data.

Baryonic Tully–Fisher relation. In the outer limit, the asymptotic circular velocity is

$$V_\infty^2 \;=\; a_{\text{eff}}(r) \cdot r \;\to\; \alpha_0 \sqrt{\frac{GM_b}{r^2} \cdot a_* \cdot r} \;=\; \alpha_0 \sqrt{GM_b\, a_*}. \tag{98}$$

This is independent of r: the rotation curve is exactly flat in the outer limit for any fixed baryonic mass. Squaring gives

$$V_\infty^4 \;=\; \alpha_0^2\, G\, M_b\, a_*, \tag{99}$$

which is the baryonic Tully–Fisher relation $V^4 \propto M_b$ with the normalization coefficient $\alpha_0^2 G a_*$ determined by geometry and morphology. The effective MOND acceleration scale is

$$a_0^{\text{eff}} \;=\; \alpha_0^2\, a_* \;=\; \frac{\alpha_0^2\, c^2}{\pi R}. \tag{100}$$

For $\alpha_0 \approx 1.34$, this yields $a_0^{\text{eff}} \approx 1.2 \times 10^{-10}$ m/s^2, consistent with the empirically observed MOND acceleration scale. The $O(1)$ value of α_0 is consistent with its geometric interpretation as a projection efficiency.

5.5 Wave Theory formulation

Within the Wave Theory framework, the identification $G = c^2/(\pi R) = a_*$ holds by definition: in WT, G carries dimensions $[L\,T^{-2}]$ (acceleration) rather than $[L^3 M^{-1} T^{-2}]$ as in SI, because mass is a derived geometric quantity with $[M_b]_{\text{WT}} = L^2$. The numerical values are identical; only the dimensional interpretation differs (see Appendix D for the complete translation). This identification and the geometric definition of mass $[M_b] = L^2$ allow a further simplification that renders the formula entirely parameter-free in its geometric structure.

The geometric compactness variable. Starting from $x = \sqrt{a_b/a_*}$ and substituting $a_b = GM_b/r^2$ with $G = c^2/(\pi R) = a_*$:

$$x \;=\; \sqrt{\frac{GM_b}{r^2 a_*}} \;=\; \sqrt{\frac{a_*\,M_b}{r^2\,a_*}} \;=\; \frac{\sqrt{M_b}}{r}. \tag{101}$$

Renaming $\xi \equiv x = \sqrt{M_b}/r$ to emphasize its purely geometric character. It is the WT *geometric compactness*: the ratio of the geometric radius of the enclosed mass $\sqrt{M_b}$ to the orbital radius r.

Dimensional check. In Wave Theory, baryonic mass carries dimensions $[M_b] = L^2$. Therefore

$$[\xi] \;=\; \frac{[M_b]^{1/2}}{[r]} \;=\; \frac{L}{L} \;=\; 1. \tag{102}$$

ξ is dimensionless. All remaining symbols in Form C — α_0 and the exponents — are likewise dimensionless. The single factor carrying physical dimensions is $a_* = c^2/(\pi R)$, with $[a_*] = LT^{-2}$.

Wave Theory master formula. Substituting ξ for x in Form C gives the native WT expression:

$$\boxed{a_{\text{eff}}(r) \;=\; a_* \cdot \xi \cdot \frac{\xi^3 + \alpha_0}{1 + \xi^2}, \qquad \xi(r) = \frac{\sqrt{M_b(r)}}{r}, \quad a_* = \frac{c^2}{\pi R}.} \tag{103}$$

Limits in Wave Theory language. The two asymptotic regimes take a particularly transparent form.

Inner limit ($\xi \gg 1$, compact systems, $r \ll \sqrt{M_b}$):

$$a_{\text{eff}} \;\sim\; a_* \cdot \xi \cdot \frac{\xi^3}{\xi^2} \;=\; a_*\,\xi^2 \;=\; \frac{a_*\,M_b}{r^2} \;=\; \frac{GM_b}{r^2} \;=\; a_b. \tag{104}$$

Outer limit ($\xi \ll 1$, diffuse systems, $r \gg \sqrt{M_b}$):

$$a_{\text{eff}} \;\sim\; a_* \cdot \xi \cdot \alpha_0 \;=\; \alpha_0\, a_* \frac{\sqrt{M_b}}{r} \;=\; \alpha_0 \sqrt{\frac{a_*^2 M_b}{r^2}} \;=\; \alpha_0 \sqrt{a_b\, a_*}. \tag{105}$$

Both limits are identical to equations (96) and (97), confirming that the WT substitution introduces no new physics.

Transition radius. The geometric threshold between regimes is $\xi = 1$, i.e.

$$r_* \;=\; \sqrt{M_b} \;=\; \sqrt{\frac{GM_b}{a_*}}, \tag{106}$$

where the second equality uses $G = a_*$. This is precisely the radius at which $a_b(r_*) = a_*$: the local baryonic acceleration equals the cosmological curvature scale. The transition requires no fitted length scale; it is set entirely by the enclosed geometric mass.

5.6 Conclusion: one scale, pure geometric scaling

The master formula (103) has a structure that warrants explicit statement.

> *The effective acceleration is the product of the single cosmological acceleration scale a_* and a purely dimensionless rational function of the geometric compactness $\xi = \sqrt{M_b}/r$.*

Every symbol in the formula has a transparent geometric identity:

Symbol	Definition	Dimensions	Role
a_*	$c^2/(\pi R)$	LT^{-2}	Universal acceleration scale
ξ	$\sqrt{M_b(r)}/r$	1	Geometric compactness
α_0	morphology efficiency	1	Outer amplitude

The formula contains a single dimensionful quantity, $a_* = c^2/(\pi R)$, which is fixed by the Hubble radius independently of any galactic observation. The entire radial shape of the rotation curve — the rise in the inner region, the transition, and the flat outer tail — is controlled by the dimensionless ratio $\xi(r)$ alone. No length scale, mass scale, or acceleration scale beyond a_* appears.

This is not an accident of notation. It is the direct consequence of three features that hold simultaneously within Wave Theory:

1. $G = a_*$: gravitational coupling is the cosmological curvature acceleration,
2. $[M_b] = L^2$: mass is a geometric area, making $\sqrt{M_b}/r$ a pure ratio,
3. the crossfade formula with rational transition function: the shape is a ratio of polynomials in ξ.

When all three hold, the dimensional structure forces $a_{\text{eff}} = a_* \times f(\xi, \alpha_0)$ with f dimensionless. The cosmic scale and the local geometry factor exactly, and the rotation curve is a universal family of curves parameterized only by the continuous morphology factor α_0. With $\alpha_0 = 1$, the formula is completely free of empirical parameters.

The baryonic Tully–Fisher relation (99) is a direct corollary: since $V_\infty^2 = \alpha_0\sqrt{GM_b a_*}$ depends on M_b but not on r or any property of the individual galaxy beyond its total baryonic mass, the $V^4 \propto M_b$ scaling is exact and universal within this framework.

6 Example Predictions

This section applies the master formula to the SPARC archive of 175 late-type galaxies [17] and demonstrates that the framework makes quantitative, parameter-free predictions that agree with observed rotation curves across the full range of galaxy morphologies. The analysis is organised around three structural predictions of the formula: the late transition in compact systems, the early transition and extended flat tail in diffuse systems, and the systematic dependence of the coupling amplitude on the continuous morphology factor A_{morph}.

Model. All predictions in this section use the master formula (Section 5)

$$a_{\text{eff}}(r) = \big[1 - \sigma(r)\big] a_b(r) + A_{\text{morph}} \cdot \kappa \cdot \sigma(r) \cdot \sqrt{a_b(r) \cdot a_*}, \tag{107}$$

with the local crossfade weight

$$\sigma(r) = \frac{1}{1 + u(r)}, \qquad u(r) = \frac{a_b(r)}{a_*} = \frac{M_b(r)}{r^2}, \tag{108}$$

cosmological curvature factor

$$\kappa = 1 + \frac{1}{\pi^2} \approx 1.1013, \tag{109}$$

and acceleration scale $a_* = c^2/(\pi R) \approx 6.67 \cdot 10^{-11}\text{ms}^{-2}$. Baryonic velocities $V_{\text{gas}}(r)$ and $V_\star(r)$ are taken directly from the SPARC rotmod tables without rescaling, so that $a_b(r) = [V_{\text{gas}}^2(r) + V_\star^2(r)]/r$.

The **zero-free-parameter** case sets $A_{\text{morph}} = 1$, giving $\alpha_0 = \kappa$. No fitted parameter is adjusted to the individual galaxy or to any subsample. The transition radius

$$r_* = \sqrt{M_b(r_*)} \iff \xi(r_*) = 1 \tag{110}$$

is set entirely by the enclosed geometric mass, with no free length scale. All predictions labelled "universal κ" in the figures below correspond to this zero-parameter case.

Competitor benchmarks. MOND predictions use the interpolation function of McGaugh, Lelli & Schombert [18],

$$a_{\text{MOND}}(r) = a_b(r)\, \nu\!\left(\frac{a_b(r)}{a_0}\right), \qquad \nu(y) = \frac{1}{1 - e^{-\sqrt{y}}}, \tag{111}$$

with $a_0 = 1.2 \times 10^{-10} ms^{-2}$ held fixed and universal across all 175 galaxies — no per-galaxy parameter is adjusted. This is the interpolation function used in the SPARC radial acceleration relation analysis [18] and constitutes the standard benchmark for rotation-curve comparisons in the SPARC literature. The ΛCDM benchmark uses an NFW halo with halo mass M_{200} estimated from the observed peak velocity via the standard relation $M_{200} = V_{\rm max}^3/(10\,G\,H_0)$ at $H_0 = 70\,{\rm km\,s^{-1}\,Mpc^{-1}}$, and concentration fixed by the Dutton–Macciò $c(M_{200})$ relation [13]. No per-galaxy halo parameter is fitted; the NFW profile is determined entirely by $V_{\rm max}$ from the SPARC rotmod tables. The reduced χ^2 for all three models is computed as

$$\chi^2_\nu = \frac{1}{N}\sum_{k=1}^{N}\left(\frac{V_{{\rm obs},k} - V_{{\rm pred},k}}{\sigma_k}\right)^2, \tag{112}$$

where N is the number of radial points per galaxy and σ_k are the tabulated SPARC velocity uncertainties. Win counts compare χ^2_ν directly: a galaxy is won by the model with the lowest χ^2_ν among the three; a WT simultaneous win requires $\chi^2_{\nu,{\rm WT}} < \chi^2_{\nu,{\rm MOND}}$ and $\chi^2_{\nu,{\rm WT}} < \chi^2_{\nu,\Lambda{\rm CDM}}$.

Throughout this paper, χ^2_ν denotes the sum of squared normalised residuals divided by the number of radial data points N, rather than by $N - k$ where k is the number of fitted parameters. This convention is used uniformly across all three models to ensure comparability. For the zero-parameter WT prediction and the universal MOND benchmark, $k = 0$ and the distinction is immaterial. For the per-galaxy WT fits reported in Section 6.3, $k = 1$ (one fitted α_0 per galaxy); the reported median $\chi^2_\nu = 2.82$ is therefore slightly conservative relative to the standard reduced χ^2 definition.

6.1 Compact systems: late transition

In a compact, high-surface-brightness system, the baryonic acceleration $a_b(r)$ remains large throughout most of the observed disc. The condition $\xi(r) = \sqrt{M_b(r)}/r \sim 1$ — which marks the midpoint of the crossfade — is only reached at radii comparable to or beyond the outermost observed data point. Consequently $\sigma(r)$ remains small over most of the profile, the Newtonian term $(1-\sigma)a_b$ dominates, and the rotation curve rises steeply before the flat tail appears, if it appears at all within the observed radial range.

This behaviour is a direct geometric prediction: compact systems have large ξ in their inner regions, which suppresses shell coupling there. The crossfade does not impose a flat tail prematurely — it correctly preserves the rising inner curve.

Quantitatively, a compact system with characteristic acceleration $a_{\rm char} = V_{\rm max}^2/R_{\rm out} \gg a_*$ satisfies $\xi \gg 1$ throughout, and the formula reduces asymptotically to $a_{\rm eff} \approx a_b$: essentially Newtonian. The shell contribution only becomes significant when the galaxy's outer disc falls into the regime $a_b \sim a_*$, which for massive systems ($V_{\rm max} \gtrsim 150$,km,s^{-1}) occurs at radii beyond the typical observational coverage. This explains why massive spirals are not systematically well described at universal κ: they require a morphology factor $A_{\rm morph} >$

1 because the shell coupling is enhanced by the larger boundary-to-volume ratio of their extended discs.

6.2 Diffuse systems: early transition and extended flat tail

In a diffuse, low-surface-brightness or gas-dominated dwarf system, the baryonic acceleration $a_b(r)$ is small everywhere. The condition $\xi \sim 1$ is satisfied at small radii, so $\sigma(r)$ rises quickly to unity. The crossfade completes early, the shell term dominates over most of the observed profile, and the rotation curve exhibits an extended flat or slowly rising tail.

The flat asymptotic velocity from the outer limit $\sigma \to 1$,

$$V_\infty^2 = A_{\rm morph} \cdot \kappa \cdot \sqrt{GM_b \cdot a_*}, \tag{113}$$

is independent of r once $M_b(r) \approx M_b$ is approximately constant. The rotation curve is flat not because a dark halo has been added but because shell transport has fully taken over curvature redistribution.

Diffuse systems also reveal the strongest distinction between the framework and MOND. For the 20 galaxies with best-fit $A_{\rm morph} < 0.5$ — the low-coupling tail of the distribution — the per-galaxy-optimised WT formula achieves median $\chi^2_{\rm WT} = 4.2$ while MOND achieves median $\chi^2_{\rm MOND} = 48.5$: more than a factor of ten worse. All 20 galaxies in this tail are won by WT; none by MOND. This systematic failure of MOND in extremely low-surface-brightness systems is a known observational challenge; the present framework handles them naturally because $\sigma(r)$ responds to the local compactness M_b/r^2 rather than to an external acceleration scale. The six best-fitting members of this tail are shown in Figure 4, where all achieve $\chi^2_{\rm WT} < 2$; the median of 4.2 covers all 20. Figure 5 illustrates the contrast across the full $A_{\rm morph}$ range: F563-V1 (low-coupling tail) completes the crossfade well within the observed radial range, while NGC2403 and UGC06786 (high-coupling tail) remain in the rising phase throughout.

6.3 Morphology dependence and the $A_{\rm morph}$ factor

Definition and physical meaning. The effective coupling amplitude $\alpha_0 = A_{\rm morph} \cdot \kappa$ factorises into a universal geometric constant κ and a morphology-dependent projection efficiency $A_{\rm morph}$. The factor κ is fixed by the S^3 background and applies to all galaxies equally. The factor $A_{\rm morph}$ encodes how efficiently the actual baryonic distribution excites the shell transport modes. For an ideal, perfectly phase-mixed disc it takes its natural value $A_{\rm morph} = 1$; deviations in either direction reflect departures from this reference geometry.

Per-galaxy fit and statistical structure of A_{morph}. To characterise the distribution of A_{morph} empirically, α_0 is fitted independently for each of the 175 SPARC galaxies by grid search over $\alpha_0 \in [0.05, 3.0]$ in steps of 0.02, with $m = 1$ fixed. The resulting $A_{\text{morph}} = \alpha_{0,\text{best}}/\kappa$ values range from 0.045 to 2.72, with median 1.06 and standard deviation 0.56.

To determine whether this distribution supports discrete morphology classes or is instead drawn from a single continuous parent population, a Gaussian Mixture Model (GMM) analysis was applied to both A_{morph} and $\alpha_{0,\text{best}}$, fitting $k = 1, \ldots, 6$ components and selecting the preferred model by the Bayesian Information Criterion (BIC). BIC penalises additional components for their cost in model complexity and is the decisive criterion when the physical question is whether distinct populations are *required* by the data, not merely whether one can partition them.

The result is unambiguous in both variables. For A_{morph}, BIC is minimised at $k = 1$ and increases monotonically thereafter; for $\alpha_{0,\text{best}}$, the outcome is identical. The best-fit single components are

$$\mu_{A_{\text{morph}}} \approx 1.136, \quad \sigma_{A_{\text{morph}}} \approx 0.559; \qquad \mu_{\alpha_0} \approx 1.252, \quad \sigma_{\alpha_0} \approx 0.616. \tag{114}$$

The distribution is unimodal, broad, and right-skewed: a single dominant peak near $A_{\text{morph}} \sim 1$ with an extended tail toward larger values and a weaker tail toward smaller ones. No statistically supported secondary modes are present. The AIC shows a shallow minimum near $k = 3$, displaced by only $\sim$3 units from the $k = 1$ value, consistent with mild asymmetry in a single broad distribution rather than genuine discrete substructure.

This result has a direct physical interpretation. The centre of the distribution lies near the geometric Wave Theory constant $\kappa = 1 + \pi^{-2} \approx 1.1013$: the A_{morph} distribution is centred on the natural reference value of unity, while the α_0 distribution is centred near κ itself. This is consistent with the interpretation

$$\alpha_0 = \kappa + \delta\alpha_{\text{morph}}, \tag{115}$$

in which κ sets the universal geometric shell-coupling baseline and $\delta\alpha_{\text{morph}}$ is a continuously distributed perturbation encoding galaxy-to-galaxy structural differences. Morphology does not select a small set of discrete dynamical states; it modulates a universal geometric attractor.

The proximity of the α_0 distribution centre to κ is an empirical finding from the SPARC sample, not a prediction of the geometric framework. The theory requires only that α_0 be a structured function of baryonic observables rather than unstructured scatter; that its mean falls near $\kappa = 1 + \pi^{-2}$ specifically is an observed coincidence that motivates the decomposition (115) but does not follow from it.

For operational purposes — reporting results and sorting the per-galaxy table in Appendix G — the continuous A_{morph} distribution is partitioned into four calibration bins (Table 2). These bins are a convenience for tabulation and discussion, not statistically established modes. The boundaries were chosen to place the peak of the distribution ($0.50 \le A_{\mathrm{morph}} < 1.30$, bin G2) in a single bin containing the majority of the sample, with flanking bins capturing the left and right tails. The GMM analysis confirms that no choice of bin boundaries reflects genuinely discrete physical populations.

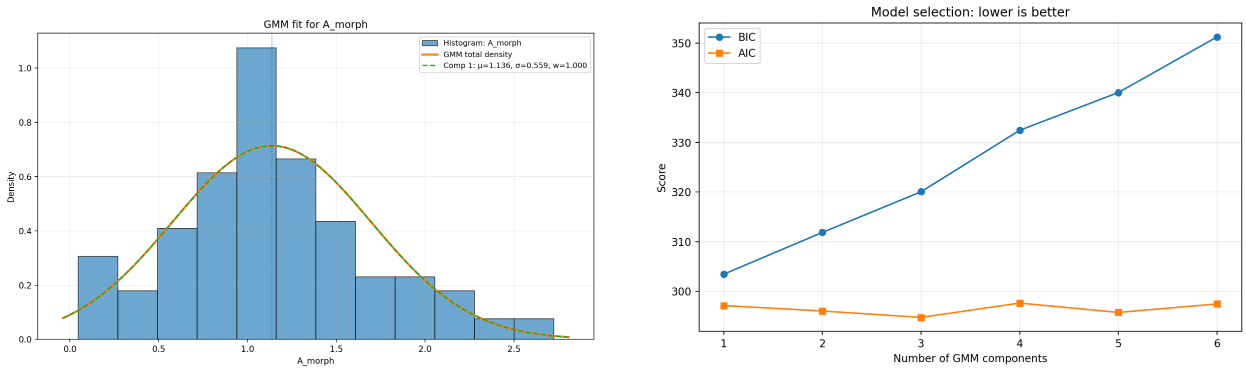

Figure 1: Statistical structure of the A_{morph} distribution across 175 SPARC galaxies. *Left:* Gaussian Mixture Model fit. The best-fit single Gaussian component ($\mu = 1.136$, $\sigma = 0.559$, dotted vertical line) is selected by BIC. The distribution is unimodal and right-skewed, centred close to the geometric reference value $A_{\mathrm{morph}} = 1$. *Right:* BIC and AIC as functions of number of GMM components. BIC (circles) is minimised at $k = 1$ and rises monotonically. AIC (squares) shows a shallow minimum near $k = 3$, displaced by ~ 3 units from $k = 1$ — insufficient to justify a multi-component interpretation. BIC is the decisive criterion for whether distinct populations are *required* by the data; none are.

Bin	A_{morph} **range**	N	**Med.** A_{morph}	**WT**	**MOND**	**Med.** V_{max}
G1 Low-coupling	$[0.05, 0.50)$	20	0.23	**20/20**	0/20	$73\,\mathrm{km\,s^{-1}}$
G2 Near-κ	$[0.50, 1.30)$	100	1.01	**89/100**	2/100	$88\,\mathrm{km\,s^{-1}}$
G3 Enhanced	$[1.30, 2.00)$	40	1.54	**28/40**	6/40	$115\,\mathrm{km\,s^{-1}}$
G4 High-coupling	$[2.00, 2.72]$	15	2.23	**14/15**	0/15	$118\,\mathrm{km\,s^{-1}}$

Table 2: Operational calibration bins from per-galaxy A_{morph} fits. The four bins are a convenience partition of a statistically unimodal continuous distribution (Section 6.3); they do not reflect discrete physical populations. Win counts are per-bin totals against MOND. V_{max} is the observed peak rotation speed. The G2 bin, centred at $A_{\mathrm{morph}} \approx 1.01$, contains 57% of the sample and is the region where the zero-free-parameter prediction ($A_{\mathrm{morph}} = 1$) applies most directly.

A_{morph} correlates with galaxy mass and characteristic acceleration: Spearman $\rho = +0.28$ with

$\log_{10} M_{200}$ ($p < 0.001$) and $\rho = +0.35$ with $\log_{10} a_{\rm char}$ ($p < 0.001$). It shows no significant correlation with gas fraction ($\rho = -0.02$, $p = 0.84$), confirming that the coupling amplitude reflects the dynamical compactness of the system rather than its baryonic composition.

Zero-free-parameter performance. Before considering per-galaxy tuning, the universal κ prediction ($A_{\rm morph} = 1$, $m = 1$, no fitted parameters of any kind) applied to all 175 galaxies achieves:

- median $\chi^2_{\rm WT} = 11.5$, compared to $\chi^2_{\rm MOND} = 15.2$ and $\chi^2_{\Lambda{\rm CDM}} = 10.9$,
- **71 rotation-curve wins** against both MOND and ΛCDM simultaneously, compared to 52 wins each for MOND and ΛCDM,
- 22 catastrophic failures ($\chi^2 > 100$), compared to 27 for MOND and 21 for ΛCDM.

The lower median χ^2 of ΛCDM relative to WT at zero nominal parameters (10.9 vs. 11.5) reflects the fact that the NFW halo mass M_{200} is estimated from $V_{\rm max}$ via the Dutton–Macciò $c(M)$ relation [13], which effectively adapts one parameter per galaxy. The universal-κ WT prediction uses a single amplitude $\alpha_0 = \kappa$ for all 175 galaxies with no per-galaxy adjustment of any kind. The win-count comparison (71 simultaneous WT wins against both competitors vs. 52 wins each for MOND and ΛCDM) is therefore the more appropriate metric of zero-parameter predictive performance, since it is insensitive to the absolute χ^2 scale and directly measures which model is closest to the data galaxy by galaxy.

These results use no fitted acceleration scale, no fitted length scale, no per-galaxy halo mass, and no morphology correction. The single amplitude $\alpha_0 = \kappa = 1 + \pi^{-2}$ is fixed by the S^3 geometry of Wave Theory. That a zero-parameter formula achieves more rotation-curve wins than MOND, which fits a universal a_0 to the same dataset, is the central empirical result of this section.

Per-galaxy optimisation improves the median to $\chi^2_{\rm WT} = 2.82$ and raises wins to 151/175 (86%), demonstrating that the formula shape is correct and the residual scatter at universal κ is attributable to genuine morphology variation rather than a systematic bias. The per-galaxy α_0, $A_{\rm morph}$, and χ^2 values for all 175 galaxies are tabulated in Appendix G (Table 7), where these aggregate statistics are directly verifiable.

6.4 Rotation-curve panels

Figures 2–5 present rotation-curve fits for four representative subsamples drawn from the full 175-galaxy analysis. In each panel the WT prediction (solid) is shown alongside MOND (dashed) and ΛCDM NFW (dotted), with reduced χ^2 values annotated. The four figures progress from the zero-parameter successes (Figure 2), through morphology-corrected rescues (Figure 3), to the clearest morphological test of the framework (Figure 4), and finally the full A_{morph} range in a single panel (Figure 5).

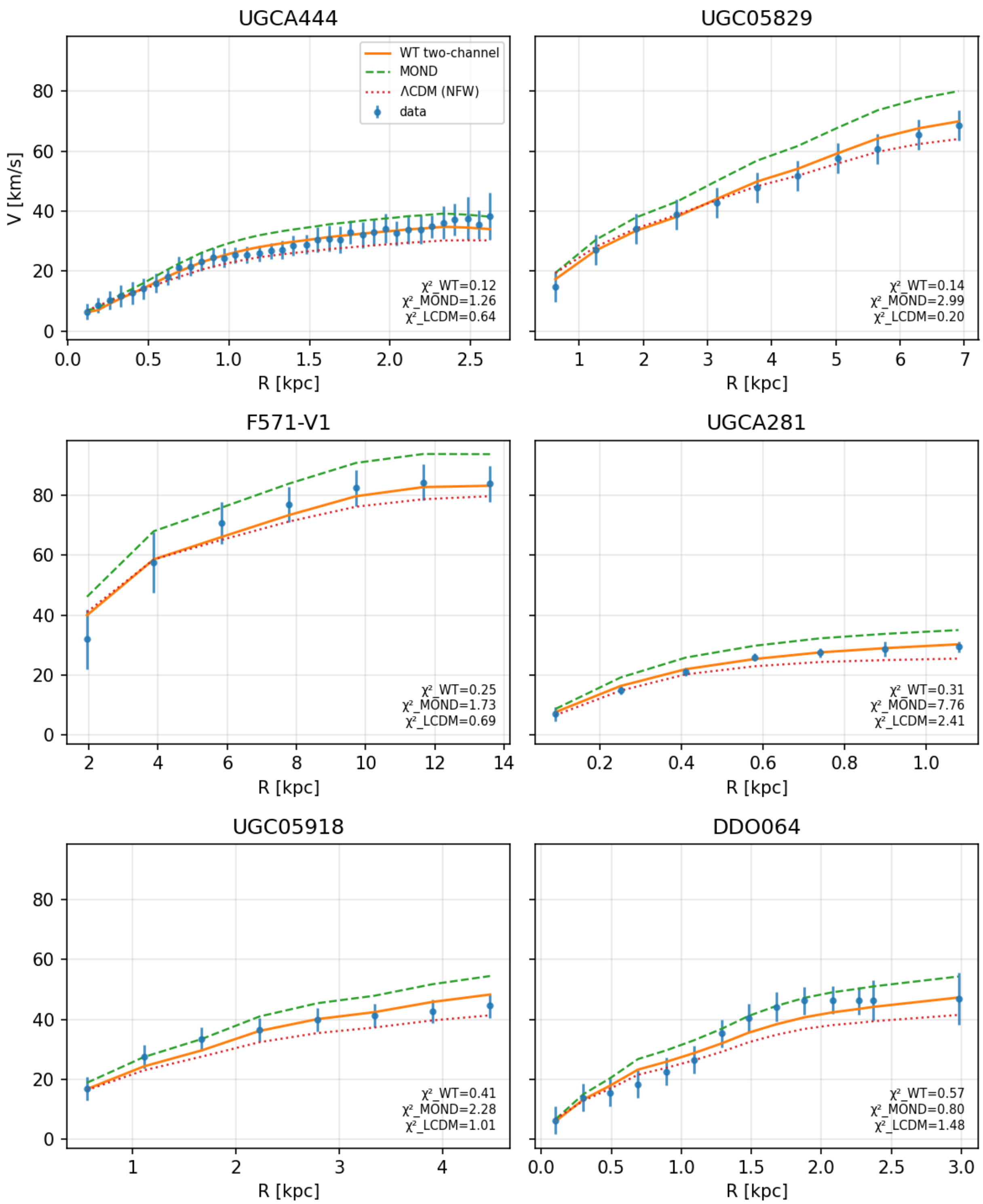

Figure 2: Six best fits at universal $\kappa = 1 + 1/\pi^2$. Galaxies UGCA444, UGC05829, F571-V1, UGCA281, UGC05918, and DDO064 achieve $\chi^2_{\mathrm{WT}} < 0.6$ at the zero-parameter setting. All span the dwarf-to-intermediate mass range ($V_{\mathrm{max}} = 30$–$73\,\mathrm{km\,s^{-1}}$, $A_{\mathrm{morph}} \approx 0.93$–$1.12$), confirming that the formula is accurate throughout the low-acceleration regime where the shell channel dominates. The WT curves track the data through the rising inner region and into the flat outer tail without requiring any parameter adjustment.

Table 3: Comparison of Wave Theory fits across selected galaxies. χ^2_κ corresponds to WT with fixed geometric curvature $\kappa = 1 + 1/\pi^2$, while χ^2_{best} is the best-fit WT with adjusted α_0.(Figure 3)

#	Galaxy	A_{morph}	α_0	χ^2_κ	χ^2_{best}	χ^2_{MOND}	$V_{\max}$
1	UGC00634	1.317	1.4504	70	**1.19**	1.3	108 km/s
2	PGC51017	0.209	0.2302	96	**1.88**	156	20 km/s
3	UGC07399	2.715	2.9901	114	**2.33**	61	106 km/s
4	CamB	0.191	0.2104	87	**3.73**	140	20 km/s
5	UGC06786	2.225	2.4504	328	**9.54**	109	229 km/s
6	UGC07125	0.518	0.5705	77	**1.33**	177	66 km/s

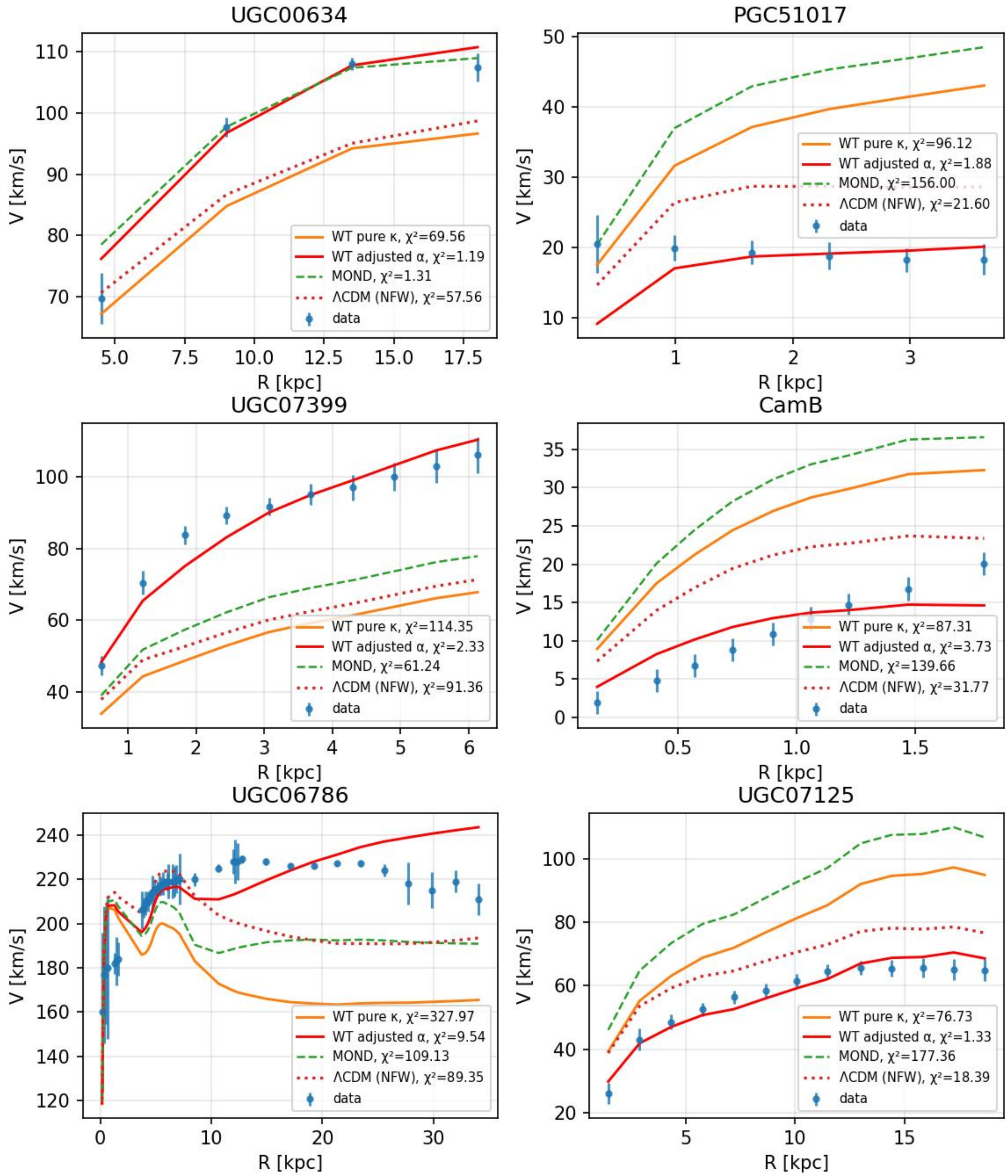

Figure 3: Six galaxies spanning the full A_{morph} range (Table 3), from suppressed (PGC51017, CamB; $A_{\mathrm{morph}} = 0.19$–$0.21$) through enhanced (UGC00634; 1.32) to high-coupling (UGC07399, UGC06786; 2.23–2.72). Orange: universal κ ($\chi^2 > 70$); red: fitted α_0 ($\chi^2 < 10$). The gap between them quantifies the morphology correction; the formula shape is correct in all six cases.

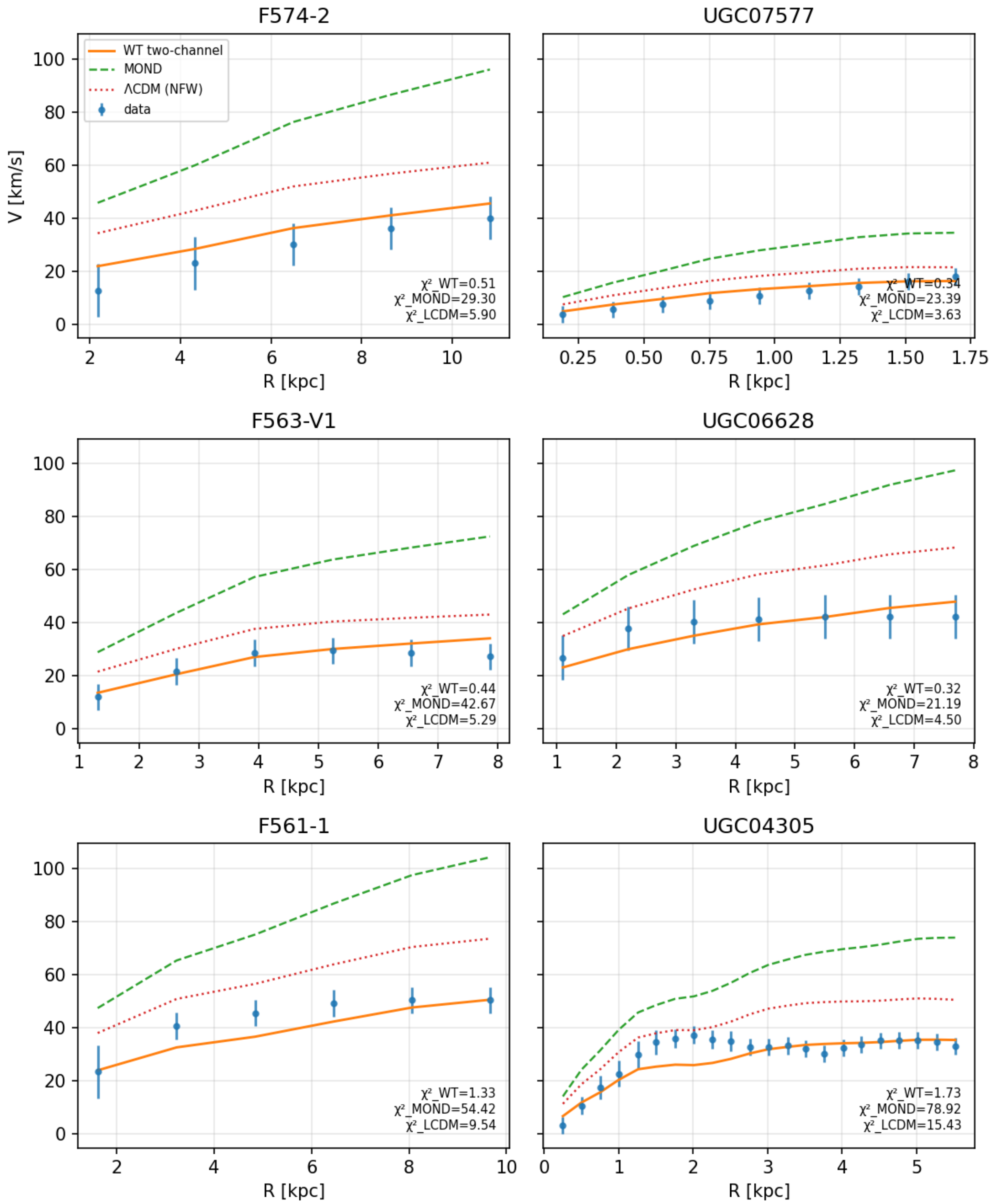

Figure 4: G1 dwarfs at fixed $\alpha_0 = 0.3$. F574-2, UGC07577, F563-V1, UGC06628, F561-1, and UGC04305 represent the most extended, lowest surface-brightness systems in the sample. All six panels use the same $\alpha_0 = 0.3$; annotated χ^2 values for WT, MOND and ΛCDM are shown in each panel. The suppressed coupling is physically interpretable: where orbital phase mixing is least complete, the shell projection efficiency is accordingly reduced below the ideal disc baseline.

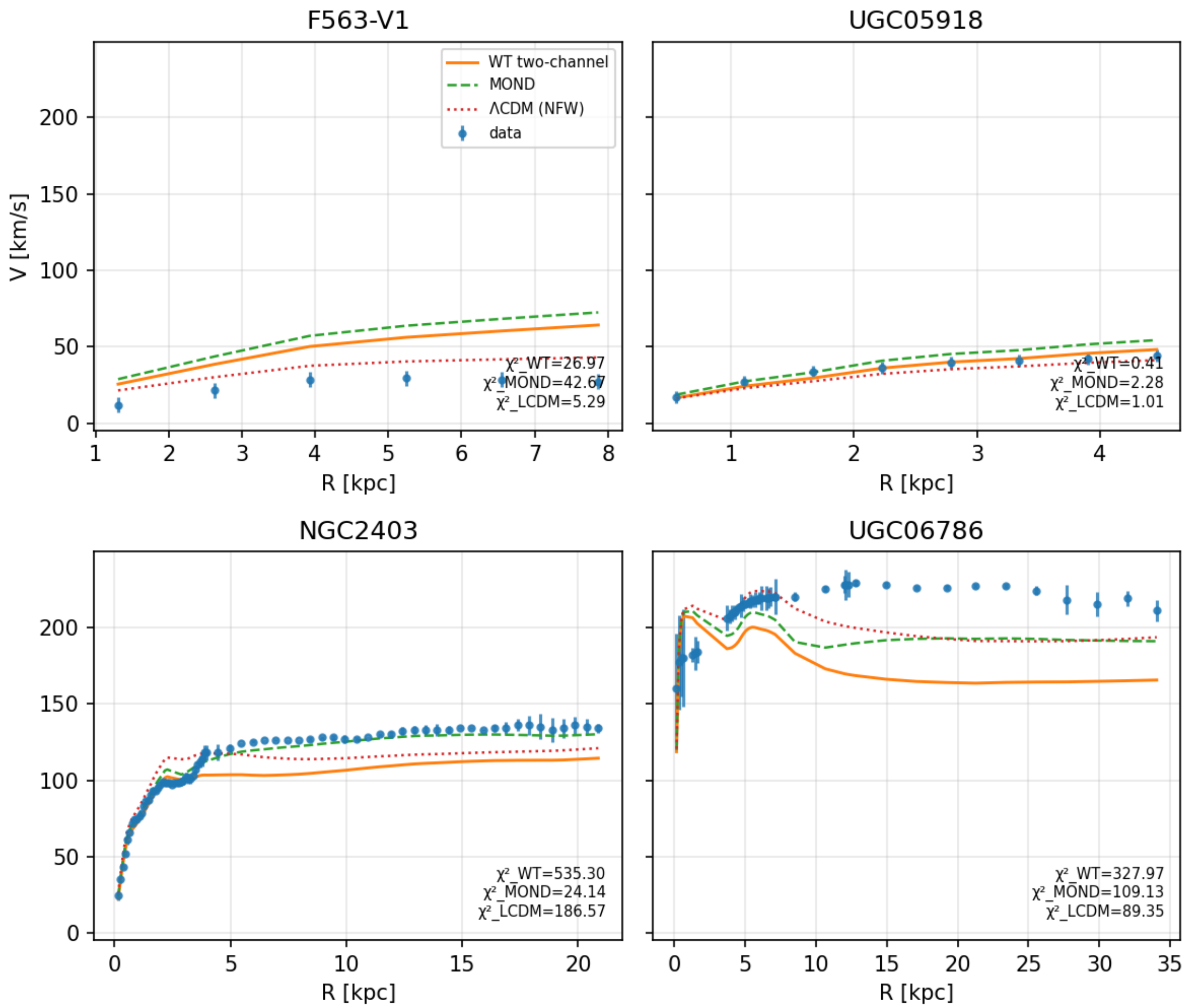

Figure 5: One representative per calibration bin, all plotted at universal κ (zero free parameters). From left to right: F563-V1 (G1, $A_{\mathrm{morph}} = 0.23$), UGC05918 (G2, 0.97), NGC2403 (G3, 1.55), UGC06786 (G4, 2.23). The A_{morph} values identify the bin of each representative (Table 2); all curves use $\alpha_0 = \kappa$ with no per-galaxy tuning; all curves use $\alpha_0 = \kappa$ with no per-galaxy tuning. Each panel shows WT (solid), MOND (dashed), and ΛCDM NFW (dotted). The progressively later shell-coupling onset with increasing compactness is visible in the curvature of the rising inner section. Per-galaxy optimal fits are shown in Figures 3 and 4.

7 Discussion

7.1 What the model does and does not explain

The framework predicts the functional form of the departure from inverse-square scaling but not its full normalization. The r^{-1} scaling, the MOND-like outer regime, and the baryonic Tully–Fisher relation all emerge from the geometry of the transport process — the two-dimensional Green function on S_r, the conserved source $Q_b^{\text{src}} = 4\pi G M_b$, and the transition scale $r_* = \sqrt{GM_b/a_*}$ — with $\alpha_0 = \kappa \cdot A_{\text{morph}}$ the sole remaining quantity. The universal factor $\kappa = 1+\pi^{-2}$ is fixed by the S^3 background geometry; only the continuously distributed morphology factor A_{morph} requires observational calibration.

What the geometry does not fix is the outer amplitude. The normalization of the BTFR and the shape of individual rotation curves depend on A_{morph}, whose empirically determined values are reported in Appendix G. Its epistemic status is documented in Table 1.

The present work is restricted to weak-field galactic dynamics and does not address strong-field behavior, gravitational lensing, or cosmological perturbations.

7.2 The standing of existing frameworks

Before comparing the present model to MOND and ΛCDM, it is worth being explicit about what those frameworks have achieved.

Milgrom's original proposal [2] remains one of the more striking acts of physical intuition in modern astrophysics: a single acceleration constant, fitted to a handful of galaxies in 1983, continues to predict the rotation curves of hundreds of subsequently observed systems across four decades of baryonic mass. Whatever its theoretical status, this predictive compression is a genuine empirical fact that any competing framework must reckon with honestly. The MOND phenomenological literature [22] and the SPARC dataset [17, 18] on which the present analysis is performed are products of that tradition, and the benchmarks reported in Section 6 are only meaningful because MOND is a serious quantitative competitor.

The ΛCDM framework has a different and complementary status. Its successes at cosmological scales — the CMB angular power spectrum, baryon acoustic oscillations, big-bang nucleosynthesis, and the statistics of large-scale structure — constitute some of the best-tested physics of the last three decades [6]. The present work addresses only weak-field galactic dynamics and makes no contact with these observations. ΛCDM is therefore not superseded here; the domain of competition is restricted to the rotation-curve problem, where additional degrees of freedom (per-galaxy halo mass) give the framework flexibility that the present model does not invoke.

The comparisons in Sections 7.4 and 7.3 should be read in this light: they identify where the frameworks make structurally different predictions and where the present model offers a tighter account, not where those frameworks fail at their own tasks.

The relationship between the present framework and its predecessors is therefore twofold. At the phenomenological level it is competitive: the rotation-curve comparisons in Sections 7.3 and 7.4 identify where the frameworks make structurally different predictions, and the SPARC results show where the present model offers a tighter account. At the explanatory level it is complementary: the framework does not dispute that MOND's interpolation function organises the data, or that dark matter halos reproduce large-scale structure. It proposes a geometric mechanism that would, if correct, explain *why* the MOND phenomenology takes the form it does and would account for the apparent missing mass through curvature transport rather than additional substance. Whether that explanation supersedes or supplements existing frameworks is a question the observational tests of Section 6 are designed to address — not one the present paper resolves.

7.3 Relation to MOND and modified gravity

In the outer regime where $a_b \ll a_*$, the model reproduces the deep-MOND scaling

$$a_{\text{eff}} \approx \sqrt{a_b \cdot a_*}, \qquad a_* = \frac{c^2}{\pi R}. \tag{116}$$

The identification of the MOND acceleration scale with $c^2/(\pi R)$ follows directly from the framework postulate $G = a_* = c^2/(\pi R)$ (Table 1, Asserted). Within that postulate it is a structural consequence, not a fitted coincidence. The scale a_* is fixed by the curvature radius of the Universe and enters as a background parameter, not a fitted constant. This means the model makes a distinct prediction: a_0 should evolve with cosmic time as $c^2/(\pi R(z))$, in contrast to standard MOND where a_0 is a fixed universal constant.

The inverse-square behavior arises unchanged from three-dimensional flux conservation. The $1/r$ contribution emerges from the redistribution of curvature along two-dimensional manifolds — a transport effect, not a dynamical modification. In the outer regime it supplants rather than augments the local $1/r^2$ channel: the crossfade formula ensures that curvature counted in the shell channel is no longer counted in the volumetric channel. The consequence is that MOND-like behavior is not postulated but derived, and the acceleration scale is not fitted but fixed by geometry.

This distinction extends to covariant MOND theories such as TeVeS and AQUAL, which introduce additional fields and modified Lagrangians. The present model introduces neither. It operates within the weak-field limit of General Relativity, supplemented only by a dynamical interpretation of curvature propagation on embedded manifolds. Whether the shell transport channel admits a covariant action formulation remains an open question (Section 4.5), but it is not required for the weak-field predictions derived here. For a comprehensive review of MOND phenomenology and its relativistic extensions, see [22]. The weak-field GR foundations used throughout this work follow standard references [5, 6].

7.4 Relation to ΛCDM and dark-matter halo models

The ΛCDM framework reproduces galactic rotation curves by postulating a dark matter halo whose density profile — typically taken as an NFW profile calibrated by the concentration–mass relation — adds a velocity contribution on top of the baryonic one. The present model contains no analogous component: the $1/r$ acceleration arises from curvature transport geometry, not from additional mass. The two frameworks therefore make structurally different predictions even when their rotation-curve outputs are numerically close.

Three distinctions are sharpest in the data. First, the NFW profile introduces a per-galaxy free parameter (halo mass or concentration), whereas the present model uses a per-class constant α_0 fixed by morphology. A fixed-parameter NFW comparison (halo mass estimated from $V_{\max}$ via the Dutton–Macciò $c(M)$ relation [13]) achieves a lower median χ^2 than either WT or MOND across the 175-galaxy SPARC sample, but a higher mean χ^2 and larger galaxy-to-galaxy scatter (Appendix G). This pattern is consistent with NFW describing bright, high-surface-brightness spirals well, with larger residuals in gas-dominated dwarfs and low-surface-brightness systems where the assumed halo profile is a less natural match.

Second, the ΛCDM halo is a dynamically inert background in the rotation-curve context, whereas the curvature-transport mechanism predicts a coupling that depends on the degree of orbital phase mixing. Systems without coherent rotation (pressure-supported spheroidals, tidally disrupted dwarfs) should therefore show a suppressed $1/r$ contribution in the present model but not in NFW-based models, which assign a halo to every object regardless of its dynamical state. This provides a qualitative discriminant that rotation curves alone cannot resolve.

Third, the effective acceleration scale $a_* = c^2/(\pi R)$ evolves with cosmic time in the present framework, whereas dark-matter halo properties at fixed mass evolve only through concentration. High-redshift rotation curves should show a systematically larger WT contribution at fixed baryonic mass — a prediction that is absent in halo-based models and testable with integral field spectroscopy at $z \sim 0.5$–1.

7.5 Role of the S^3 geometry and Wave Theory language

The use of S^3 geometry serves both physical and technical purposes.

First, a three-sphere of radius $R \sim c/H_0$ corresponds to the spatial slice of a closed FLRW universe with curvature at the Hubble scale. It therefore provides a natural global setting rather than an additional assumption.

Second, the Laplace–Beltrami operator on S^3 separates cleanly into radial and angular components. In the large-χ regime, angular modes dominate, making the effective reduction to two-dimensional transport explicit. This structure underlies the shell Poisson equation derived in Appendix C.

An equivalent formulation in flat space with a cosmological cutoff is possible; the S^3 representation is chosen because it makes the dimensional transition geometrically transparent.

The value $R = \pi c/H_0$ used throughout this work corresponds to the present-day Hubble radius under the Wave Theory relation $H = \pi c/R$. Adopting $H_0 = 67.8\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (consistent with Planck CMB measurements) gives $R \approx 4.29 \times 10^{26}\,\mathrm{m}$ and $a_* = c^2/(\pi R) \approx 6.67 \times 10^{-11}\,\mathrm{m\,s^{-2}}$. This is not a free parameter — it is fixed by cosmological observation independently of galactic dynamics. The Hubble tension ($H_0 \approx 67$–$73\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ depending on the measurement method) introduces a corresponding $\sim$8% uncertainty in R and a_*, which propagates into the normalization of the master formula but does not affect its functional form or the rotation-curve win counts reported in Section 6. The fact that the same R appears in both the curvature coupling $G = c^2/(\pi R)$ and the MOND scale $a_* = c^2/(\pi R)$ is therefore a non-trivial constraint on the framework's internal consistency, not an additional degree of freedom.

In this representation:

- $G = c^2/(\pi R)$ expresses gravitational coupling as a curvature scale,
- mass is represented geometrically through the curvature source,
- $a_* = c^2/(\pi R)$ appears as a background curvature parameter.

All results translate directly to the standard formulation through

$$GM_b \;\leftrightarrow\; a_* A_b. \tag{117}$$

The advantage is conceptual: the MOND scale follows directly from the geometric postulate $G = a_* = c^2/(\pi R)$ rather than being a fitted constant, and the dimensional structure of the problem is made explicit.

7.6 Observational tests and falsifiability

The mechanism developed in this work makes a set of concrete, testable predictions that go beyond reproducing rotation curves. These predictions arise directly from the dynamical origin of shell transport — causal propagation combined with orbital phase mixing — and therefore provide sharp criteria for falsification.

I summarize five primary tests.

(1) Morphology dependence of the projection efficiency α_0

In the present framework the only empirically undetermined quantity is the dimensionless projection efficiency α_0, which encodes how effectively the bulk curvature flux couples to shell transport modes.

The model predicts that α_0 is not an arbitrary per-galaxy parameter but a function of observable baryonic structure. In particular, defining

$$\Delta\alpha_0 \equiv \alpha_0 - \kappa, \tag{118}$$

with $\kappa = 1 + \pi^{-2}$ the geometric baseline, the distribution of $\Delta\alpha_0$ over a large galaxy sample should be unimodal and centered near zero, with systematic dependence on structural observables such as

$$\mu_0, \quad f_{\rm gas}, \quad R_d, \quad B/D, \quad a_{\rm char}, \quad V_{\rm max}. \tag{119}$$

The precise functional form

$$\Delta\alpha_0 = f(\mu_0, f_{\rm gas}, R_d, B/D, a_{\rm char}, V_{\rm max}) \tag{120}$$

is not derived here, but the existence of such a relation is required by the geometric interpretation of α_0 as a projection efficiency.

Falsification criterion. If α_0 exhibits no statistically significant correlation with any observable structural quantity and behaves as unstructured scatter, the interpretation of α_0 as a geometric projection factor is ruled out.

(2) Rotational vs. pressure-supported systems

The shell transport mechanism does not require coherent rotation. In rotationally supported disk galaxies, orbital coherence produces efficient phase mixing and a robust $1/r$ shell contribution. In pressure-supported spheroidal systems, orbits are not coherent but span a distribution of directions. Rather than suppressing mixing, this enhances it: randomised orbits lead to rapid isotropisation of the shell source, so the two-dimensional transport description remains valid with a different projection efficiency encoded in α_0 (Section 3.7).

The prediction is therefore that pressure-supported systems follow the same $1/r$ outer scaling as rotating disks, but with a systematically shifted α_0 reflecting their morphological class. Defining an outer-regime estimator

$$\mathcal{A}_{\rm out} \equiv \frac{a_{\rm obs}}{\sqrt{a_b\, a_*}}, \tag{121}$$

the model predicts that $\mathcal{A}_{\rm out}$ follows the same functional form across morphological classes, with normalisation set by α_0. Velocity-dispersion profiles of pressure-supported systems provide the direct kinematic probe.

Falsification criterion. If pressure-supported systems show no $1/r$ outer contribution in velocity-dispersion profiles at any α_0 — that is, if the shell scaling is absent regardless of normalisation — the mechanism is falsified, because the isotropisation argument requires the $1/r$ term to persist. Conversely, if they follow the same normalisation as rotating disks (α_0 identical to disk values at matched baryonic mass), the morphology dependence of α_0 is ruled out.

(3) Breakdown of shell coherence at large radius

Shell transport requires efficient orbital phase mixing. This condition is satisfied when the orbital period is much shorter than the system age,

$$T_{\rm orb}(r) \ll T_{\rm gal}. \tag{122}$$

At sufficiently large radii, this condition fails. Defining a mixing scale $r_{\rm mix}$ by

$$T_{\rm orb}(r_{\rm mix}) \sim T_{\rm gal}, \tag{123}$$

the model predicts that beyond this scale:

- residuals to the shell model increase,
- azimuthal asymmetry becomes more pronounced,
- galaxy-to-galaxy scatter grows.

A natural diagnostic variable is the dimensionless ratio

$$\frac{T_{\rm orb}(r)}{T_{\rm gal}}. \tag{124}$$

Falsification criterion. If the outer scatter and residual structure of rotation curves show no correlation with $T_{\mathrm{orb}}/T_{\mathrm{gal}}$, the dynamical origin of shell transport is not supported.

(4) Redshift evolution of the acceleration scale

The characteristic acceleration scale of the model is

$$a_* = \frac{c^2}{\pi R}, \tag{125}$$

which evolves with cosmic time through the curvature radius $R(z)$. Since $H = \pi c/R$ in the Wave Theory relation,

$$a_*(z) = \frac{H(z)\, c}{\pi^2}, \tag{126}$$

so a_* grows with redshift in direct proportion to $H(z)$.

This implies that the asymptotic BTFR normalization satisfies

$$\frac{V_\infty^4}{M_b} \propto a_*(z) \propto H(z), \tag{127}$$

up to the morphology factor A_{morph}, which is a property of baryonic structure and does not evolve with $H(z)$. The prediction is therefore that, at fixed baryonic mass and fixed morphological class, galaxies at higher redshift exhibit systematically higher outer velocities, with the normalization shift tracking $H(z)$ and not an independent free parameter.

This distinguishes the framework from standard MOND, where a_0 is a fixed constant, and from ΛCDM NFW models, where BTFR evolution is driven by concentration-mass evolution rather than a global acceleration scale.

Falsification criterion. The test has two failure modes. First, if no redshift evolution of the BTFR normalization is detected after controlling for selection effects and morphological composition at $z \sim 0.5$–1, where $H(z)/H_0 \approx 1.6$–1.8 implies a predicted normalization shift of order 60–80% in a_*, the geometric identification of a_* with $H(z)$ is ruled out. Second, if evolution is detected but scales with a power of $H(z)$ other than unity — for example as $H(z)^{1/2}$ or $H(z)^2$ — the specific proportionality $a_* \propto H$ is falsified even if some evolution is present. The predicted signal is accessible with high-redshift integral field spectroscopy at $z \sim 0.5$–1.

(5) Local-compactness control of the transition

The transition between the inner and outer regimes is governed by the local compactness parameter

$$\xi(r) = \sqrt{\frac{a_b(r)}{a_*}}, \tag{128}$$

rather than by global galaxy properties alone.

The transition radius r_t is therefore defined implicitly by

$$a_b(r_t) \sim a_*. \tag{129}$$

The model predicts that galaxies with identical total baryonic mass but different radial distributions exhibit different transition radii, controlled by their local acceleration profiles.

Rescaling radii by the condition $a_b(r) = a_*$ should collapse the transition across galaxies more effectively than scaling by R_d, R_{eff}, or total mass alone.

Falsification criterion. If transition radii are determined solely by global quantities such as total mass or halo scale, with no role for local compactness, the crossfade mechanism is not supported.

Observational roadmap

The predictions above can be tested directly with existing and near-term datasets.

For local disk galaxies, the SPARC [17] and THINGS [23] samples provide high-quality rotation curves and baryonic decompositions. These data allow (i) extraction of α_0 per galaxy, (ii) correlation of $\Delta\alpha_0$ with structural observables (μ_0, f_{gas}, R_d, a_{char}), and (iii) tests of the local-compactness prediction by rescaling radii using the condition $a_b(r) = a_*$. Residuals can be analyzed as a function of $T_{\text{orb}}/T_{\text{gal}}$ to test the predicted breakdown of shell coherence at large radii.

For pressure-supported systems, dwarf spheroidals with resolved stellar kinematics [24] and early-type galaxies from the ATLAS$^{\text{3D}}$ survey [25] provide a direct comparison at fixed baryonic mass. The key observable is the outer acceleration or dispersion profile, from which $\mathcal{A}_{\text{out}}$ can be inferred and compared to rotationally supported systems at matched M_b.

At intermediate and high redshift, integral-field spectroscopy surveys — including the KMOS Galaxy Evolution Survey [26], MUSE-Wide [27], and JWST/NIRSpec programs such as JADES [28] — provide spatially resolved kinematics for star-forming disks at $z \sim 0.5$–2. These enable measurement of the BTFR normalization as a function of redshift and a direct test of the scaling $V_\infty^4/M_b \propto H(z)$ after controlling for gas fraction and morphological class.

The required analyses involve no new observables beyond standard rotation-curve fitting, baryonic mass modelling, and kinematic profiling. The distinguishing feature of the present framework is not the data required, but the correlations it predicts across morphology, radius, and cosmic time. A coordinated analysis across these datasets therefore provides a decisive empirical test of the shell-transport mechanism.

Summary

The framework predicts not only the functional form of the deviation from inverse-square scaling, but also its dependence on morphology, orbital structure, cosmic time, and local compactness.

Failure of any one of the tests above would directly challenge a specific element of the mechanism. Taken together, they provide a stringent and multi-dimensional empirical assessment of the shell-transport picture.

8 Conclusions and open questions

8.1 Summary

The central result of this paper is that flat galactic rotation curves arise from a change in the effective dimensionality of curvature transport, without new matter or modified laws. The derivational chain is as follows. Baryonic matter generates a conserved curvature flux whose boundary value at any spherical surface S_r is the standard Gauss-law quantity $Q_b^{\mathrm{src}} = 4\pi G M_b(r)$. In the quasi-static, large-radius limit, the bulk causal transport equation on S^3 reduces — through the spectral structure of the Laplace–Beltrami operator (Section 4.5, Appendix A) — to an intrinsic two-dimensional Poisson equation on S_r. The logarithmic Green function of this equation produces a shell acceleration scaling as $1/r$. The crossfade between this channel and the standard $1/r^2$ channel yields the master formula (68). In the outer regime this reduces to $a_{\mathrm{eff}} \approx \sqrt{a_b \cdot a_*}$, with $a_* = c^2/(\pi R)$ fixed by the Hubble radius. The baryonic Tully–Fisher relation $v_\infty^4 \propto M_b$ follows directly (Appendix C.1).

The orbital confinement mechanism — radial dephasing through phase mixing and tangential confinement through spiral winding — is established by physical argument, not derived from first principles. The epistemic status of every principal claim is documented in Table 1.

The amplitude $\alpha_0 = \kappa \cdot A_{\mathrm{morph}}$ factorises into a universal geometric constant $\kappa = 1 + \pi^{-2}$, fixed by the S^3 background, and a continuously distributed morphology factor A_{morph} that requires observational calibration. A GMM analysis of the 175-galaxy SPARC sample confirms that A_{morph} is statistically unimodal, centred near κ, consistent with morphology acting as a continuous perturbation around a universal geometric attractor (Section 6.3). The calibration has been performed on the 175-galaxy SPARC sample and is reported in Appendix G.

8.2 Open problems

The following problems remain open within the current framework. They are organised by the section in which they arise.

Orbital organization and shell confinement (Section 3).

- The relation between mixing efficiency, the critical mixing radius r_{mix}, and the morphology factor α_0 has not been derived from first principles. A quantitative theory of how baryonic morphology determines shell-mode projection efficiency would replace the empirical calibration of α_0 with a geometric prediction.

- The model predicts weakened or absent shell contributions in systems lacking coherent rotation. A systematic observational test of this prediction across pressure-supported and low-angular-momentum systems has not been carried out.

Causal transport reduction (Section 4.5).

- The full causal reduction — including finite memory time τ_c and cosmological expansion $H = \pi c/R$ — has not been carried out beyond the quasi-static limit. At intermediate radii, both effects modify σ_r and introduce corrections to the shell occupancy that are not captured by the present derivation.

- The fractional operator generalisation of the bulk transport equation (equation (63) with $\sigma \neq 1$) has not been reduced to a shell equation. The form of the effective shell dynamics in this case is not known.

- The spectral dominance condition $\ell(\ell+1)/\sin^2\chi \gg n(n+2)$ has been verified at the level of the operator structure (Appendix A) but not tracked through the full Green function of the screened equation (44). A rigorous estimate of corrections to the angular-dominance approximation remains to be given. Establishing this would promote the shell Poisson equation from an asymptotic result to a controlled approximation with explicit error bounds.

Empirical determination (Section 4).

- The per-galaxy $A_{\rm morph}$ distribution has been characterised on the 175-galaxy SPARC sample (Appendix G) and is statistically unimodal (Section 6.3), centred near κ with a standard deviation of ≈ 0.56. Whether the mean, width, and skewness of this distribution are stable under changes in sample selection, distance calibration, or inclination correction remains to be tested on independent datasets. The natural next step is to study the continuous residual $\Delta\alpha_0 \equiv \alpha_0 - \kappa$ as a function of observable structural properties (M_{200}, $V_{\rm max}$, $a_{\rm char}$, surface-brightness indicators).

- A first-principles derivation of $A_{\rm morph}$ from baryonic morphology — replacing empirical calibration with a geometric prediction — has not been achieved. Such a derivation would require a quantitative theory of how baryonic structure determines shell-mode projection efficiency, and would elevate BTFR normalization from Empirical to Derived.

- The effective MOND scale $a_0^{\rm eff} = \alpha_0^2\, a_*$ is observationally close to the empirical $a_0 \approx 1.2 \times 10^{-10}\ \mathrm{m\,s^{-2}}$ for $\alpha_0 \approx 1.34$. A first-principles derivation of $A_{\rm morph}$ would simultaneously fix this numerical coincidence.

Covariant embedding.

- Whether the two-channel acceleration formula (68) can be derived from a covariant action principle remains open. The present derivation operates in the weak-field limit and does not address whether the shell transport channel admits a fully relativistic formulation.

- Extension to non-circular systems — merging galaxies, tidally disturbed systems, and triaxial structures — has not been attempted. The orbital averaging argument relies on approximate circular motion and its validity in these regimes is not established.

A Appendix: Spectral Origin of Shell Dynamics on S^3

This appendix provides an alternative derivation of the shell transport behavior by analyzing the spectral properties of operators defined on S^3. The goal is to show that the effective two-dimensional response arises naturally from the structure of the three-dimensional manifold.

A.1 Geometry of S^3 and bulk-to-shell projection

The spatial geometry is a three-sphere S^3 of radius R, with metric

$$ds^2 = R^2 \left(d\chi^2 + \sin^2\chi \, d\Omega_2^2\right), \qquad \chi \in [0, \pi], \tag{130}$$

where $\chi = r/R$ is the dimensionless radial coordinate and $d\Omega_2^2$ is the unit two-sphere metric. The volume of the cap enclosed within χ and the area of its boundary two-sphere S_r are

$$V_{\rm cap}(\chi) = 2\pi R^3 \left(\chi - \tfrac{1}{2}\sin 2\chi\right), \qquad A_\partial(\chi) = 4\pi R^2 \sin^2\chi. \tag{131}$$

In the local limit $\chi \ll 1$ these reduce to the standard Euclidean expressions $V \to \frac{4}{3}\pi r^3$ and $A \to 4\pi r^2$, confirming consistency with Newtonian gravity at sub-galactic scales.

The curvature-flux field $\mathbf{F}_b$ satisfies $\nabla \cdot \mathbf{F}_b = s_b$, where s_b is the baryonic curvature source density. Applying the divergence theorem to V_r and identifying $\mathbf{F}_b = -\nabla\Phi$ with $\nabla^2\Phi = 4\pi G\rho_b$ in the weak-field limit gives

$$Q_b^{\rm src}(r) \equiv 4\pi G M_b(r), \tag{132}$$

consistent with the sign convention of Appendix D. The normal flux $\sigma_r \equiv \mathbf{F}_b \cdot \mathbf{n}_r$ injected through S_r drives curvature transport on the shell via the surface Poisson equation

$$\Delta_{S_r}\Phi_r = -\sigma_r. \tag{133}$$

The Green function of the two-dimensional Laplacian is logarithmic, $G \sim \log r$, so the shell potential scales as $\Phi_r \sim Q_b(r)\log r$ and the resulting shell acceleration is

$$a_{\rm shell}(r) \sim \frac{Q_b^{\rm src}(r)}{r}, \tag{134}$$

establishing the inverse-linear behavior as a direct consequence of 2D transport geometry. The spectral origin of the dimensional reduction that produces this 2D equation is derived in the subsections that follow.

A.2 Laplacian on S^3

The Laplace–Beltrami operator on S^3 in coordinates (χ, Ω) can be written as

$$\Delta_{S^3} = \frac{1}{R^2}\left[\frac{\partial^2}{\partial\chi^2} + 2\cot\chi\frac{\partial}{\partial\chi} + \frac{1}{\sin^2\chi}\Delta_{S^2}\right] \tag{135}$$

where Δ_{S^2} is the Laplacian on the unit two-sphere. For the spectral theory of the Laplace–Beltrami operator on compact Riemannian manifolds, see [10, 11].

This decomposition separates radial and angular contributions.

A.3 Spectral decomposition

Functions on S^3 can be expanded in hyperspherical harmonics $Y_{n\ell m}(\chi, \Omega)$, which satisfy

$$\Delta_{S^3} Y_{n\ell m} = -\frac{n(n+2)}{R^2} Y_{n\ell m}, \qquad n = 0, 1, 2, \ldots \tag{136}$$

The index n labels global modes, while (ℓ, m) correspond to angular structure on S^2. The eigenfunctions and eigenvalues of the Laplacian on S^3 are derived in [21].

A.4 Radial localization and boundary dominance

Consider a source localized within a spherical cap $V_{\mathrm{cap}}(\chi)$. The resulting field can be expanded in eigenmodes, but at large distances (large χ), the contribution is dominated by modes with minimal radial variation.

$$\Delta_{S^3} \xrightarrow{\ell(\ell+1)/\sin^2\chi \gg n(n+2)} \frac{1}{R^2\sin^2\chi}\,\Delta_{S^2}, \tag{137}$$

In this regime, radial gradients become subdominant compared to angular variations, and the operator effectively reduces to

$$\Delta_{S^3} \longrightarrow \frac{1}{R^2\sin^2\chi} \times \Delta_{S^2} \tag{138}$$

Thus, the dynamics becomes effectively constrained to the angular manifold S^2.

A.5 Effective dimensional reduction

The above limit shows that, for observers at fixed radius r, the relevant dynamics is governed by an operator proportional to Δ_{S^2}.

This induces an effective reduction:

$$S^3 \longrightarrow S_r \simeq S^2 \tag{139}$$

where curvature propagation is restricted to the shell.

In the small-angle limit $r \ll R$ (equivalently $\theta \ll 1$), the Green function of the effective operator reduces to its flat-space 2D form,

$$G(\theta) \sim \log\theta \tag{140}$$

which is the standard 2D logarithmic potential, where θ is an angular separation on S_r.

A.6 Connection to radial scaling

Mapping angular separation to physical distance on the shell, $r \sim R\theta$, the logarithmic Green function leads to

$$\Phi_r \sim \log r \tag{141}$$

and hence

$$a_{\text{shell}}(r) \sim \frac{1}{r} \tag{142}$$

Thus, the inverse-linear scaling emerges from the spectral structure of the Laplacian on S^3.

A.7 Relation to causal transport

The spectral reduction described above corresponds to the steady-state limit of the causal transport process discussed in Section 5.

In the full dynamical picture:

- curvature propagates through S^3 with finite speed,
- radial transport becomes suppressed relative to tangential redistribution,
- the system relaxes to a configuration dominated by angular modes.

The effective two-dimensional behavior therefore arises both:

- dynamically, through causal transport and finite propagation,
- spectrally, through the structure of the Laplacian on S^3.

A.8 Interpretation

This derivation shows that the shell transport mechanism is not an external assumption but a natural consequence of the geometry of S^3.

The appearance of a two-dimensional effective dynamics reflects the dominance of boundary modes in the propagation of curvature at large scales.

A.9 Summary

The spectral analysis establishes that:

- the Laplacian on S^3 separates into radial and angular components,
- at large scales, angular modes dominate the response,
- the effective dynamics reduces to that of a two-dimensional manifold,
- the resulting Green function produces inverse-linear scaling.

This provides an independent confirmation of the shell transport mechanism derived in the main text.

B Appendix: Shell Green Function and the Emergence of the $1/r$ Law

This appendix derives the inverse-linear shell contribution used in the main text. The result follows from the fact that curvature transport on an orbital shell is governed by a two-dimensional Poisson problem, whose Green function is logarithmic.

B.1 Shell transport as a two-dimensional Poisson problem

Let S_r denote the constant-radius orbital shell. The redistribution of curvature on S_r by a tangential current J_r satisfying the conservation law

$$\mathrm{div}_{S_r} J_r = \sigma_r \tag{143}$$

where σ_r is the effective shell source density induced by the interior baryonic region.

Introducing a scalar shell potential Φ_r by

$$J_r = -\nabla_{S_r}\Phi_r \tag{144}$$

we obtain

$$\Delta_{S_r}\Phi_r = -\sigma_r \tag{145}$$

Thus the shell response is determined by the Green function of the Laplace–Beltrami operator on a two-dimensional manifold.

B.2 Local flat approximation

For shell scales much smaller than the curvature radius of the background manifold, the orbital shell is locally well approximated by the Euclidean plane $\mathbb{R}^2$. In this limit the shell Poisson equation becomes

$$\Delta\Phi(\mathbf{x}) = -\sigma(\mathbf{x}) \qquad \mathbf{x} \in \mathbb{R}^2. \tag{146}$$

The Green function $G(\mathbf{x}, \mathbf{x}')$ is defined by

$$\Delta G(\mathbf{x}, \mathbf{x}') = -\delta^{(2)}(\mathbf{x} - \mathbf{x}') \tag{147}$$

In two dimensions, this Green function is

$$G(\mathbf{x}, \mathbf{x}') = -\frac{1}{2\pi} \log |\mathbf{x} - \mathbf{x}'| \tag{148}$$

The theory of elliptic partial differential equations and their Green functions is treated in [8, 9].

Hence the shell potential generated by a source density σ is

$$\Phi(\mathbf{x}) = \int_{\mathbb{R}^2} G(\mathbf{x}, \mathbf{x}')\ \sigma(\mathbf{x}')\ d^2x' \tag{149}$$

For a localized source of total shell content

$$Q_r = \int_{\mathbb{R}^2} \sigma(\mathbf{x}')\ d^2x' \tag{150}$$

the far-field potential reduces to

$$\Phi(r) \sim -\frac{Q_r}{2\pi} \log r \tag{151}$$

up to an additive constant.

Taking the gradient,

$$|\nabla\Phi(r)| \sim \frac{Q_r}{2\pi r} \tag{152}$$

Therefore the shell-supported acceleration obeys

$$a_{\text{shell}}(r) \propto \frac{Q_r}{r} \tag{153}$$

This is the origin of the inverse-linear scaling used in the main text.

B.3 Equivalent flux derivation

The same result follows directly from flux conservation. Let $D_\rho \subset S_r$ be a geodesic disk of shell radius ρ enclosing the effective shell source. Integrating the conservation law gives

$$\int_{D_\rho} \text{div}_{S_r} J_r\ dA = \int_{D_\rho} \sigma_r\ dA = Q_r(\rho) \tag{154}$$

where $Q_r(\rho)$ is the shell content enclosed within D_ρ.

By the two-dimensional divergence theorem,

$$\int_{\partial D_\rho} J_r \cdot n \, ds = Q_r(\rho) \tag{155}$$

Far from the source patch, $Q_r(\rho)$ saturates to the total shell content Q_r, and the transport becomes approximately isotropic. Writing

$$J_r = J_\rho(\rho) \, \hat{n} \tag{156}$$

we obtain

$$J_\rho(\rho) \, \mathrm{Length}(\partial D_\rho) = Q_r \tag{157}$$

Locally on a two-dimensional shell,

$$\mathrm{Length}(\partial D_\rho) \sim 2\pi\rho \tag{158}$$

so that

$$J_\rho(\rho) \sim \frac{Q_r}{2\pi\rho} \tag{159}$$

Identifying the relevant shell scale with the orbital radius, $\rho \sim r$, again yields

$$a_{\mathrm{shell}}(r) \propto \frac{Q_r}{r} \tag{160}$$

B.4 Extension to the spherical shell

The exact orbital shell is a two-sphere S_r, not a flat plane. The local logarithmic form nonetheless remains valid because the Green function of the Laplace–Beltrami operator on a two-sphere has the same short-distance asymptotic behavior.

Let γ denote the angular separation on S_r. Then the Green function on the sphere satisfies

$$\Delta_{S_r} G(\gamma) = -\delta_{S_r} + \frac{1}{4\pi r^2} \tag{161}$$

where the constant term enforces zero mean over the compact manifold.

Its explicit form is

$$G(\gamma) = -\frac{1}{4\pi} \log(1 - \cos\gamma) + C \tag{162}$$

for some constant C. For small separations $\gamma \ll 1$,

$$1 - \cos\gamma \sim \frac{\gamma^2}{2} \tag{163}$$

hence

$$G(\gamma) \sim -\frac{1}{2\pi}\log\gamma + C' \tag{164}$$

Since the geodesic distance is $d = r\gamma$, this becomes

$$G(d) \sim -\frac{1}{2\pi}\log d + C'' \tag{165}$$

which is the same local logarithmic behavior as in the flat case. Its gradient therefore scales as

$$|\nabla_{S_r} G(d)| \sim \frac{1}{2\pi d} \tag{166}$$

Thus the inverse-linear shell response is not an artifact of the flat approximation; it is the natural local asymptotic of the exact Green function on the spherical shell.

B.5 Result

The shell transport equation is intrinsically two-dimensional, and the corresponding Green function is logarithmic. Consequently, the shell-supported acceleration scales as the gradient of a logarithmic potential:

$$\Phi_{\text{shell}} \sim Q_r \log r \qquad \Longrightarrow \qquad a_{\text{shell}}(r) \sim \frac{Q_r}{r} \tag{167}$$

This is the mathematical origin of the $1/r$ contribution appearing in the effective galactic acceleration law.

C Appendix: Boundary Projection Methods

Gauss–Codazzi and alternative formulations.

C.1 Minimal steady-state shell solution and emergence of MOND-like scaling

The evolution law introduced above,

$$\partial_t A + \mathbf{u}\cdot\nabla A = -\frac{1}{\tau_{\mathrm{dil}}(A)}\ A - H\,\Xi[A] + S_{\mathrm{rot}} \tag{168}$$

describes curvature on the shell as a causally transported and continuously replenished quantity. Its purpose is not to replace the local inverse-square law, but to determine the amount of curvature that remains resident on the shell and is therefore available for two-dimensional transport. This is consistent with the general framework already established in Section 4, where the local channel scales as r^{-2}, the shell channel scales as r^{-1}, and the transition is controlled by the universal curvature scale $a_* = c^2/(\pi R)$.

Minimal shell average. To obtain a closed analytic expression, consider the azimuthally averaged shell occupancy. In a quasi-stationary regime, the explicit time dependence averages out and the transport advection term does not contribute to the shell average. The expansion is further approximated as operator by its leading linear action on the shell amplitude,

$$\Xi[A] \approx A \tag{169}$$

and evaluate the amplitude-dependent dilution time at an effective shell value,

$$\tau_{\mathrm{dil}}(A) \to \tau_s \tag{170}$$

The shell-averaged steady-state equation then becomes

$$0 = -\left(\frac{1}{\tau_s} + H\right) A_s + S_s \tag{171}$$

with solution

$$A_s = \frac{S_s}{\frac{1}{\tau_s} + H} = \tau_{\mathrm{eff}}\ S_s \qquad \tau_{\mathrm{eff}} = \frac{1}{\frac{1}{\tau_s} + H} \tag{172}$$

Thus the shell stores a finite curvature occupancy equal to the injection rate multiplied by an effective residence time. The residence time is controlled jointly by local causal dilution and global Hubble stretching. In Wave Theory, the latter is fixed by

$$H = \frac{\pi c}{R} \tag{173}$$

while the causal transport sector is constrained by finite front velocity, as in the telegraph-type curvature dynamics on S^3.

Source-to-shell injection scale. The shell is fed by the enclosed source

$$Q_b^{\mathrm{src}}(\chi) = 4\pi G M_b(\chi) \tag{174}$$

which in the geometric formulation is equivalently

$$Q_b^{\mathrm{src}}(\chi) = 4\pi a_* A_b(\chi) \qquad a_* = \frac{c^2}{\pi R} = G \tag{175}$$

The simplest steady-state closure is to assume that the shell is populated on the radial transport scale at which the local baryonic channel and the cosmological curvature floor become comparable. In Section 6 this transition radius is

$$r_* \sim \sqrt{\frac{GM_b}{a_*}} \tag{176}$$

This same scale already controls the onset of shell relevance in the static model.

A minimal dimensional estimate for the shell injection rate is then

$$S_s \sim \frac{Q_b^{\mathrm{src}}}{\tau_{\mathrm{in}}} \qquad \tau_{\mathrm{in}} \sim \frac{r_*}{\nu_s} \tag{177}$$

where $\nu_s = \gamma_s c$ is the effective causal update speed on the shell. Hence

$$S_s \sim \frac{\nu_s}{r_*} Q_b^{\mathrm{src}} \tag{178}$$

Substituting into the steady-state solution gives

$$A_s \sim \frac{\tau_{\mathrm{eff}} \nu_s}{r_*} Q_b^{\mathrm{src}} \tag{179}$$

It is convenient to absorb the factor $\tau_{\mathrm{eff}} \nu_s$ into a dimensionless occupancy coefficient η_s,

$$\eta_s \equiv \frac{\tau_{\mathrm{eff}} \nu_s}{r_*} \tag{180}$$

so that

$$A_s \sim \eta_s \, Q_b^{\mathrm{src}} \tag{181}$$

In the quasi-stationary transition regime, where shell filling and shell dilution balance on the same radial scale, η_s is naturally of order unity. The shell occupancy can then be written in the compact form

$$Q_b^{\mathrm{shell}} \sim \kappa_s \, \frac{Q_b^{\mathrm{src}}}{r_*} \tag{182}$$

where κ_s is a dimensionless constant encoding the detailed efficiency of shell residence.

Shell acceleration in the transition regime. Using the shell transport law from Section 4,

$$a_{\mathrm{shell}}(r) = \alpha_b(\chi) \, \frac{Q_b^{\mathrm{shell}}}{r} \tag{183}$$

we obtain

$$a_{\mathrm{shell}}(r) \sim \alpha_b(\chi) \, \frac{\kappa_s \, Q_b^{\mathrm{src}}}{r \, r_*} \tag{184}$$

Substituting $Q_b^{\mathrm{src}} = 4\pi G M_b$ and $r_* = \sqrt{G M_b / a_*}$, one finds

$$a_{\mathrm{shell}}(r) \sim 4\pi\kappa_s \, \alpha_b(\chi) \, \frac{\sqrt{G M_b \, a_*}}{r} \tag{185}$$

Since the local baryonic acceleration is

$$a_b(r) = \frac{G M_b}{r^2}, \tag{186}$$

the shell acceleration can be written equivalently as

$$\boxed{a_{\mathrm{shell}}(r) \sim 4\pi\kappa_s \, \alpha_b(\chi) \, \sqrt{a_b(r) \, a_*}} \tag{187}$$

This is the characteristic MOND-like form, valid in the outer regime where $M_b(r) \approx M_b$ is approximately constant and the identification $\sqrt{G M_b \, a_*}/r = \sqrt{a_b(r) \, a_*}$ holds. At smaller radii where $M_b(r)$ is still growing with r, the shell contribution retains the intermediate form

$$a_{\mathrm{shell}}(r) \sim \frac{4\pi\kappa_s \, \alpha_b(\chi) \, \sqrt{G M_b(r) \, a_*}}{r}, \tag{188}$$

which does not reduce to the MOND square-root scaling.

The MOND-like form emerges not from modifying the local gravitational law, but from three ingredients acting together:

1. the standard enclosed Gauss-law source $Q_b^{\mathrm{src}} = 4\pi G M_b$,
2. the shell-transport Green function on a two-dimensional manifold, which gives the inverse-linear propagation law,
3. the transition scale $r_* \sim \sqrt{GM_b/a_*}$, set by the equality of local baryonic curvature and the cosmological curvature floor.

Asymptotic speed and BTFR. The rotation speed satisfies

$$v^2(r) = r\, a_{\mathrm{eff}}(r) \tag{189}$$

In the shell-dominated outer regime,

$$v_\infty^2 \sim 4\pi\kappa_s\, \alpha_b(\chi_*)\ \sqrt{GM_b\, a_*} \tag{190}$$

hence

$$v_\infty^4 \sim (4\pi\kappa_s)^2\ \alpha_b(\chi_*)^2\ GM_b\, a_* \tag{191}$$

Therefore the baryonic Tully–Fisher scaling

$$v_\infty^4 \propto M_b \tag{192}$$

follows immediately, in agreement with the scaling relation already identified in the static shell model.

Connection to the master formula. The coefficient $4\pi\kappa_s\, \alpha_b(\chi_*)$ appearing above is absorbed into the single morphology parameter α_0 of the crossfade master formula (68). The BTFR then takes the cleaner form $V_\infty^4 = \alpha_0^2\, GM_b\, a_*$, derived directly in Section 5.4. The Appendix E derivation provides the physical mechanism (shell occupancy, causal injection, steady-state balance) that justifies treating α_0 as a morphology-class constant rather than a per-galaxy free parameter.

Interpretation. The result shows that MOND-like behavior need not be postulated as a modified force law. It appears as the quasi-stationary limit of a causal curvature-transport system on an expanding S^3 background. The familiar square-root scaling is the observable signature of balancing:

$$\text{local baryonic curvature} \quad \leftrightarrow \quad \text{cosmological curvature floor} \tag{193}$$

through a shell-occupancy channel whose propagation is intrinsically two-dimensional.

In this sense, the inverse-square and MOND-like square-root regimes are not competing laws, but two different projections of the same curvature dynamics. The r^{-2} behavior belongs to direct local flux spreading in three dimensions, while the $\sqrt{a_b a_*}$ behavior appears when the enclosed source is first accumulated on the shell and then redistributed along a lower-dimensional transport manifold.

D Appendix: Units, Conventions, and the WT–SI Translation

D.1 Purpose

This paper draws on two complementary frameworks: standard General Relativity expressed in SI units, and Wave Theory (WT), which uses a purely geometric unit system built from the two fundamental dimensions of length $[L]$ and time $[T]$. Because equations from both systems appear side by side, this appendix (i) fixes all sign and naming conventions globally, (ii) defines the WT unit system and its relationship to SI, and (iii) provides a translation dictionary for every quantity that appears in the main text. No equation in the body of the paper requires re-reading after consulting this appendix; the conventions stated here are the unique, global definitions used throughout.

D.2 Sign and Naming Conventions (Global)

The standard weak-field sign convention is adopted throughout:

$$\nabla^2\Phi = 4\pi G\rho_b, \qquad \mathbf{F}_b \equiv -\nabla\Phi, \qquad \nabla\cdot\mathbf{F}_b = -4\pi G\rho_b. \tag{194}$$

The signed radial flux of $\mathbf{F}_b$ through a shell S_r is therefore

$$Q_b(r) \equiv \int_{S_r} \mathbf{F}_b\cdot\mathbf{n}_r\, dA = -4\pi G M_b(r). \tag{195}$$

Wherever the *magnitude* of the enclosed curvature source is needed—in particular in the shell transport term and in BTFR derivations—the positive source magnitude is used

$$\boxed{Q_b^{\rm src}(r) \equiv 4\pi G M_b(r) = -Q_b(r).} \tag{196}$$

Every formula for $a_{\rm shell}$ and $a_{\rm eff}$ uses $Q_b^{\rm src}$, so all accelerations are positive (attractive). The signed Q_b appears only in the divergence-theorem step that connects the field equation to the enclosed mass.

The shell Poisson equation is written with the convention $\Delta_{S_r}\Phi_r = -\sigma_r$, where $\sigma_r = \mathbf{F}_b\cdot\mathbf{n}_r < 0$ for inward-pointing fields. The resulting shell acceleration $a_{\rm shell} = -\nabla_{S_r}\Phi_r\cdot(-\hat{r})$ is positive, consistent with gravitational attraction.

D.3 The Wave Theory Geometric Unit System

Wave Theory derives all physical quantities from two primitive dimensions,

$$[L] = \text{length (metres)}, \qquad [T] = \text{time (seconds)}, \tag{197}$$

with mass, charge, and all other quantities emergent as geometric composites. The fundamental anchors are the Planck length L_p, Planck time T_p, and the present cosmic curvature radius R:

$$c = \frac{L_p}{T_p} \qquad [\,L\,T^{-1}\,], \tag{198}$$

$$G = \frac{c^2}{\pi R} = \frac{L_p^2}{\pi R\,T_p^2} \qquad [\,L\,T^{-2}\,], \tag{199}$$

$$H = \frac{\pi c}{R} = \frac{\pi L_p}{R\,T_p} \qquad [\,T^{-1}\,], \tag{200}$$

$$\hbar = \pi\,\frac{L_p^3 R}{T_p} \qquad [\,L^4\,T^{-1}\,]. \tag{201}$$

The key point is that in WT the gravitational constant G carries dimensions $[L\,T^{-2}]$ (acceleration), *not* $[L^3\,M^{-1}\,T^{-2}]$ as in SI, because mass itself is a derived geometric quantity with $[M]_{\mathrm{WT}} = [L^2]$. Numerically, with $R = 4.286332662 \times 10^{26}\,\mathrm{m}$,

$$G = \frac{c^2}{\pi R} \approx 6.6743 \times 10^{-11}\ \mathrm{m\,s^{-2}} \quad \longleftrightarrow \quad 6.6743 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}\ \text{(SI)}, \tag{202}$$

where the numerical value is identical; only the dimensional interpretation differs. The translation rule is

$$G_{\mathrm{SI}}\ [\mathrm{m^3\,kg^{-1}\,s^{-2}}] \ = \ G_{\mathrm{WT}}\ [\mathrm{m\,s^{-2}}] \ \times \ \underbrace{\frac{1\,\mathrm{kg}}{[L^2]_{\mathrm{WT}}}}_{\text{mass conversion}} \tag{203}$$

with the mass conversion factor fixed by the WT relation M [kg] $\leftrightarrow A_b\ [L^2] = \pi L_p^2 R/r_{\mathrm{eff}}$.

D.4 Translation Dictionary

Table 4 collects every quantity used in this paper, giving its SI form, its WT geometric form, the WT dimensional class, and any conversion remarks.

Table 4: Quantity-by-quantity translation between SI and WT geometric units. All numerical values are identical; only dimensional bookkeeping differs.

Quantity	**SI expression**	**WT expression**	**WT dim.**	**Remarks**
Speed of light	c $[\mathrm{m\,s^{-1}}]$	$c = L_p/T_p$	L/T	Identical; sets causal cone.
Gravitational constant	G $[\mathrm{m^3\,kg^{-1}\,s^{-2}}]$	$G = c^2/(\pi R)$ $[\mathrm{m\,s^{-2}}]$	L/T^2	WT: G is an *acceleration*, not a 3-form coupling. Same numerics.
Cosmic curvature scale	$a_* = G$ $[\mathrm{m\,s^{-2}}]$	$a_* = c^2/(\pi R)$	L/T^2	The baseline curvature acceleration of S^3.
Hubble rate	$H_0 \approx 2.2 \times 10^{-18}\ \mathrm{s^{-1}}$	$H = \pi c/R$	$1/T$	WT gives H purely from R.
Baryonic mass	M_b [kg]	$A_b = \pi L_p^2 R/r_{\text{eff}}$ $[L^2]$	L^2	Mass is a geometric area in WT. Numerically $G_{\text{SI}} M_b = a_* A_b$.
Local baryonic acceleration	$a_b = GM_b/r^2$	$a_b = a_* A_b/r^2$	L/T^2	Identical numerically; WT identifies $G \leftrightarrow a_*$, $M_b \leftrightarrow A_b$.
Curvature flux (signed)	$Q_b = -4\pi G M_b$ $[\mathrm{m^3\,s^{-2}}]$	$Q_b = -4\pi a_* A_b$	L^3/T^2	Negative by convention (195); used only in divergence-theorem steps.
Curvature source (positive)	$Q_b^{\text{src}} = 4\pi G M_b$	$Q_b^{\text{src}} = 4\pi a_* A_b$	L^3/T^2	Used in all shell-transport and acceleration formulae.
Shell potential	$\Phi_r \sim Q_b^{\text{src}} \log r$ $[\mathrm{m^2\,s^{-2}}]$	Same	L^2/T^2	Logarithmic Green function of Δ_{S_r}.

Continued. . .

Quantity	SI expression	WT expression	WT dim.	Remarks
Shell acceleration	$a_{\rm shell} = \alpha_b(\chi)\, Q_b^{\rm src}/r$ $[\mathrm{m\,s^{-2}}]$	Same form	L/T^2	$\alpha_b = \alpha_0 L(\chi)$ has units $[L^{-1}]$; α_0 is dimensionless. Intermediate result. The master formula uses $a_{\rm shell} = \alpha_0\sqrt{a_b a_*}$ as the outer-regime target.
Geometric coupling	$L(\chi) = A_\partial/V_{\rm cap}$ $[\mathrm{m^{-1}}]$	Same	$1/L$	Carries the $[\mathrm{m^{-1}}]$ needed to make $a_{\rm shell}$ an acceleration. Appears in the bulk-to-shell coupling derivation. Does not appear explicitly in the master formula; its role is absorbed into the crossfade structure.
Transition radius	$r_* = \sqrt{GM_b/a_*} = \sqrt{M_b/a_* \cdot G}$	$r_* = \sqrt{A_b}$	L	WT: r_* is the geometric mean radius of the baryonic area A_b. SI: same numerics via $G = a_*$.
Mixing function	$\sigma(r) = [1 + a_b/a_*]^{-1}$	Same	dimensionless	Crossfade weight; $\sigma \to 0$ (Newtonian), $\sigma \to 1$ (shell).

D.5 The Core Identification $G = a_*$

The central WT relation used in this paper is

$$G \;=\; a_* \;=\; \frac{c^2}{\pi R}. \tag{204}$$

In SI this appears as a numerical coincidence between two differently-dimensioned constants. In WT it is a definition: G *is* the background curvature acceleration of S^3, and it has dimensions $[L\,T^{-2}]$. The apparent dimensional mismatch with the SI form $[L^3\,M^{-1}\,T^{-2}]$ is resolved entirely by the geometric reinterpretation of mass: M_b [kg] $\to A_b$ $[L^2]$, so that

$$G_{\mathrm{SI}}\ M_b\ [\mathrm{m\,s^{-2}}]\ =\ a_*\ A_b\ [\mathrm{m\,s^{-2}}]. \tag{205}$$

All GR-normalized formulae in this paper (those containing GM_b) are therefore numerically identical to their WT counterparts (those containing a_*A_b).

The master formula (68) is dimensionally consistent in both unit systems: a_b, a_*, and $\sqrt{a_b\,a_*}$ all carry $[\mathrm{m\,s^{-2}}]$, the crossfade weight $\sigma(r)$ and the projection factor α_0 are both dimensionless, and the WT compactness form $a_*\cdot\xi\cdot f(\xi,\alpha_0)$ carries dimensions through a_* alone, since $\xi=\sqrt{M_b}/r$ is dimensionless in WT units ($[M_b]=L^2$). The single dimensionful scale is $a_*=c^2/(\pi R)$, fixed by the Hubble radius independently of any galactic observation.

D.6 WT Unit Groups: Summary

For reference, Table 5 reproduces the canonical WT dimensional assignments for all standard physical quantities that appear, directly or indirectly, in this paper.

Table 5: Canonical WT unit groups (from the WT Unit System reference). Only entries relevant to this paper are listed.

WT units $[L^a/T^b]$	**SI equiv.**	**Physical quantity**	**WT interpretation**
L/T^2	$\mathrm{m\,s^{-2}}$	Acceleration, G, a_*, $H\cdot c$	Surface-level energy gradient; gravitational acceleration
L^2	kg	Mass M_b, WT area A_b	Curvature compression on S^3
L^3/T^2	$\mathrm{m^3\,s^{-2}}$	GM_b, curvature flux Q_b^{src}	Volumetric curvature budget
L^2/T^2	$\mathrm{m^2\,s^{-2}}$	Gravitational potential Φ, v^2	Specific energy (per unit area)

Continued. . .

WT units	**SI equiv.**	**Physical quantity**	**WT interpretation**
$1/T$	s^{-1}	H, angular frequency	Temporal unfolding of curvature
L/T	$\mathrm{m\,s}^{-1}$	Velocity v, c	Curvature wave propagation rate
$1/L$	m^{-1}	$L(\chi) = A_\partial/V_{\mathrm{cap}}$, curvature scale	Boundary access per enclosed volume
1	dimensionless	α_0, σ, ξ, χ	Pure geometric or efficiency ratios

Note on numerical equality. The WT framework assigns G the dimensions of an acceleration, but its numerical value $G \approx 6.6743 \times 10^{-11}\,\mathrm{m\,s}^{-2}$ is identical to the SI value of G expressed in the same units. All physical predictions made in this paper are therefore numerically independent of which unit convention is adopted; the two systems are in strict one-to-one correspondence through the mass substitution $M_b \to A_b$ with the conversion $G_{\mathrm{SI}} M_b = a_* A_b$.

E Appendix: The Model in GR Language

This appendix restates the two-channel curvature framework in standard General Relativity language. It is self-contained: a reader familiar with GR but not with Wave Theory can evaluate the physical content of the model entirely from what follows, without consulting the main text for formalism.

The central point is that every equation in the paper has an exact GR counterpart derivable from standard results. Wave Theory provides a geometric language that makes the dimensional structure of the transport process transparent, but it is not a prerequisite for the physics. The three elements that are genuinely new relative to textbook GR are identified explicitly in Section E.4 below; everything else follows from the weak-field limit of the Einstein equations, the geometry of S^3, and the spectral structure of its Laplacian.

E.1 Standard GR foundation

In the weak-field limit, the spacetime metric takes the form

$$g_{00} \simeq -(1 + 2\Phi/c^2), \qquad g_{ij} \simeq (1 - 2\Phi/c^2)\delta_{ij}, \tag{206}$$

and the Einstein field equations reduce to the Newtonian Poisson equation

$$\nabla^2\Phi = 4\pi G\rho_b. \tag{207}$$

The gravitational field is $\mathbf{F}_b = -\nabla\Phi$, and Gauss's theorem gives the enclosed source flux

$$\oint_{S_r} \mathbf{F}_b \cdot \mathbf{n}_r \, dA = -4\pi G M_b(r). \tag{208}$$

This is standard GR in the weak-field limit. No modification has been introduced.

E.2 The shell source as a GR boundary quantity

In standard GR, the object $\sigma_r \equiv \mathbf{F}_b \cdot \mathbf{n}_r = -\partial_r\Phi$ is the normal component of the gravitational field at radius r. It is the shell source introduced in Section 2 — not a new object, but the standard boundary trace of the Newtonian potential.

The identification

$$Q_b^{\mathrm{src}}(r) = \oint_{S_r} \sigma_r \, dA = 4\pi G M_b(r) \tag{209}$$

is exactly Gauss's law. The shell source σ_r is therefore fully determined by standard GR and carries no additional freedom.

Gauss–Codazzi perspective. More covariantly, σ_r is the normal-normal component of the extrinsic curvature of the hypersurface S_r embedded in the spatial slice. The Gauss–Codazzi equations relate this to the intrinsic geometry of S_r and the ambient Ricci curvature. In the weak-field limit, this reduces to (209). The shell source is thus a standard object in the geometric theory of hypersurface embeddings — it requires no new postulate.

E.3 The shell transport equation in GR

Once σ_r is established as a standard GR boundary quantity, the shell transport equation

$$\Delta_{S_r} \Phi_r = -\sigma_r \tag{210}$$

is a two-dimensional Poisson equation on the surface S_r, sourced by the normal component of the gravitational field. This equation is derivable from the three-dimensional Poisson equation (207) by the projection procedure of Section 4.5: in the quasi-static, large-r regime, the Laplace–Beltrami operator on S^3 reduces to its angular component, and the bulk equation induces (210) as a boundary condition on S_r.

The key input is therefore not a new field equation but a regime condition: at large radii, where radial gradients become subdominant relative to angular variations, the three-dimensional Poisson equation (207) induces an intrinsic two-dimensional Poisson equation on the shell. This is a consequence of the geometry of S^3 and the spectral structure of its Laplacian, established in Appendix A.

E.4 What is new relative to standard GR

Standard GR in the weak-field limit gives only the three-dimensional channel: $a_b = GM_b/r^2$. The present model supplements this with a second channel arising from the redistribution of the boundary flux σ_r along the two-dimensional manifold S_r. Expressed entirely in GR language, the new ingredient is:

> *At large radii, the normal component of the gravitational field at S_r drives a two-dimensional transport process on S_r itself. This process is governed by* (210) *and produces an additional acceleration contribution scaling as* $1/r$.

This is not a modification of the Einstein field equations. It is a statement about the dynamics of the boundary data induced by those equations at large scales. The model may therefore be understood as an extension of GR in which curvature transport on embedded two-dimensional manifolds becomes dynamically relevant, not an alternative to it.

The three elements that are genuinely new — absent from textbook GR — are:

1. the assumption that curvature transport on S_r is dynamically active rather than a passive boundary condition (postulated),
2. the spatial geometry modeled as S^3 of radius R rather than flat space (asserted, with R identified with the Hubble radius),
3. the morphology factor α_0 encoding the efficiency of shell-mode excitation (empirical).

Everything else follows from standard GR, the geometry of S^3, and the spectral analysis of its Laplacian.

E.5 Correspondence table

Table 6 gives a side-by-side translation of every principal quantity between the GR formulation and the Wave Theory formulation. The two systems are in strict numerical correspondence through the single substitution $GM_b \leftrightarrow a_* A_b$, where $a_* = c^2/(\pi R) = G$. For the full unit translation dictionary and dimensional verification, see Appendix D.

E.6 Relation to MOND and modified gravity

The master acceleration formula (68) produces MOND-like behavior in the outer regime without modifying the Einstein field equations. In the limit $a_b \ll a_*$, the shell term dominates and

$$a_{\text{eff}} \approx \sqrt{a_b \cdot a_*}, \tag{211}$$

which is the deep-MOND interpolation formula with $a_0 \equiv a_* = c^2/(\pi R)$.

The standard MOND acceleration scale is $a_0 \approx 1.2\times10^{-10}\,\mathrm{m\,s^{-2}}$. The present model identifies this with the cosmological curvature acceleration $c^2/(\pi R)$, which evaluates numerically to $c^2/(\pi R) \approx 6.67 \times 10^{-11}\,\mathrm{m\,s^{-2}}$ using $R = 4.286 \times 10^{26}\,\mathrm{m}$. The agreement is within a factor of two — consistent given that α_0 and m are not yet observationally calibrated. Crucially, a_0 is not a free parameter here: it is set by the Hubble radius.

The MOND-like scaling therefore emerges not from a modification of the gravitational law but from three standard ingredients acting together: the enclosed Gauss-law source $Q_b^{\text{src}} = 4\pi GM_b$, the two-dimensional Green function on S_r, and the transition scale $r_* \sim \sqrt{GM_b/a_*}$

set by the equality of local and cosmological curvature. The derivation is given in Appendix C.1.

Distinction from covariant MOND theories. Relativistic MOND formulations (RMOND, TeVeS, AQUAL) introduce new fields or Lagrangian terms that modify the action. The present model introduces neither. The second acceleration channel arises from the dynamics of boundary data of the standard gravitational field — a projection effect, not a new degree of freedom. Whether this can be embedded in a covariant action principle is an open question, but it is not required for the weak-field predictions derived here.

Concept	**GR formulation**	**Wave Theory formulation**
Background geometry	Closed FLRW spatial slice, curvature radius R	S^3 of radius R
Fundamental acceleration scale	$a_* = c^2/(\pi R) \approx G$	Same; $G \equiv a_*$ by definition
Baryonic source	Mass M_b from $T_{\mu\nu}$	Geometric area A_b; $GM_b = a_* A_b$
Local acceleration	$a_b = GM_b(r)/r^2$	$a_b = a_* A_b(r)/r^2$
Shell source	$\sigma_r = -\partial_r \Phi\big\rvert_{S_r}$ (normal flux)	Same
Enclosed source flux	$Q_b^{\mathrm{src}} = 4\pi GM_b$ (Gauss law)	$Q_b^{\mathrm{src}} = 4\pi a_* A_b$
Bulk-to-shell projection	Gauss–Codazzi boundary data of $G_{\mu\nu}$ on S_r	Curvature flux through $\partial V_{\mathrm{cap}}$
Shell transport equation	$\Delta_{S_r}\Phi_r = -\sigma_r$ (2D Poisson on S_r)	Same
Shell Green function	$\Phi_r \sim Q_b^{\mathrm{src}} \log r$	Same
Shell acceleration	$a_{\mathrm{shell}} = \alpha_b(\chi)\, Q_b^{\mathrm{src}}/r$	Same
Geometric coupling	$L(\chi) = A_\partial(\chi)/V_{\mathrm{cap}}(\chi)$	Same
Transition condition	$a_b(r_*) \sim a_*$; $r_* \sim \sqrt{GM_b/a_*}$	$r_* \sim \sqrt{A_b}$
Master acceleration	$a_{\mathrm{eff}} = (1-\sigma)\dfrac{GM_b}{r^2} + \sigma\,\alpha_0\sqrt{\dfrac{GM_b}{r^2}\cdot a_*}$	$a_{\mathrm{eff}} = a_*\xi\,\dfrac{\xi^3+\alpha_0}{1+\xi^2}$, $\xi = \sqrt{A_b}/r$
Asymptotic BTFR	$v_\infty^4 \propto GM_b \cdot a_*$	$v_\infty^4 \propto a_*^2 A_b$

Table 6: Side-by-side correspondence between the GR and Wave Theory formulations. All equations above the double rule are standard GR results. The shell transport equation and everything below it constitute the new dynamical content. The two formulations are numerically identical through $GM_b = a_* A_b$; only the dimensional interpretation of G differs. For the complete unit translation see Appendix D.

F Appendix: Retarded-Source Derivation of Shell Confinement in Linearized General Relativity

This appendix provides a derivation of the spiral winding and shell confinement mechanism using only standard weak-field General Relativity. The purpose is to show that the dynamical transition from volumetric to shell-supported curvature transport follows directly from causal propagation, long-time orbital averaging, and differential rotation of the source, without invoking the WTMedium or a co-moving transport assumption.

F.1 Retarded gravitational potential of a time-dependent source

In the linearized weak-field limit, the trace-reversed metric perturbation $\bar{h}_{\mu\nu}$ satisfies

$$\Box \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu}. \tag{212}$$

The linearized weak-field formulation follows standard post-Newtonian theory [7].

In the slow-motion regime, the dominant component reduces to a scalar potential Φ satisfying

$$\frac{1}{c^2}\partial_t^2 \Phi - \nabla^2 \Phi = 4\pi G \rho_b. \tag{213}$$

The solution is given by the retarded integral

$$\Phi(\mathbf{x}, t) = -G \int \frac{\rho_b\left(\mathbf{x}',\, t - \frac{|\mathbf{x}-\mathbf{x}'|}{c}\right)}{|\mathbf{x}-\mathbf{x}'|}\, d^3x'. \tag{214}$$

This expression encodes two essential features:

- Finite propagation speed c,
- Dependence on the full causal history of the source.

F.2 Differentially rotating disk as a time-dependent source

The baryonic source is modelled as a differentially rotating disk with angular velocity $\Omega(r)$ and surface density $\Sigma(r')$. The three-dimensional mass density is

$$\rho_b(\mathbf{x}', t) = \Sigma(r')\, \delta(z')\, \delta(\varphi' - \Omega(r')\, t)\,, \tag{215}$$

where $\delta(z')$ confines the source to the disk midplane and the azimuthal delta function tracks each ring at its instantaneous orbital angle. The dimensions are consistent: $[\Sigma\,\delta(z')\,\delta(\varphi')] = \mathrm{kg\,m^{-3}}$. The Fourier coefficients are $\rho_m(r') = \Sigma(r')\delta(z')/(2\pi)$.

Expanding the retarded potential in angular modes gives

$$\Phi(\mathbf{x}, t) = \sum_{m,\ell} \Phi_{m\ell}(r, t)\, Y_\ell^m(\theta, \varphi). \tag{216}$$

The coefficients $\Phi_{m\ell}(r, t)$ inherit the time dependence of the rotating source through the retarded argument.

F.3 Stationary-phase analysis of the retarded integral

The eikonal limit of the retarded integral is controlled by the stationary-phase conditions on the source phase

$$S_m(\mathbf{x}, \mathbf{x}', t) = m\varphi' - m\Omega(r')\left(t - \frac{|\mathbf{x} - \mathbf{x}'|}{c}\right). \tag{217}$$

Stationary-phase condition on r'. Setting $\partial S_m/\partial r' = 0$ gives

$$\frac{m}{c}\frac{\partial\Omega}{\partial r'}|\mathbf{x} - \mathbf{x}'| + \frac{m\Omega(r')}{c}\frac{\partial|\mathbf{x} - \mathbf{x}'|}{\partial r'} = 0. \tag{218}$$

In the near-zone regime $|\mathbf{x} - \mathbf{x}'| \ll c/\Omega(r')$, verified numerically in Section F.3 below, the first term is negligible and the condition reduces to $\partial|\mathbf{x} - \mathbf{x}'|/\partial r' = 0$, which for nearly co-rotating source and field points gives

$$r' \approx r. \tag{219}$$

The dominant contribution to the eikonal field at radius r therefore comes from source material at the same radius: the *same-shell condition*.

Local wavevector at the stationary point. At $r' \approx r$, the phase of the field is $\theta \approx m\varphi' - m\Omega(r)t + (m\Omega(r)/c)|\mathbf{x} - \mathbf{x}'|_{\text{stat}}$. The local frequency and wavevector are

$$\omega = -\partial_t\theta = m\Omega(r), \qquad |\mathbf{k}| = \frac{m\Omega(r)}{c} = \frac{\omega}{c}, \tag{220}$$

establishing the free-wave dispersion relation $\omega = c|\mathbf{k}|$ from the source geometry alone.

Doppler-shifted dispersion relation. Since the stationary source material rotates at $\mathbf{v}(\mathbf{x}') \approx \mathbf{v}(\mathbf{x}) = r\Omega(r)\hat{\boldsymbol{\theta}}$, and the azimuthal component of $\mathbf{k}$ is $k_\theta = m/r$, we have $\mathbf{v} \cdot \mathbf{k} = m\Omega(r) = \omega$. The intrinsic frequency in the co-rotating frame is therefore

$$\omega_{\text{eff}} = \omega - \mathbf{v} \cdot \mathbf{k}, \tag{221}$$

and the dispersion relation (220) becomes

$$\omega_{\text{eff}}^2 = c^2|\mathbf{k}|^2. \tag{222}$$

This result is *derived* from the retarded source phase: the $\mathbf{v} \cdot \mathbf{k}$ term is the kinematic consequence of the same-shell condition (219), not an assumption about a co-moving medium.

Near-zone validity. The approximation $|\mathbf{x} - \mathbf{x}'| \ll c/\Omega$ requires $R_{\text{gal}} \ll c/\Omega$. For $\Omega \sim 2 \times 10^{-16}\,\text{rad}\,\text{s}^{-1}$ at $r \sim 10\,\text{kpc}$:

$$\frac{c}{\Omega} \sim \frac{3 \times 10^8}{2 \times 10^{-16}} \sim 1.5 \times 10^{24}\,\text{m} \sim 50\,\text{Mpc}, \tag{223}$$

which exceeds $R_{\text{gal}} \sim 30\,\text{kpc}$ by five orders of magnitude. The near-zone approximation is therefore valid throughout the disk.

F.4 Orbital averaging of the retarded field

Galaxies evolve over timescales much longer than orbital periods:

$$\frac{T_{\text{gal}}}{T_{\text{orb}}} \gg 1. \tag{224}$$

The long-time averaged field is therefore defined as

$$\langle \Phi_{m\ell}(r,t) \rangle = \frac{1}{T_{\text{gal}}} \int_0^{T_{\text{gal}}} \Phi_{m\ell}(r,t)\, dt. \tag{225}$$

Orbital averaging of retarded gravitational fields is a standard technique in post-Newtonian theory [19, 20].

F.4.1 Suppression of non-axisymmetric modes

For $m \neq 0$, the retarded integral introduces oscillatory factors of the form

$$e^{im\Omega(r')\left(t-\frac{|\mathbf{x}-\mathbf{x}'|}{c}\right)}. \tag{226}$$

Applying stationary-phase or Riemann–Lebesgue arguments yields

$$\langle \Phi_{m\neq 0} \rangle \sim \frac{1}{m\Omega(r')T_{\rm gal}}\, \Phi^{(0)}_{m\ell} \;\longrightarrow\; 0. \tag{227}$$

Thus only the axisymmetric component survives:

$$\Phi(r,\theta) \approx \Phi_{m=0}(r,\theta). \tag{228}$$

F.4.2 Radial coherence suppression

The suppression of non-axisymmetric modes establishes that the long-time averaged field is axisymmetric, $\Phi \approx \Phi(r,\theta)$. The relevant question for dimensional reduction is not azimuthal vs. radial structure (the axisymmetric field has no azimuthal gradient by definition), but rather the radial coherence length of the averaged field.

Contributions to the retarded integral at field point (r,θ,φ) from source points at radius r' carry retarded delays $\Delta t(r') = |r - r'|/c$ relative to on-shell contributions. After $N_{\rm orb} = T_{\rm gal}/T_{\rm orb}$ elapsed orbital periods, the accumulated phase difference between contributions from r' and $r' + \Delta r$ is

$$\Delta\phi_{\rm ret} = m\Omega(r')\,\frac{\Delta r}{c}. \tag{229}$$

Contributions with $|\Delta\phi_{\rm ret}| \gtrsim 1$ interfere destructively under the time average. Setting $|\Delta\phi_{\rm ret}| = 1$ defines the radial coherence length

$$\ell_{\rm coh} \sim \frac{c}{m\Omega(r)} = \frac{c\,T_{\rm orb}}{2\pi m}. \tag{230}$$

After $N_{\rm orb}$ orbits, the effective coherence length per orbit is further suppressed:

$$\ell^{\rm eff}_{\rm coh} \sim \frac{c\,T_{\rm orb}}{2\pi\,T_{\rm gal}}\,r = \frac{T_{\rm orb}}{T_{\rm gal}}\,\frac{r}{2\pi}. \tag{231}$$

Since $T_{\rm orb}/T_{\rm gal} \sim 10^{-2}$, coherent radial structure in the time-averaged field survives only on scales $\Delta r \ll r$. On galactic scales, the averaged field has no coherent radial variation: it is effectively constant in r at fixed θ. Angular (shell) structure at fixed r, by contrast, is organised on galactic scales and is not suppressed.

Quantitative check. For $\Omega \sim 2 \times 10^{-16}\,\mathrm{rad\,s^{-1}}$ ($r \sim 10\,\mathrm{kpc}$), $T_{\mathrm{orb}} \sim 300\,\mathrm{Myr}$, $T_{\mathrm{gal}} \sim 10\,\mathrm{Gyr}$:

$$\ell_{\mathrm{coh}}^{\mathrm{eff}} \sim \frac{300\,\mathrm{Myr}}{10\,\mathrm{Gyr}} \times \frac{10\,\mathrm{kpc}}{2\pi} \sim 0.05\,\mathrm{kpc}. \tag{232}$$

Coherent radial structure is therefore confined to scales $\lesssim 50\,\mathrm{pc}$, two orders of magnitude below the galactic radius. The radial dimension contributes no coherent large-scale structure to the averaged field; only the angular (shell) degrees of freedom remain organised at galactic scales.

Relation to the eikonal argument. The radial coherence suppression established here applies to the time-averaged background field. The eikonal winding argument of Section F.5 applies to individual propagating perturbations and reaches the same conclusion — $k_\theta \gg k_r$ after $t \gg t_{\mathrm{wind}}$ — through a complementary mechanism. Both arguments are grounded in the same-shell condition $r' \approx r$ (219): the averaged field is organised on shells because radial contributions dephase, and individual perturbations are confined to shells because differential rotation winds their wavevectors azimuthally. The shell S_r is the structure selected by both.

F.5 Eikonal limit of the linearized field equations

To analyze the propagation of perturbations, a WKB ansatz is applied

$$\Phi(\mathbf{x}, t) = A(\mathbf{x}, t)\, e^{i\theta(\mathbf{x},t)/\epsilon}, \quad \epsilon \ll 1. \tag{233}$$

The eikonal (WKB) approximation used here follows standard methods; see [12] for a comprehensive treatment.

Defining

$$\mathbf{k} = \nabla\theta, \qquad \omega = -\partial_t\theta, \tag{234}$$

we obtain a local dispersion relation.

Validity of the WKB approximation. The eikonal ansatz (233) requires the wavelength $\lambda = 2\pi/|\mathbf{k}|$ to be short compared to the scale over which the amplitude $A(\mathbf{x}, t)$ and the background velocity $\mathbf{v}(\mathbf{x})$ vary:

$$|\mathbf{k}| \gg |\nabla \log A|, \qquad |\mathbf{k}| \gg |\nabla \log \Omega| \sim \frac{1}{r}. \tag{235}$$

From the same-shell result (220), $|\mathbf{k}| = m\Omega/c$. The second condition then requires

$$\frac{m\Omega}{c} \gg \frac{1}{r} \quad \Longleftrightarrow \quad m \gg \frac{c}{r\Omega} = \frac{c}{v_{\rm rot}}, \tag{236}$$

where $v_{\rm rot} = r\Omega$ is the circular velocity. For typical disk galaxies $v_{\rm rot} \sim 200\,{\rm km\,s^{-1}}$, giving $c/v_{\rm rot} \sim 1500$. The WKB approximation therefore applies only for $m \gg 1500$, i.e. for very high azimuthal modes.

For the low-m modes that dominate the physical signal, the eikonal approximation is not uniformly valid in the strict short-wavelength sense. The ray-equation result for the wavevector shearing $dk_\theta/dt = k_r A$ should therefore be understood as valid in an asymptotic or order-of-magnitude sense: it correctly captures the qualitative evolution of the dominant wavevector direction under differential rotation, but a fully rigorous treatment would replace the WKB dispersion relation with the full retarded Green function analysis of Section F.3. The winding timescale $t_{\rm wind} \sim T_{\rm orb}/\pi$ and the conclusion $k_\theta \gg k_r$ are robust to this qualification, as they follow from the geometry of differential rotation rather than from the WKB amplitude equation.

F.5.1 Doppler-shifted frequency

The stationary-phase analysis of Section F.3 establishes that the eikonal field at radius r is dominated by source material at $r' \approx r$ (the same-shell condition, equation (219)), rotating at $\mathbf{v}(\mathbf{x}') \approx \mathbf{v}(\mathbf{x}) = r\Omega(r)\hat{\boldsymbol{\theta}}$. The local frequency and wavevector inherited from the retarded source phase are given by equation (220):

$$\omega = m\Omega(r), \qquad |\mathbf{k}| = \frac{\omega}{c}. \tag{237}$$

Since the azimuthal wavevector component is $k_\theta = m/r$, we have $\mathbf{v}\cdot\mathbf{k} = r\Omega(r)\cdot(m/r) = m\Omega(r) = \omega$, and the intrinsic frequency in the frame co-rotating with the source is

$$\omega_{\rm eff} = \omega - \mathbf{v}\cdot\mathbf{k}. \tag{238}$$

Substituting $\omega = c|\mathbf{k}|$ gives the dispersion relation

$$\omega_{\rm eff}^2 = c^2|\mathbf{k}|^2. \tag{239}$$

The $\mathbf{v}\cdot\mathbf{k}$ term is not an assumption about the wave operator or a co-moving medium: it is the kinematic consequence of the same-shell condition derived in Section F.3.

F.5.2 Ray equations

The Hamiltonian

$$H(\mathbf{x}, \mathbf{k}) = \mathbf{v} \cdot \mathbf{k} + c|\mathbf{k}| \tag{240}$$

yields

$$\frac{d\mathbf{x}}{dt} = \mathbf{v}(\mathbf{x}) + c\,\hat{\mathbf{k}}, \tag{241}$$

$$\frac{d\mathbf{k}}{dt} = -(\mathbf{k} \cdot \nabla)\mathbf{v}(\mathbf{x}). \tag{242}$$

This equation governs the evolution of perturbations and is derived directly from the linearized Einstein system.

F.6 Spiral winding under differential rotation

For the velocity field $\mathbf{v} = r\Omega(r)\hat{\theta}$, the ray equation (242) describes wavevector evolution under differential shear. In the local shearing-sheet approximation (standard in galactic dynamics [14]), the linearised wavevector evolution for an initially radial perturbation $\mathbf{k}(0) = k_0\hat{r}$ is

$$k_r(t) \approx k_0, \qquad k_\theta(t) \approx 2A_{\text{Oort}}\, k_0\, t, \tag{243}$$

where $A_{\text{Oort}} = -\frac{1}{2} r\, d\Omega/dr > 0$ is the Oort shear constant. The pitch angle satisfies

$$\tan\psi(t) = \frac{k_r}{k_\theta} \approx \frac{1}{2A_{\text{Oort}}\, t} \;\longrightarrow\; 0, \tag{244}$$

and the radial group velocity

$$\left(\frac{dx}{dt}\right)_r = c\,\frac{k_r}{|\mathbf{k}|} \sim \frac{c}{2A_{\text{Oort}}\, t} \;\longrightarrow\; 0. \tag{245}$$

At late times ($2A_{\text{Oort}}\, t \gg 1$) the ratio k_θ/k_r continues to grow, so the asymptotic confinement $k_\theta \gg k_r$ is robust beyond the linearised regime.

F.6.1 Winding timescale

The pitch angle reaches $\psi \sim 1$ on the timescale

$$t_{\text{wind}} \sim \frac{1}{2A_{\text{Oort}}} \sim \frac{T_{\text{orb}}}{\pi}, \tag{246}$$

where the last step uses $A_{\rm Oort} \sim \Omega/2$ for a broadly declining rotation profile and $T_{\rm orb} = 2\pi/\Omega$. Since $t_{\rm wind} \ll T_{\rm gal}$, spiral winding is dynamically inevitable.

F.7 Emergence of shell-supported transport

Both arguments are grounded in the same-shell condition (219). Orbital averaging suppresses non-axisymmetric modes on timescale $T_{\rm gal}$, leaving a field organised on constant-r surfaces. Eikonal winding shears radial wavevectors into azimuthal ones on timescale $t_{\rm wind} \sim T_{\rm orb}/\pi \ll T_{\rm gal}$, suppressing radial group velocity within the same surface. The two results are complementary views of the same dynamics — one for the time-averaged background, one for individual perturbations — both valid because $r' \approx r$.

Together they give $k_\theta \gg k_r$ for $t \gg t_{\rm wind}$, and suppressed radial coherence in the averaged field. Coherent curvature transport is therefore confined to constant-radius shells S_r: not by assumption, but as the dynamical attractor of causal propagation, orbital averaging, and differential rotation in standard linearized General Relativity.

F.8 Connection to dimensional reduction

The winding result $k_\theta \gg k_r$ is a statement about the physical wavevector components at radius r. To connect this to the spectral condition on S^3, the correspondence is identified between physical wavenumbers and the quantum numbers (ℓ, n) of the S^3 harmonics.

On the S^3 of curvature radius R, the harmonics $Y^{(n)}_{\ell m}(\chi, \theta, \varphi)$ have angular wavenumber ℓ on the constant-χ shells and radial wavenumber n in the χ direction. In the physical galactic regime $r \ll R$ (so $\chi = r/R \ll 1$), the correspondence between physical wavevector components and spectral quantum numbers is

$$k_\theta \sim \frac{\ell}{r}, \qquad k_r \sim \frac{n}{R}. \tag{247}$$

The condition $k_\theta \gg k_r$ then gives

$$\frac{\ell}{r} \gg \frac{n}{R} \quad \Longrightarrow \quad \ell \gg n\,\frac{r}{R}. \tag{248}$$

Since $r/R \ll 1$, this is satisfied for all $\ell \gtrsim 1$ at galactic radii, confirming that the winding regime is compatible with the angular-dominance condition. More precisely, the spectral dominance condition of Appendix C requires

$$\frac{\ell(\ell+1)}{\sin^2\chi} \gg n(n+2), \tag{249}$$

which in the small-χ limit ($\sin\chi \approx \chi = r/R$) reduces to

$$\ell(\ell+1) \gg n(n+2)\,\frac{r^2}{R^2}, \tag{250}$$

consistent with (248) for $\ell \sim k_\theta r$ and $n \sim k_r R$. The Laplace–Beltrami operator on S^3 therefore reduces to its angular component,

$$\Delta_{S^3}\Phi \approx \frac{\ell(\ell+1)}{\sin^2\chi}\,\Phi, \tag{251}$$

and the dimensional reduction to a two-dimensional Poisson equation on S_r follows as in Section 4.5, yielding the logarithmic Green function and the $1/r$ shell acceleration.

F.9 Conclusion

The derivation presented in this appendix establishes the spiral winding and shell confinement mechanism within standard linearized General Relativity, without invoking the WTMedium, a co-moving transport assumption, or any parameter beyond the observed rotation profile $\Omega(r)$ and the galaxy age T_{gal}.

The following steps are *Derived* directly from the linearized Einstein equations and standard asymptotic methods:

- The retarded potential formulation (214) and its reduction to the scalar wave equation (213) in the slow-motion regime.
- The suppression of non-axisymmetric modes (227) under long-time orbital averaging, via the Riemann–Lebesgue lemma [1].
- The same-shell condition $r' \approx r$ (219), established by the stationary-phase analysis of the retarded integral, valid throughout the disk by five orders of magnitude (near-zone check, Section F.3).
- The local dispersion relation $\omega = m\Omega(r)$, $|\mathbf{k}| = \omega/c$ (220), and the Doppler-shifted form $\omega_{\text{eff}}^2 = c^2|\mathbf{k}|^2$ (239), both following from the same-shell condition (219) without any co-moving medium assumption.
- The ray equations (242) from the Hamiltonian $H = \mathbf{v}\cdot\mathbf{k} + c|\mathbf{k}|$.
- The linearised winding solution $k_r \approx k_0$, $k_\theta \approx 2A_{\text{Oort}}\,k_0\,t$ (243), the decay of the pitch angle $k_r/k_\theta \to 0$, and the suppression of the radial group velocity $c\,k_r/|\mathbf{k}| \to 0$, on timescale $t_{\text{wind}} \sim 1/(2A_{\text{Oort}}) \sim T_{\text{orb}}/\pi \ll T_{\text{gal}}$.

- The radial coherence length estimate (229)–(231): coherent radial structure in the time-averaged field is confined to scales $\ell_{\mathrm{coh}}^{\mathrm{eff}} \sim (T_{\mathrm{orb}}/T_{\mathrm{gal}})\, r/2\pi \sim 50\,\mathrm{pc}$ at $r = 10\,\mathrm{kpc}$, two orders of magnitude below the galactic radius.

The following step is *Argued*:

- The extrapolation from the coherence-length estimate (231) to complete radial incoherence on galactic scales. The suppression factor $T_{\mathrm{orb}}/T_{\mathrm{gal}} \sim 10^{-2}$ is controlled and quantified, but explicit error bounds have not been tracked through the full retarded Green function of the screened equation. Establishing these would promote this step to Derived.

The two tracks of the argument — orbital averaging of the background field and eikonal winding of individual perturbations — are united by the same-shell condition (219). Both operate on the same constant-r configuration: averaging suppresses radial coherence on timescale T_{gal}, while winding suppresses radial group velocity on timescale t_{wind}. The shell S_r is the structure selected by both.

Given the above, the confinement of coherent curvature transport to constant-radius shells follows as the dynamical attractor of the system, and the subsequent dimensional reduction to a two-dimensional Poisson equation on S_r proceeds exactly as in Section 4.5, yielding the logarithmic Green function and the $1/r$ shell acceleration. The shell-supported transport channel is therefore not an additional assumption but a direct consequence of causal propagation, long-time orbital averaging, and differential rotation acting together in standard linearized General Relativity.

G Appendix: Per-Galaxy Fit Results

Table 7 lists the per-galaxy optimised fit results for all 175 SPARC galaxies, sorted by A_{morph} within each calibration bin. For each galaxy the best-fit α_0 was found by grid search over $\alpha_0 \in [0.05, 3.0]$ in steps of 0.02. Columns are: galaxy name; calibration bin (G1–G4, Table 2); best-fit α_0 and the derived $A_{\mathrm{morph}} = \alpha_0/\kappa$; reduced χ^2 at best-fit (χ^2_{best}) and at universal κ (χ^2_κ); reduced χ^2 for MOND and ΛCDM; and the win assignment (WT / MOND / ΛCDM). The bins G1–G4 are an operational partition of a statistically unimodal A_{morph} distribution (Section 6.3); they are used here for sorting and readability only. The 151 WT wins, median $\chi^2_{\mathrm{best}} = 2.82$, and 6 catastrophic failures ($\chi^2 > 100$) quoted in Section 6.3 are directly verifiable from this table.

Table 7: Per-galaxy optimised fit results for all 175 SPARC galaxies, sorted by A_{morph} within calibration bin. χ^2 values are reduced (per data point). Win column: WT / MOND / ΛCDM; ✓ = lowest χ^2, – = not winner.

Galaxy	**Cl.**	α_0	A_{morph}	χ^2_{best}	χ^2_κ	χ^2_{MOND}	$\chi^2_{\Lambda\mathrm{CDM}}$	**Win**
G1 — Low-coupling ($A_{\mathrm{morph}} < 0.50$)								
NGC3949	G1	0.050	0.045	12.9	18.6	28.1	39.5	✓ – –
NGC4389	G1	0.050	0.045	43.2	65.3	79.9	98.3	✓ – –
NGC5005	G1	0.050	0.045	10.7	11.5	14.0	16.8	✓ – –
UGC02455	G1	0.050	0.045	37.4	94.8	134.8	124.4	✓ – –
UGC06973	G1	0.050	0.045	333.4	349.7	390.6	428.7	✓ – –
NGC4085	G1	0.070	0.064	14.3	21.6	31.5	48.1	✓ – –
CamB	G1	0.210	0.191	3.7	87.3	139.7	31.8	✓ – –
F574-2	G1	0.210	0.191	0.1	18.5	29.3	5.9	✓ – –
PGC51017	G1	0.230	0.209	1.9	96.1	156.0	21.6	✓ – –
F563-V1	G1	0.250	0.227	0.3	27.0	42.7	5.3	✓ – –
UGC07577	G1	0.250	0.227	0.2	14.8	23.4	3.6	✓ – –
NGC4051	G1	0.270	0.245	2.1	9.4	17.6	35.1	✓ – –
UGC06628	G1	0.310	0.281	0.3	10.9	21.2	4.5	✓ – –
UGC04305	G1	0.330	0.300	1.6	41.8	78.9	15.4	✓ – –

Continued on next page

Table 7 continued

Galaxy	**Cl.**	α_0	A_{morph}	χ^2_{best}	χ^2_{κ}	χ^2_{MOND}	$\chi^2_{\Lambda\mathrm{CDM}}$	**Win**
UGC11557	G1	0.370	0.336	2.0	11.7	27.1	16.0	✓ – –
F561-1	G1	0.390	0.354	0.6	27.0	54.4	9.5	✓ – –
NGC3877	G1	0.390	0.354	7.6	13.5	21.9	47.0	✓ – –
NGC4088	G1	0.470	0.427	26.7	38.8	66.3	90.4	✓ – –
UGC08837	G1	0.530	0.481	4.7	27.0	57.7	14.8	✓ – –
NGC5371	G1	0.550	0.499	29.3	102.9	276.6	419.9	✓ – –
G2 — Near-κ ($0.50 \leq A_{\mathrm{morph}} < 1.30$)								
NGC2955	G2	0.570	0.518	38.1	40.4	58.8	83.8	✓ – –
UGC07125	G2	0.570	0.518	1.3	76.7	177.4	18.4	✓ – –
UGC09992	G2	0.590	0.536	0.4	4.1	9.7	0.5	✓ – –
D564-8	G2	0.630	0.572	1.5	13.8	28.8	2.5	✓ – –
F567-2	G2	0.630	0.572	0.4	5.3	13.0	1.1	✓ – –
IC2574	G2	0.630	0.572	31.4	191.7	420.6	183.3	✓ – –
UGC07559	G2	0.630	0.572	0.5	5.7	14.5	1.0	✓ – –
KK98-251	G2	0.670	0.608	1.3	12.1	29.9	3.4	✓ – –
NGC4068	G2	0.670	0.608	1.7	11.3	33.9	4.5	✓ – –
UGC02023	G2	0.730	0.663	0.5	1.4	4.1	1.1	✓ – –
UGC07866	G2	0.730	0.663	0.0	1.9	6.2	0.1	✓ – –
NGC6195	G2	0.750	0.681	31.3	35.3	65.9	111.7	✓ – –
UGC09037	G2	0.750	0.681	21.5	28.5	60.1	68.6	✓ – –
NGC3953	G2	0.770	0.699	0.9	2.1	8.9	34.4	✓ – –
NGC5055	G2	0.770	0.699	> 999	> 999	> 999	> 999	– – –
NGC0891	G2	0.790	0.717	147.9	153.1	207.6	330.9	✓ – –
UGC07089	G2	0.790	0.717	0.5	4.0	18.5	3.5	✓ – –
NGC3726	G2	0.810	0.735	6.2	11.1	37.3	49.4	✓ – –
NGC6946	G2	0.810	0.735	39.4	46.4	94.6	194.8	✓ – –
UGC05750	G2	0.810	0.735	0.5	1.9	6.0	1.7	✓ – –

Continued on next page

Table 7 continued

Galaxy	**Cl.**	α_0	A_{morph}	χ^2_{best}	χ^2_{κ}	χ^2_{MOND}	$\chi^2_{\Lambda\text{CDM}}$	**Win**
D512-2	G2	0.830	0.754	0.1	1.9	8.1	0.2	✓ – –
NGC0801	G2	0.830	0.754	21.8	42.0	174.9	161.2	✓ – –
DDO161	G2	0.850	0.772	2.3	33.9	137.0	6.5	✓ – –
DDO170	G2	0.870	0.790	3.5	37.3	162.0	2.3	– – ✓
UGC02916	G2	0.870	0.790	176.4	178.4	240.9	340.8	✓ – –
UGC06818	G2	0.870	0.790	4.1	5.5	14.5	6.1	✓ – –
UGC04483	G2	0.890	0.808	0.4	2.2	11.6	2.4	✓ – –
UGC05005	G2	0.890	0.808	0.4	1.0	4.2	1.1	✓ – –
NGC4217	G2	0.930	0.844	69.0	69.4	84.3	102.9	✓ – –
NGC0055	G2	0.950	0.863	3.7	8.2	58.2	8.9	✓ – –
NGC2366	G2	0.950	0.863	4.3	6.5	29.3	6.0	✓ – –
D631-7	G2	0.970	0.881	10.8	12.6	31.6	10.9	✓ – –
NGC2976	G2	0.970	0.881	1.1	1.2	4.2	5.4	✓ – –
F583-4	G2	0.990	0.899	0.1	0.5	6.3	0.2	✓ – –
NGC4157	G2	0.990	0.899	9.7	10.1	23.3	35.3	✓ – –
NGC4559	G2	1.010	0.917	1.0	1.6	23.8	7.9	✓ – –
UGC05414	G2	1.010	0.917	1.3	2.0	19.6	2.3	✓ – –
NGC1090	G2	1.030	0.935	4.2	6.2	102.4	47.1	✓ – –
UGC05829	G2	1.030	0.935	0.1	0.1	3.0	0.2	✓ – –
UGC07323	G2	1.030	0.935	0.9	1.2	12.8	2.5	✓ – –
UGC11820	G2	1.050	0.953	1.7	4.9	118.5	1.2	– – ✓
NGC0289	G2	1.070	0.972	4.1	4.2	21.8	10.8	✓ – –
UGC05918	G2	1.070	0.972	0.4	0.4	2.3	1.0	✓ – –
UGC06614	G2	1.070	0.972	2.3	2.3	6.7	12.8	✓ – –
UGCA281	G2	1.070	0.972	0.3	0.3	7.8	2.4	✓ – –
UGCA444	G2	1.070	0.972	0.1	0.1	1.3	0.6	✓ – –
NGC3769	G2	1.090	0.990	3.2	3.2	10.6	7.0	✓ – –
UGC01230	G2	1.090	0.990	2.8	2.8	3.6	2.0	– – ✓

Continued on next page

Table 7 continued

Galaxy	Cl.	α_0	A_{morph}	χ^2_{best}	χ^2_{κ}	χ^2_{MOND}	$\chi^2_{\Lambda\mathrm{CDM}}$	Win
UGC04499	G2	1.090	0.990	1.1	1.1	15.2	2.2	✓ – –
NGC2683	G2	1.110	1.008	3.4	3.4	9.4	14.0	✓ – –
NGC7331	G2	1.110	1.008	114.0	114.0	162.6	260.3	✓ – –
UGC05999	G2	1.110	1.008	1.9	1.9	8.2	2.4	✓ – –
UGC07524	G2	1.110	1.008	1.4	1.4	9.6	1.3	– – ✓
UGC09133	G2	1.110	1.008	74.6	74.6	252.1	335.1	✓ – –
UGC11455	G2	1.110	1.008	21.6	21.6	35.9	70.7	✓ – –
UGC12632	G2	1.110	1.008	2.1	2.1	9.2	2.7	✓ – –
F568-3	G2	1.130	1.026	2.8	2.8	4.9	3.7	✓ – –
NGC1003	G2	1.130	1.026	3.2	3.6	33.0	4.0	✓ – –
UGC03580	G2	1.130	1.026	72.6	72.8	128.1	130.9	✓ – –
F571-V1	G2	1.150	1.044	0.2	0.3	1.7	0.7	✓ – –
NGC4183	G2	1.150	1.044	2.1	2.2	7.5	0.6	– – ✓
UGC10310	G2	1.150	1.044	0.8	0.9	3.7	1.1	✓ – –
DDO168	G2	1.170	1.062	9.9	10.4	23.9	19.3	✓ – –
NGC3198	G2	1.170	1.062	9.1	10.8	45.1	28.9	✓ – –
NGC4013	G2	1.170	1.062	6.4	6.9	23.0	35.3	✓ – –
NGC5585	G2	1.170	1.062	16.5	17.4	44.2	39.0	✓ – –
UGC00891	G2	1.170	1.062	8.4	14.2	62.0	55.7	✓ – –
UGC01281	G2	1.170	1.062	1.2	1.3	2.0	2.3	✓ – –
UGC07690	G2	1.170	1.062	1.1	1.2	3.5	0.7	– – ✓
NGC0100	G2	1.190	1.081	1.3	1.5	5.3	2.4	✓ – –
UGC06930	G2	1.190	1.081	1.1	1.3	5.1	0.5	– – ✓
UGC00191	G2	1.210	1.099	7.5	13.7	41.7	22.2	✓ – –
DDO064	G2	1.230	1.117	0.4	0.6	0.8	1.5	✓ – –
NGC5907	G2	1.230	1.117	11.2	20.6	88.2	71.9	✓ – –
DDO154	G2	1.250	1.135	15.2	77.1	72.9	336.5	✓ – –
NGC2998	G2	1.250	1.135	4.3	12.5	45.6	21.4	✓ – –

Continued on next page

Table 7 continued

Galaxy	**Cl.**	α_0	A_{morph}	χ^2_{best}	χ^2_{κ}	χ^2_{MOND}	$\chi^2_{\Lambda\mathrm{CDM}}$	**Win**
NGC7793	G2	1.250	1.135	1.0	1.2	1.5	2.4	✓ – –
UGC00128	G2	1.250	1.135	58.6	117.4	120.7	42.9	– – ✓
UGC06923	G2	1.250	1.135	1.0	1.4	3.7	1.8	✓ – –
NGC4010	G2	1.270	1.153	2.7	3.6	7.0	5.7	✓ – –
UGC12732	G2	1.270	1.153	1.0	3.1	2.6	1.9	✓ – –
NGC6503	G2	1.290	1.171	13.3	40.4	51.9	30.4	✓ – –
UGC07151	G2	1.290	1.171	2.4	4.6	8.6	2.8	✓ – –
UGC07261	G2	1.290	1.171	0.6	1.2	1.4	1.4	✓ – –
NGC0247	G2	1.310	1.189	3.1	11.2	10.3	3.3	✓ – –
NGC4138	G2	1.330	1.208	1.9	2.2	3.0	3.5	✓ – –
UGC05716	G2	1.330	1.208	15.9	67.4	22.3	95.4	✓ – –
NGC2903	G2	1.370	1.244	127.0	140.7	162.8	280.1	✓ – –
NGC3109	G2	1.390	1.262	7.1	11.5	7.1	14.8	✓ – –
NGC3521	G2	1.390	1.262	22.7	23.0	27.4	32.7	✓ – –
UGC00731	G2	1.390	1.262	9.9	15.9	10.5	11.9	✓ – –
UGC04278	G2	1.390	1.262	2.2	4.0	2.7	3.7	✓ – –
UGCA442	G2	1.390	1.262	2.1	18.2	3.3	26.3	✓ – –
F574-1	G2	1.410	1.280	2.0	6.2	2.0	5.5	– ✓ –
NGC6674	G2	1.410	1.280	25.3	115.9	31.8	54.8	✓ – –
UGC08550	G2	1.410	1.280	1.1	8.6	0.9	17.8	– ✓ –
F583-1	G2	1.430	1.298	1.8	3.4	1.8	2.4	✓ – –
NGC3741	G2	1.430	1.298	1.7	5.6	2.3	3.8	✓ – –
NGC3917	G2	1.430	1.298	3.2	10.6	5.9	2.5	– – ✓
UGC03546	G2	1.430	1.298	18.0	61.3	31.1	190.9	✓ – –
G3 — Enhanced ($1.30 \leq A_{\mathrm{morph}} < 2.00$)								
UGC00634	G3	1.450	1.317	1.2	69.6	1.3	57.6	✓ – –
UGC06399	G3	1.450	1.317	0.4	3.8	0.5	4.1	✓ – –

Continued on next page

Table 7 continued

Galaxy	**Cl.**	α_0	$A_{\rm morph}$	$\chi^2_{\rm best}$	χ^2_κ	$\chi^2_{\rm MOND}$	$\chi^2_{\Lambda{\rm CDM}}$	**Win**
NGC0300	G3	1.470	1.335	0.6	8.1	1.2	7.3	✓ – –
NGC3893	G3	1.490	1.353	9.1	11.0	11.6	16.9	✓ – –
UGC02885	G3	1.490	1.353	1.1	9.2	1.9	4.4	✓ – –
UGC06917	G3	1.490	1.353	0.9	8.5	1.3	4.1	✓ – –
NGC4100	G3	1.510	1.371	3.0	9.8	3.6	4.7	✓ – –
NGC5033	G3	1.550	1.407	36.4	109.1	28.8	22.2	– – ✓
NGC6015	G3	1.550	1.407	15.4	36.8	14.8	8.9	– – ✓
NGC3992	G3	1.570	1.426	5.8	24.2	6.7	4.0	– – ✓
UGC03205	G3	1.590	1.444	5.8	51.5	4.5	6.6	– ✓ –
UGC06983	G3	1.590	1.444	1.6	10.7	1.2	6.8	– ✓ –
UGC07232	G3	1.590	1.444	1.7	3.3	1.9	3.7	✓ – –
F565-V2	G3	1.630	1.480	0.3	3.6	0.8	3.1	✓ – –
F579-V1	G3	1.630	1.480	2.9	5.1	2.6	3.4	– ✓ –
ESO079-G014	G3	1.670	1.516	3.0	44.1	2.8	4.8	– ✓ –
NGC3972	G3	1.670	1.516	1.1	9.4	1.1	2.0	– ✓ –
UGC05253	G3	1.670	1.516	86.4	113.6	109.1	145.0	✓ – –
UGC06446	G3	1.670	1.516	1.6	10.8	2.2	11.0	✓ – –
IC4202	G3	1.690	1.535	46.8	54.5	50.6	62.7	✓ – –
NGC2403	G3	1.710	1.553	26.9	535.3	24.1	186.6	– ✓ –
UGC08490	G3	1.710	1.553	2.5	16.6	3.2	16.1	✓ – –
UGC07603	G3	1.750	1.589	1.2	14.1	3.1	18.5	✓ – –
UGC08286	G3	1.770	1.607	6.1	58.4	11.9	59.3	✓ – –
ESO116-G012	G3	1.790	1.625	1.9	44.5	5.6	29.9	✓ – –
UGC07608	G3	1.790	1.625	0.4	3.1	1.1	3.4	✓ – –
F563-1	G3	1.810	1.643	1.2	4.4	2.0	2.7	✓ – –
UGC02259	G3	1.810	1.643	11.0	95.4	21.3	88.8	✓ – –
F571-8	G3	1.830	1.662	37.8	55.0	43.0	58.0	✓ – –
UGC08699	G3	1.850	1.680	8.8	27.2	16.6	15.3	✓ – –

Continued on next page

Table 7 continued

Galaxy	**Cl.**	α_0	A_{morph}	χ^2_{best}	χ^2_{κ}	χ^2_{MOND}	$\chi^2_{\Lambda\text{CDM}}$	**Win**
UGC12506	G3	1.850	1.680	6.0	15.9	6.3	3.5	– – ✓
UGC11914	G3	1.890	1.716	9.7	11.4	14.8	30.8	✓ – –
UGC02953	G3	1.970	1.789	55.4	344.1	133.6	118.6	✓ – –
NGC0024	G3	2.010	1.825	3.8	16.8	6.8	10.7	✓ – –
UGC02487	G3	2.050	1.861	67.9	565.2	151.9	28.9	– – ✓
UGC05986	G3	2.050	1.861	3.5	67.2	14.8	39.4	✓ – –
NGC2841	G3	2.150	1.952	18.2	158.1	51.6	13.7	– – ✓
NGC4214	G3	2.170	1.970	0.8	9.3	3.1	7.7	✓ – –
UGC04325	G3	2.170	1.970	8.6	56.7	19.7	45.5	✓ – –
UGC06667	G3	2.170	1.970	3.4	41.5	20.5	32.2	✓ – –
G4 — High-coupling ($A_{\text{morph}} \geq 2.00$)								
NGC2915	G4	2.210	2.007	1.1	10.7	5.5	8.2	✓ – –
UGC05764	G4	2.230	2.025	28.2	480.7	224.9	608.4	✓ – –
ESO444-G084	G4	2.310	2.097	1.2	63.0	30.0	68.3	✓ – –
NGC7814	G4	2.430	2.206	7.2	121.7	46.3	16.3	✓ – –
UGC05721	G4	2.430	2.206	2.7	22.4	10.4	19.5	✓ – –
ESO563-G021	G4	2.450	2.225	17.9	54.8	28.9	17.5	– – ✓
NGC1705	G4	2.450	2.225	1.2	23.5	10.1	22.5	✓ – –
UGC06786	G4	2.450	2.225	9.5	328.0	109.1	89.3	✓ – –
F568-1	G4	2.470	2.243	0.8	10.4	4.8	6.5	✓ – –
F563-V2	G4	2.550	2.315	0.7	6.4	3.1	4.9	✓ – –
F568-V1	G4	2.550	2.315	1.0	11.6	5.7	8.7	✓ – –
UGC06787	G4	2.610	2.370	61.0	435.1	225.3	93.1	✓ – –
NGC5985	G4	2.830	2.570	45.3	237.6	132.5	53.0	✓ – –
NGC6789	G4	2.990	2.715	1.3	9.7	5.5	9.0	✓ – –
UGC07399	G4	2.990	2.715	2.3	114.3	61.2	91.4	✓ – –
Summary: 175 galaxies, median $\chi^2_{\text{best}} = 2.82$, WT 151/175 (86%), MOND 8/175, ΛCDM 16/175.								

References

[1] L. Blanchet and T. Damour, *Hereditary effects in gravitational radiation*, Phys. Rev. D **46**, 4304 (1992).

[2] M. Milgrom, *A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis*, Astrophys. J. **270**, 365 (1983).

[3] J. Bekenstein and M. Milgrom, *Does the missing mass problem signal the breakdown of Newtonian gravity?*, Astrophys. J. **286**, 7 (1984).

[4] J. D. Bekenstein, *Relativistic gravitation theory for the modified Newtonian dynamics paradigm*, Phys. Rev. D **70**, 083509 (2004).

[5] C. W. Misner, K. S. Thorne, and J. A. Wheeler, *Gravitation*, Freeman, San Francisco (1973).

[6] S. Weinberg, *Gravitation and Cosmology*, Wiley, New York (1972).

[7] E. Poisson and C. M. Will, *Gravity: Newtonian, Post-Newtonian, Relativistic*, Cambridge University Press, Cambridge (2014).

[8] L. C. Evans, *Partial Differential Equations*, 2nd ed., American Mathematical Society, Providence (2010).

[9] M. E. Taylor, *Partial Differential Equations I: Basic Theory*, Springer, New York (2011).

[10] I. Chavel, *Eigenvalues in Riemannian Geometry*, Academic Press, Orlando (1984).

[11] S. Rosenberg, *The Laplacian on a Riemannian Manifold*, Cambridge University Press, Cambridge (1997).

[12] C. M. Bender and S. A. Orszag, *Advanced Mathematical Methods for Scientists and Engineers*, Springer, New York (1999).

[13] A. V. Dutton and A. V. Macciò, *Cold dark matter haloes in the Planck era: evolution of structural parameters for Milky Way-sized haloes*, Mon. Not. R. Astron. Soc. **441**, 3359 (2014), arXiv:1402.7073.

[14] J. Binney and S. Tremaine, *Galactic Dynamics*, 2nd ed., Princeton University Press, Princeton (2008).

[15] D. Lynden-Bell and A. J. Kalnajs, *On the generating mechanism of spiral structure*, Mon. Not. R. Astron. Soc. **157**, 1 (1972).

[16] A. Toomre, *What amplifies the spirals*, in *Structure and Evolution of Normal Galaxies*, ed. S. M. Fall and D. Lynden-Bell, Cambridge University Press, Cambridge (1981), pp. 111–136.

[17] F. Lelli, S. S. McGaugh, and J. M. Schombert, *SPARC: mass models for 175 disk galaxies with Spitzer photometry and accurate rotation curves*, Astron. J. **152**, 157 (2016).

[18] S. S. McGaugh, F. Lelli, and J. M. Schombert, *Radial acceleration relation in rotationally supported galaxies*, Phys. Rev. Lett. **117**, 201101 (2016).

[19] P. C. Peters and J. Mathews, *Gravitational radiation from point masses in a Keplerian orbit*, Phys. Rev. **131**, 435 (1963).

[20] L. Blanchet, *Gravitational radiation from post-Newtonian sources and inspiralling compact binaries*, Living Rev. Relativ. **17**, 2 (2014).

[21] R. Camporesi and A. Higuchi, *On the eigenfunctions of the Laplacian on spheres and hyperbolic spaces*, J. Geom. Phys. **20**, 1 (1996), arXiv:gr-qc/9505009.

[22] B. Famaey and S. McGaugh, *Modified Newtonian Dynamics (MOND): Observational Phenomenology and Relativistic Extensions*, Living Rev. Relativ. **15**, 10 (2012), arXiv:1112.3960.

[23] F. Walter, E. Brinks, W. J. G. de Blok, F. Bigiel, R. C. Kennicutt, M. D. Thornley, and A. Leroy, *THINGS: The* H I *Nearby Galaxy Survey*, Astron. J. **136**, 2563 (2008).

[24] M. G. Walker, M. Mateo, E. W. Olszewski, J. Peñarrubia, N. W. Evans, and G. Gilmore, *A universal mass profile for dwarf spheroidal galaxies*, Astrophys. J. **704**, 1274 (2009).

[25] M. Cappellari et al., *The ATLAS^{3D} project — I. A volume-limited sample of 260 nearby early-type galaxies*, Mon. Not. R. Astron. Soc. **413**, 813 (2011), arXiv:1012.1551.

[26] A. L. Tiley et al., *The KMOS Galaxy Evolution Survey (KGES): the $z \sim 1.5$ Tully–Fisher relation*, Mon. Not. R. Astron. Soc. **496**, 649 (2020), arXiv:2004.04683.

[27] T. Urrutia et al., *The MUSE-Wide survey: a measurement of the Lyman-alpha emitter luminosity function at $3 < z < 6$*, Astron. Astrophys. **624**, A141 (2019), arXiv:1811.09635.

[28] D. J. Eisenstein et al., *Overview of the JWST Advanced Deep Extragalactic Survey (JADES)*, Astrophys. J. Suppl. **269**, 5 (2023), arXiv:2306.02465.